

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J117 MLB

LAST MODIFIED=Thu Sep 18 13:37:48 2014
LAST MODIFICATION=Thu Sep 18 13:37:48 2014

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
3	0003501996	ENGINEERING RELEASED	2014-12-09

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DRAWING
TITLE=J16 MLB_IG
ABBREV=DRAWING
LAST_MODIFIED=Thu Sep 18 13:37:48 2014

DRAWING TITLE		SCHEM, MLB, J117	
 Apple Inc.	DRAWING NUMBER		SIZE
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	REVISION		
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Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-00081	1	SCH,MLB,J117	SCH	CRITICAL	
820-00034	1	PCBF,MLB,J117	PCB	CRITICAL	
685-00013	1	PCBA,MLB,J117,COMMON PARTS	CHNPTS		MLB_CHNPTS

Main BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-00016	PCBA,MLB,DEV,J117	DEVELOPMENT,J117_DEVEL
939-00158	PCBA,MLB,DEV,J117,CPU_INT	MLB_CMNPTS,ALTERNATE,CPU:SOCKET,DDR3:HYNIX_8GB_1866,SSD:Y
639-00910	PCBA,MLB,J117,HY_8GB_29NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_8GB_1866,SSD:N
639-00911	PCBA,MLB,J117,EL_8GB_25NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_8GB_1866,SSD:N
639-00912	PCBA,MLB,J117,SA_8GB_23NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:SAMSUNG_8GB_1866,SSD:N
639-00913	PCBA,MLB,J117,EL_16GB_25NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_16GB_1866,SSD:N
639-00914	PCBA,MLB,J117,HY_16GB_25NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_16GB_1866,SSD:N
639-00713	PCBA,MLB,J117,HY_8GB_29NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_8GB_1866,SSD:Y
639-00714	PCBA,MLB,J117,EL_8GB_25NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_8GB_1866,SSD:Y
639-00715	PCBA,MLB,J117,SA_8GB_23NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:SAMSUNG_8GB_1866,SSD:Y
639-00716	PCBA,MLB,J117,EL_16GB_25NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_16GB_1866,SSD:Y
639-00717	PCBA,MLB,J117,HY_16GB_25NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_16GB_1866,SSD:Y
685-00013	PCBA,MLB,J117,COMMON PARTS	J117_COMMON

Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S00011	377S0155		ALL	USB3 diodes
377S0184	377S0155		ALL	USB3 diodes
377S0124	377S0057		ALL	TVS
155S0578	155S0367		ALL	120OHM EMI BEAD
128S0368	128S0365		ALL	150UF AL POLY
138S0681	138S0638		ALL	Taiyo 10uf 805 alt
197S0479	197S0478		ALL	12 MHz Cam. Xtal
197S0486	197S0478		ALL	12 MHz Cam. Xtal
107S0251	107S0249		ALL	Sense resistor
197S0481	197S0480		ALL	25MHz Xtal
197S0343	197S0480		ALL	25MHz Xtal
138S0860	138S0775		ALL	Single-source 1uF 402
138S0859	138S0788		ALL	Single-source 10uF
378S0391	378S0390		ALL	Debug LEDs
341S00016	341S3912		ALL	ENET ROM,ADESTO,V1.15
138S0747	138S0773		ALL	1uF,X6S,402
197S0542	197S0544		ALL	24 MHz PCH Xtal
197S0545	197S0544		ALL	24 Mhz PCH Xtal
197S0369	197S0392		ALL	32 KHz PCH Xtal
197S0399	197S0392		ALL	32 KHz PCH Xtal
376S0572	376S0659		ALL	Single P-Ch FET
376S0972	376S00075		ALL	Single N-Ch FET
132S00012	132S0401		ALL	0.22uF,X7R,0402
138S00012	138S0771		ALL	1uF,X6S,0402
138S1103	138S0719		ALL	4.7uF,X5R,0402
155S0830	155S0316		ALL	FER BD,600 OHM,0.5A,0603
155S00076	155S0546		ALL	FER BD,600 OHM,300NA,402

Strategic Silicon

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1089	376S1128		ALL	Single N-Ch FET
155S0660	155S0513		ALL	220HM EMI BEAD
155S0694	155S0387		ALL	4700HM EMI BEAD
127S0164	127S0162		ALL	1UF 25V TANT
376S00074	376S0855		ALL	Dual N-Ch FET
376S1129	376S0855		ALL	Dual N-Ch FET
376S00074	376S0855	SSD:Y	ALL	Dual N-Ch FET
376S1129	376S0855	SSD:Y	ALL	Dual N-Ch FET
311S0649	311S0541		ALL	Single AND Gate
107S0375	107S00039		ALL	DDR Sense Res
107S00011	107S0372		ALL	CPU VR Sense Res
353S3814	353S3812		ALL	TBT mux
353S4376	353S3384		ALL	HDD OOBv1 comparator
311S00014	311S0515		ALL	TBT JTAG ISOLATION BUFFER
311S00013	311S0508		ALL	Single Buffer Driver OD
311S00019	371S0463		ALL	Rohn Schottky Barrier Diode
376S00037	376S1193		ALL	30V,64A,Single N-Channel FET
376S00036	376S1194		ALL	30V,52A,Single N-Channel FET
138S0746	138S0705		ALL	CAP,CER,X5R,100F,20%,10V,402
152S1757	152S1821		ALL	TND,0.4ohm,23A
138S00013	138S0772		ALL	CAP,CER,2.2UF,20V,10V,K66,402
371S00017	371S0749		ALL	DIODE,SCHOTTKY,30V,1A,800-323

PART#	STRATEGIC VALUE	COMMENT
337800065	08	CPU,BDW-ULT,2*GT3
33881247	02	TST,Falcon Ridge-4c
335800006	07	IC,SERIAL FLASH,Quad-IO
333S0784	07	32Gb,25nm LPDDR3-1866
333S0786	07	16Gb,29nm LPDDR3-1866
333S0790	07	32Gb,25nm LPDDR3-1866
333S0792	07	16Gb,25nm LPDDR3-1866
333S00004	07	16Gb,23nm LPDDR3-1866

BOM Groups

BOM GROUP	BOM OPTIONS
J117_COMMON	COMMON,ALTERNATE,J117_PROGPARTS,SMCREG:SUP,XDP,SMBUS1:ISOL,USB_OC_ISO:Y,RTCRST:Y,AUDIO_DP_SNS:N
J117_PROGPARTS	SMC:PROG,BOOTROM:PROG,CAMROM:PROG,TBTROM:PROG,ENETROM:PROG
J117_DEVEL	XDP_CONN,TEMPSENSDEV,SAMCONN

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337800101	1	1C,CPU,BOM-ULT,GR3T,ES,CS-82,2/3,1-6,13W,,95,1168	U0500	CRITICAL	CPU:ULT
998-7866	1	INTERPODER,80A1168,SINGLE SIDE	U0500	CRITICAL	CPU:SOCKET

ASIC Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S1247	1	1C,T8T,FR-4C,A0,P90,C10,SR1JC,PCBQ208	U2800	CRITICAL	
343S0616	1	1C,BC957166A,C1V+,A0,8x8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341800158	1	IC,EFI,V0108,J117	U5210	CRITICAL	BOOTROM:PROG
335S00006	1	IC,SERIAL_FLASH,64MBIT,1V,8P,NSOM,QB#1	U5210	CRITICAL	BOOTROM:BLANK
341S00110	1	IC,SMC-B1,EXTERNAL,V2.24A19,POC,,J117	U5000	CRITICAL	SMC:PROG
338S1214	1	IC,SMC12-B1,40MHZ/50KHZ,NCT,1579GA	U5000	CRITICAL	SMC:BLANK
34183778	1	IC,CAMERA_FLASH,V7229,J316	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK
341800154	1	IC,EPROM,T29,FALCON RIDGE,V27-1,J117	U2890	CRITICAL	TBTROM:PROG
33580915	1	IC,FLASH,SPI,4MBIT,50MHZ	U2890	CRITICAL	TBTROM:BLANK
34183912	1	IC,ENET-8P1 ROM,NYNONYE,V1.15,216/0166/0117	U3990	CRITICAL	ENETROM:PROG
335S1025	1	IC,SERIAL_FLASH,2MBIT,2.7V,REV F	U3990	CRITICAL	ENETROM:BLANK

CPU DRAM SPD Straps

BOM GROUP	BOM OPTIONS
DDR3:HYNIX_8GB_1600	RAMCFG0:L, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, RAMCFG4:L, HYNIX_8GB_1600
DDR3:HYNIX_16GB_1600	RAMCFG0:L, RAMCFG1:L, RAMCFG2:H, RAMCFG3:L, RAMCFG4:L, HYNIX_16GB_1600
DDR3:HYNIX_8GB_1866	RAMCFG0:L, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, RAMCFG4:H, HYNIX_8GB_1866
DDR3:HYNIX_16GB_1866	RAMCFG0:L, RAMCFG1:L, RAMCFG2:H, RAMCFG3:L, RAMCFG4:H, HYNIX_16GB_1866
DDR3:ELPIDA_8GB_1600	RAMCFG0:H, RAMCFG1:H, RAMCFG2:L, RAMCFG3:L, RAMCFG4:L, ELPIDA_8GB_1600
DDR3:ELPIDA_16GB_1600	RAMCFG0:H, RAMCFG1:H, RAMCFG2:H, RAMCFG3:L, RAMCFG4:L, ELPIDA_16GB_1600
DDR3:ELPIDA_8GB_1866	RAMCFG0:H, RAMCFG1:H, RAMCFG2:L, RAMCFG3:L, RAMCFG4:H, ELPIDA_8GB_1866
DDR3:ELPIDA_16GB_1866	RAMCFG0:H, RAMCFG1:H, RAMCFG2:H, RAMCFG3:L, RAMCFG4:H, ELPIDA_16GB_1866
DDR3:SAMSUNG_8GB_1600	RAMCFG0:H, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, RAMCFG4:L, SAMSUNG_8GB_1600
DDR3:SAMSUNG_8GB_1866	RAMCFG0:H, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, RAMCFG4:H, SAMSUNG_8GB_1866

CPU DRAM CFG Chart

	VENDOR	CFG 1	CFG 0
	HYNIX	0	0
	SAMSUNG	0	1
	N/A	1	0
	ELPIDA	1	1

SIZE	CFG 2
8GB	0
16GB	1

DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0783	4	1C, SDRAM, 32mB 32Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	HYNIX_16GB_1600
333S0784	4	1C, SDRAM, 25mB 32Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	HYNIX_16GB_1866
333S0785	4	1C, SDRAM, 23mB 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	HYNIX_8GB_1600
333S0786	4	1C, SDRAM, 23mB 16Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	HYNIX_8GB_1866
333S0789	4	1C, SDRAM, 25mB 32Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	ELPIDA_16GB_1600
333S0790	4	1C, SDRAM, 25mB 32Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	ELPIDA_16GB_1866
333S0791	4	1C, SDRAM, 25mB 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	ELPIDA_8GB_1600
333S0792	4	1C, SDRAM, 25mB 16Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	ELPIDA_8GB_1866
333S00003	4	1C, SDRAM, 23mB 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	SAMSUNG_8GB_1600
333S00004	4	1C, SDRAM, 23mB 16Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	SAMSUNG_8GB_1866

DIE REV	CFG 3	SPEED	CFG 4
A	0	1600	0
B	1	1866	1

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PAGE TITLE			
BOM Configuration			
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D

C

B

A

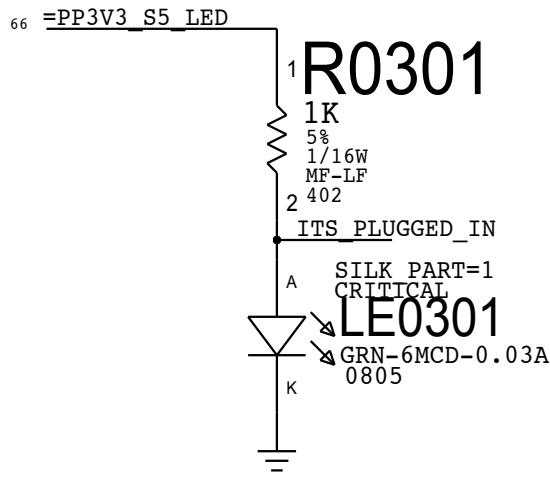
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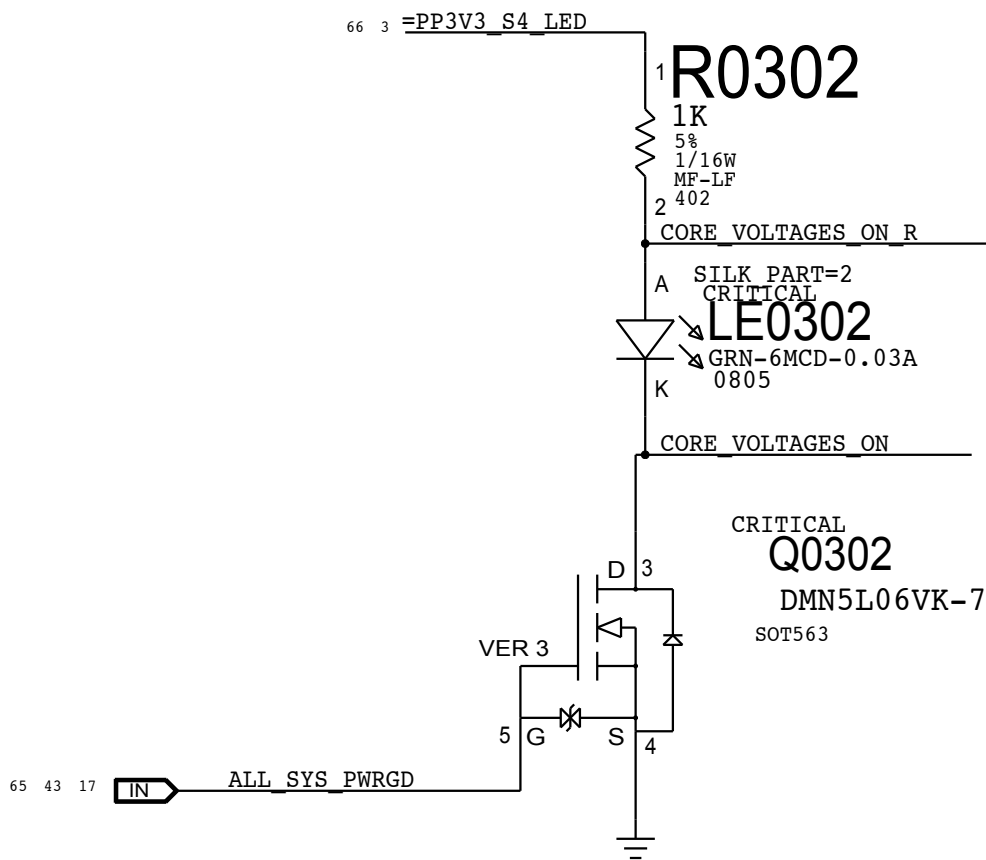
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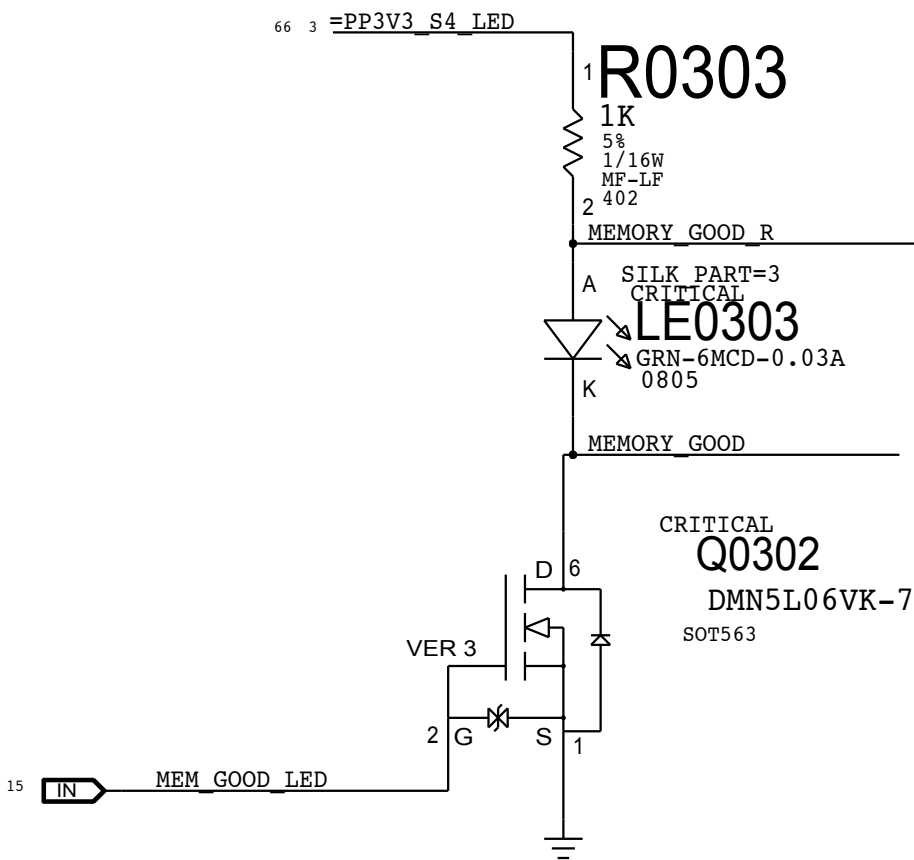
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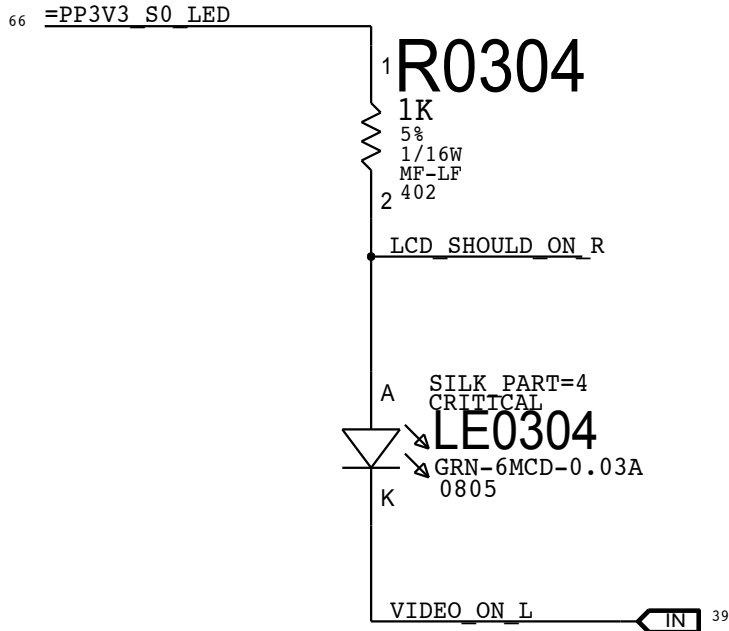
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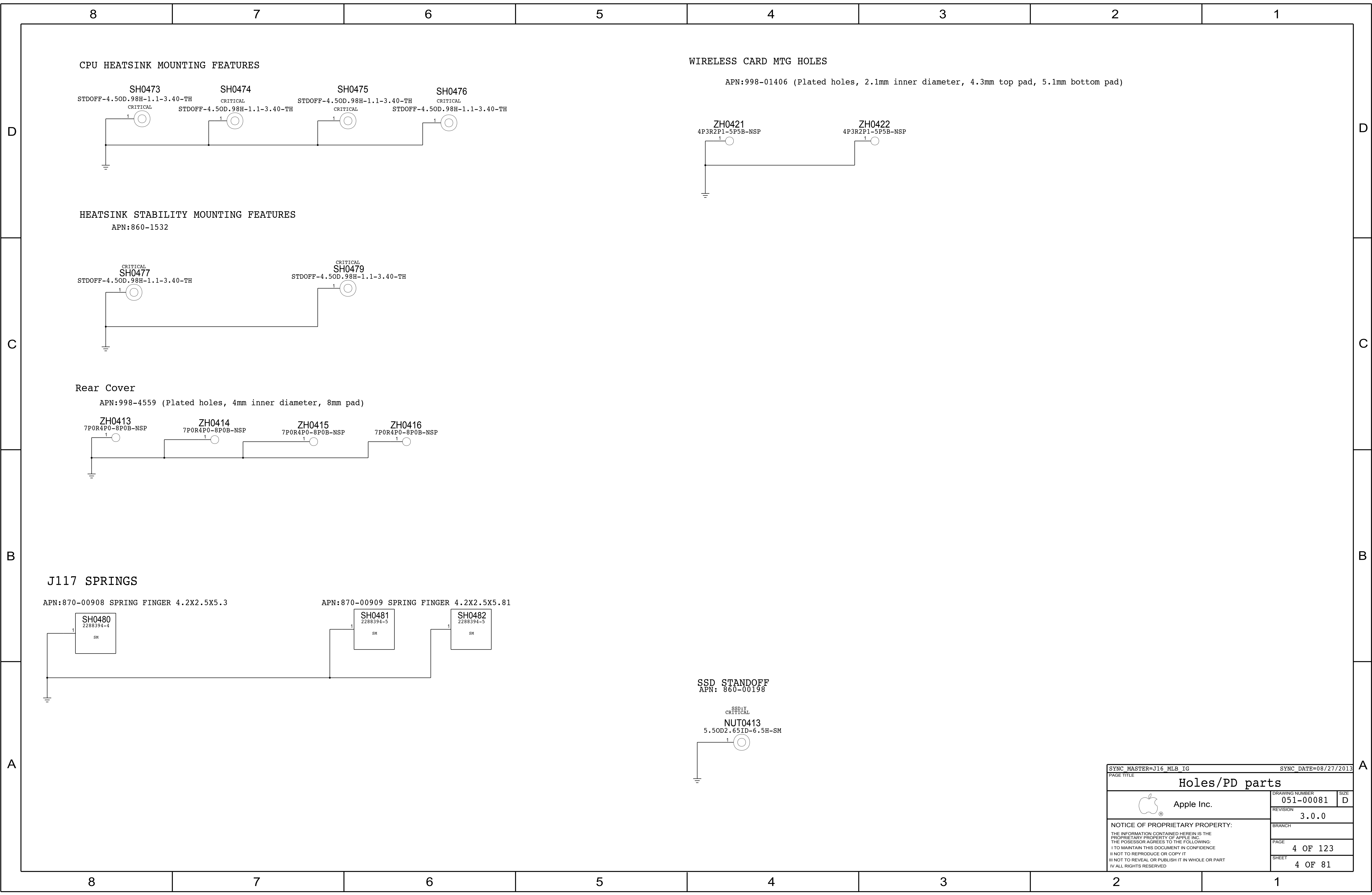


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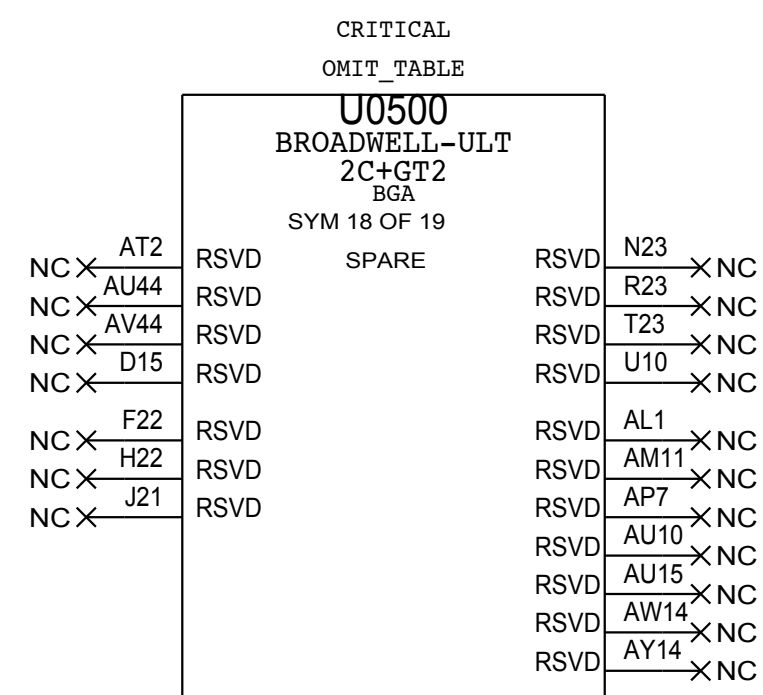
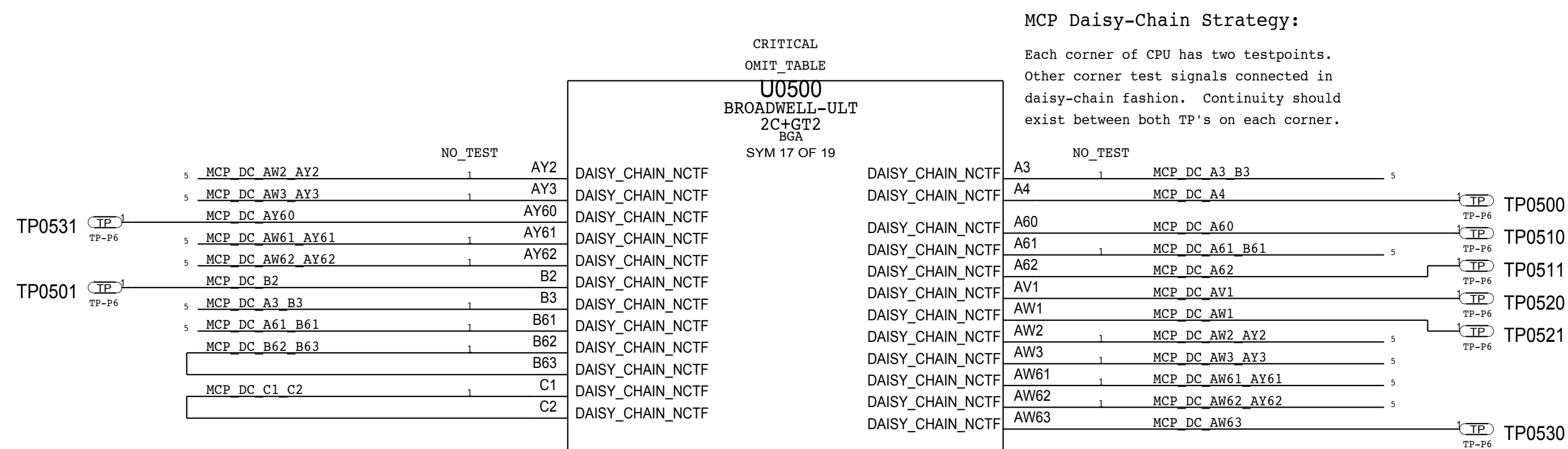
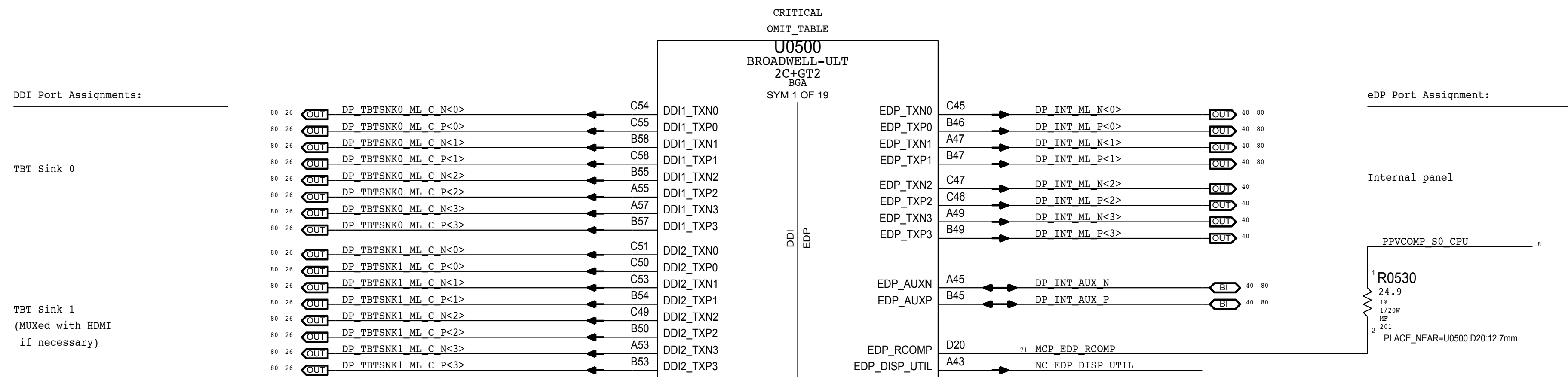


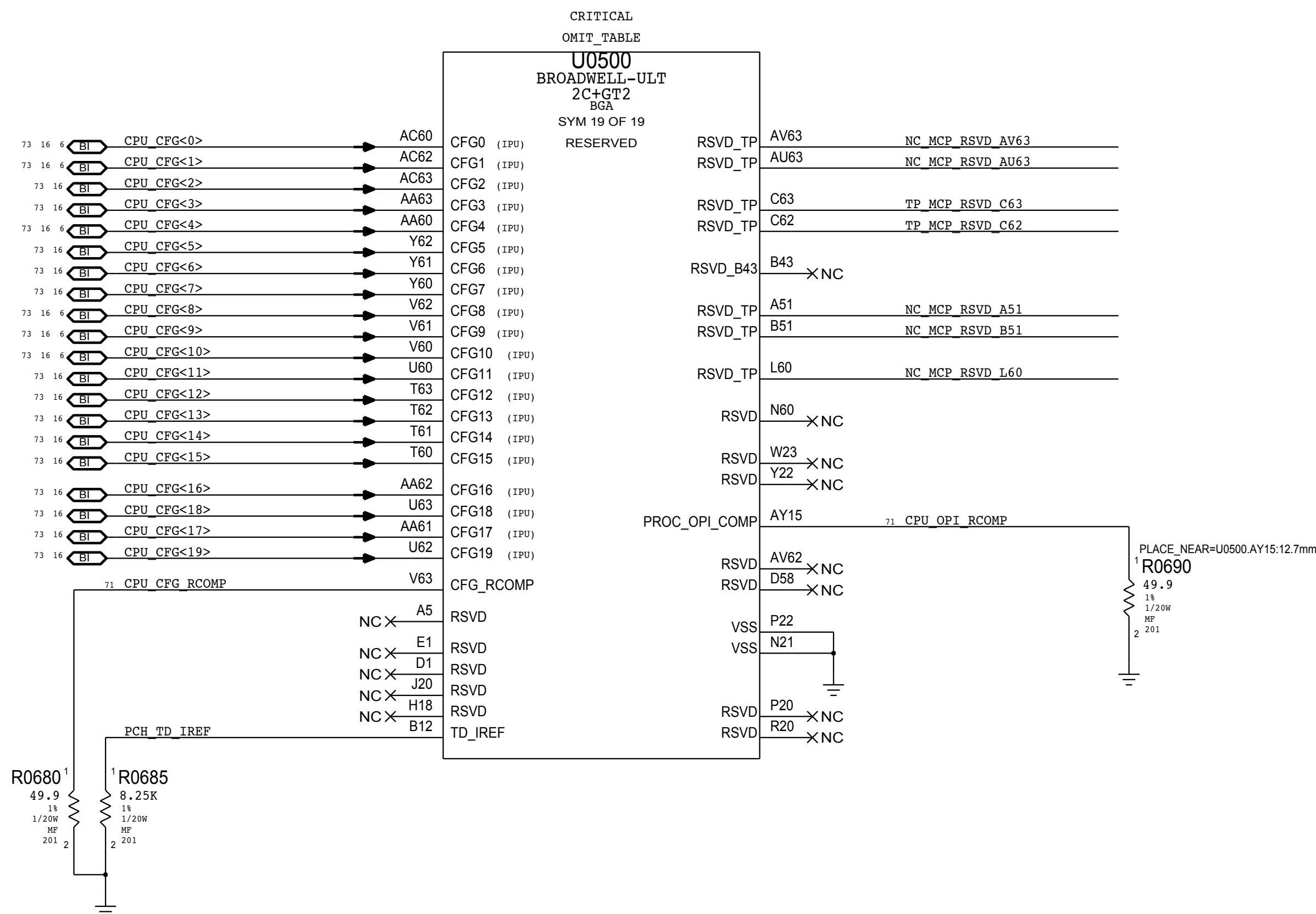
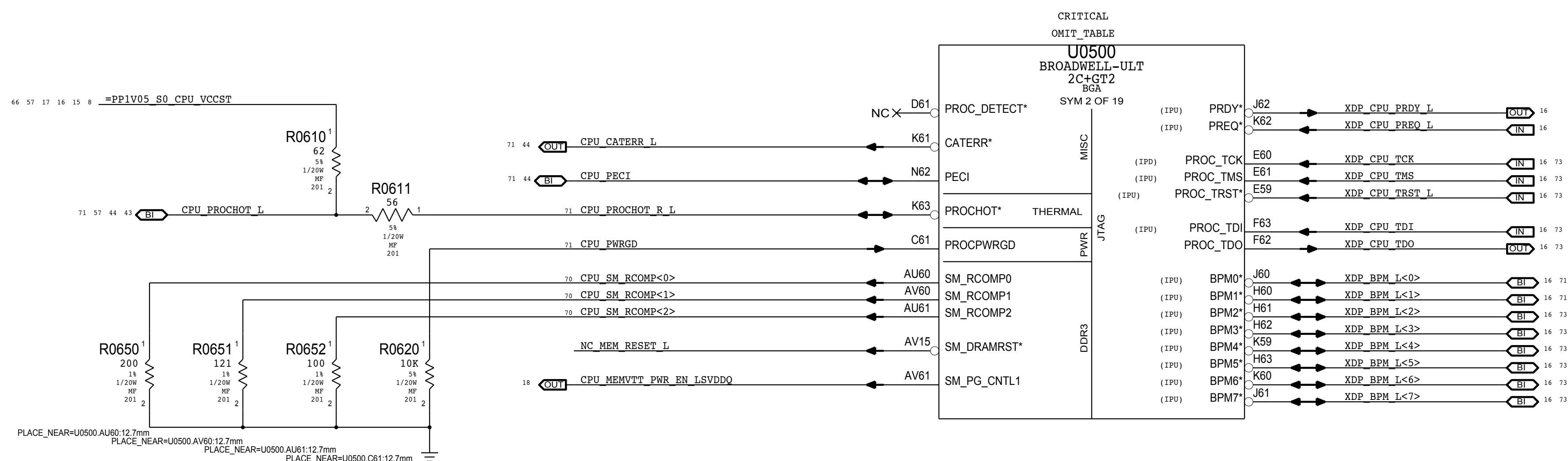
VIDEO ON Led



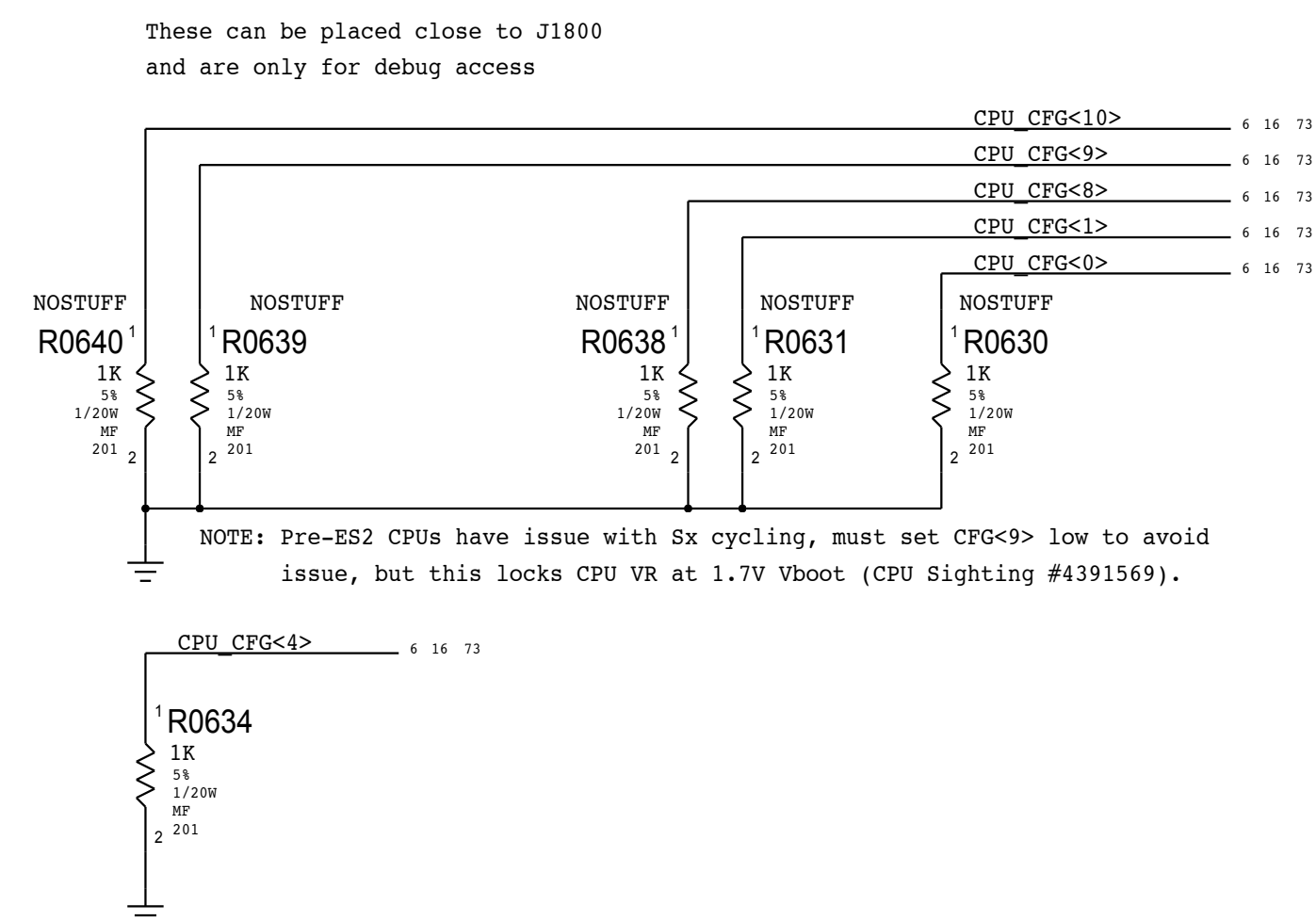


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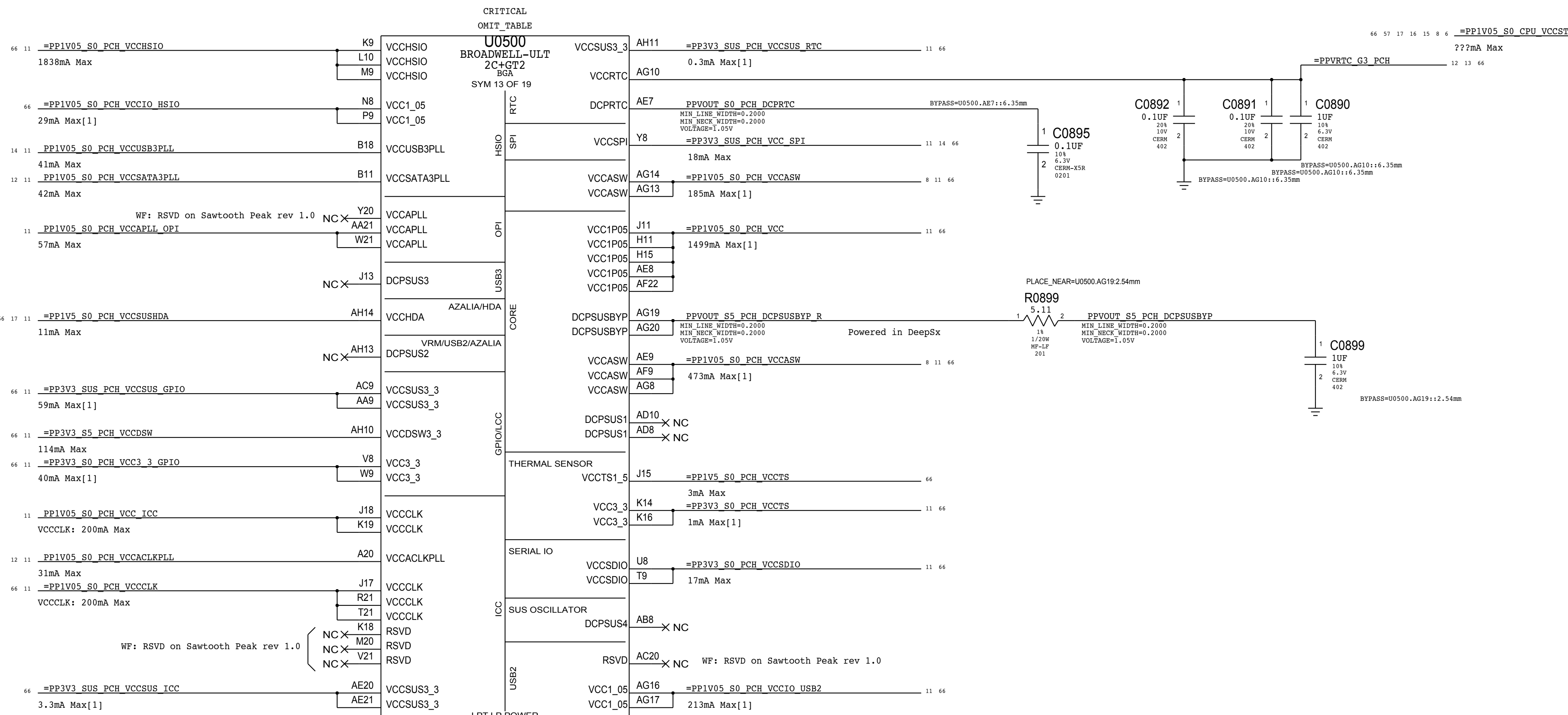
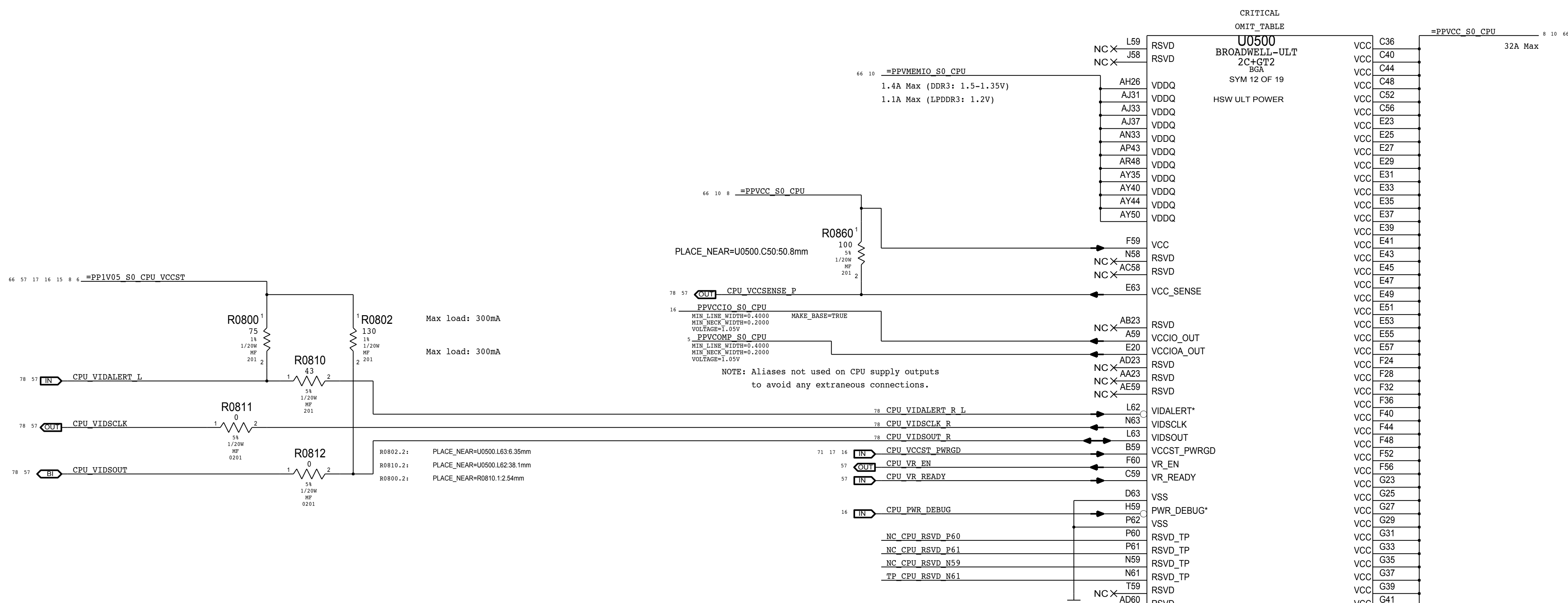


CFG<10>:SAFE MODE BOOT	1 = NORMAL OPERATION	0 = POWER FEATURES NOT ACTIVE
CFG<9>:NO SVID-CAPABLE VR	1 = VR SUPPORTS SVID	0 = VR DOES NOT SUPPORT SVID
CFG<8>:ALLOW NOA ON LOCKED UNITS	1 = NORMAL OPERATION	0 = NOA ALWAYS UNLOCKED
CFG<4>:eDP ENABLE/DISABLE	1 = DISABLED	0 = ENABLED
CFG<1>:PCH-LESS MODE	1 = NORMAL OPERATION	0 = PCH-LESS MODE
CFG<0>:RESET SEQUENCE STALL	1 = NORMAL OPERATION	0 = STALL AFTER PCU PLL LOCK





BDW-ULT current estimates from Broadwell Mobile ULT Processor EDS vol 1, doc #514405, v1.5.
WPT-LP current estimates from Wildcat Point-LP PCH EDS, doc #515621, v1.0.
Numbers may not be accurate values as of 08/26/2014



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CPU/PCH POWER			
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All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 1.0 unless stated otherwise

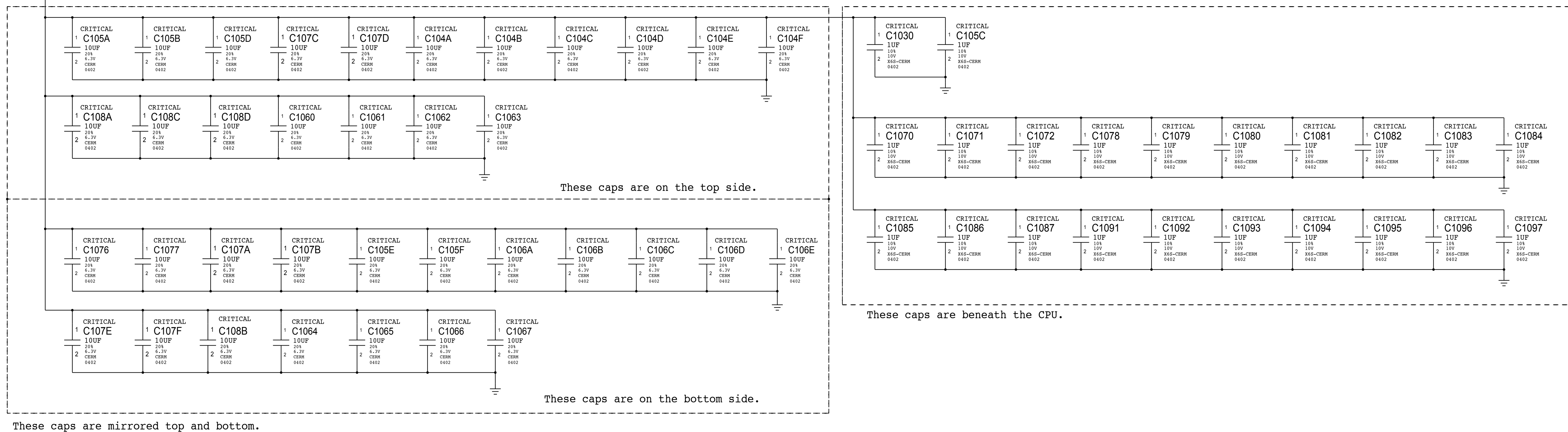
CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff

Apple implementation : 18x 10uF 0402 mirrored stuff, 1x 470uF stuff, 50x 10uF mirrored no stuff, 50x 10uF single sided no stuff

J70 implementation : 18x 10uF 0402 mirrored stuff, 32x 10uF single sided no stuff

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66 8  =PPVCC S0 CPU
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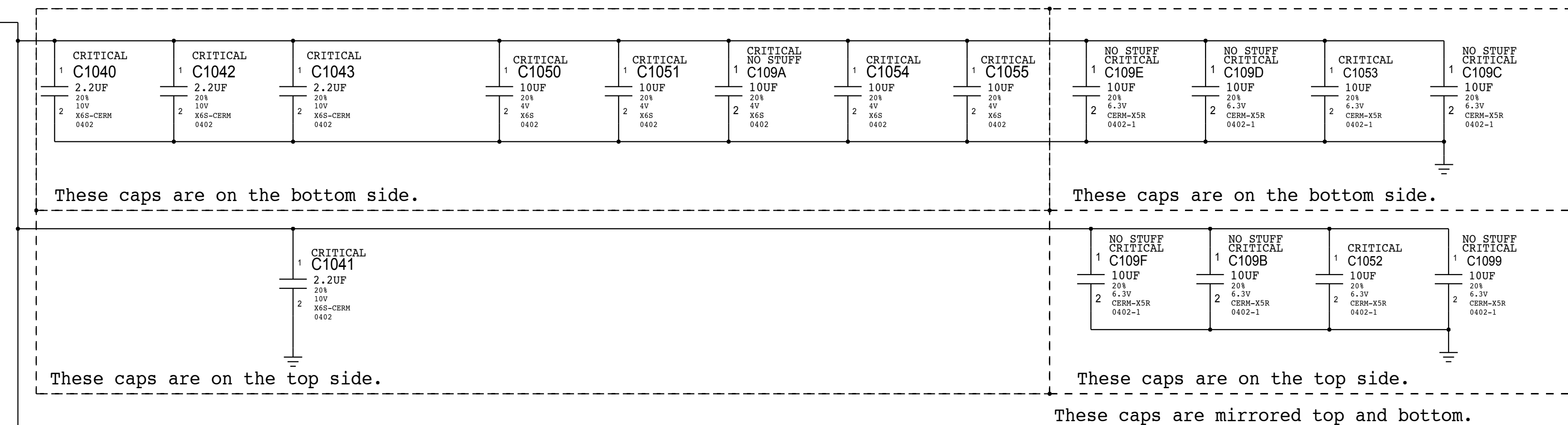


CPU VDDQ DECOUPLING

Intel recommendation (Table 5-4): 4x 2.2uF 0402, 6x 10uF 0603

Apple implementation : 4x 2.2uF 0402, 6x 10uF 0402, 6x 10uF no stuff, 1x 270 uF Bulk

```
66 8 =PPVMEMIO S0 CPU
```



CRITICAL
C1056
270UF-0.006OHM
20%
2V
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CASE-D2

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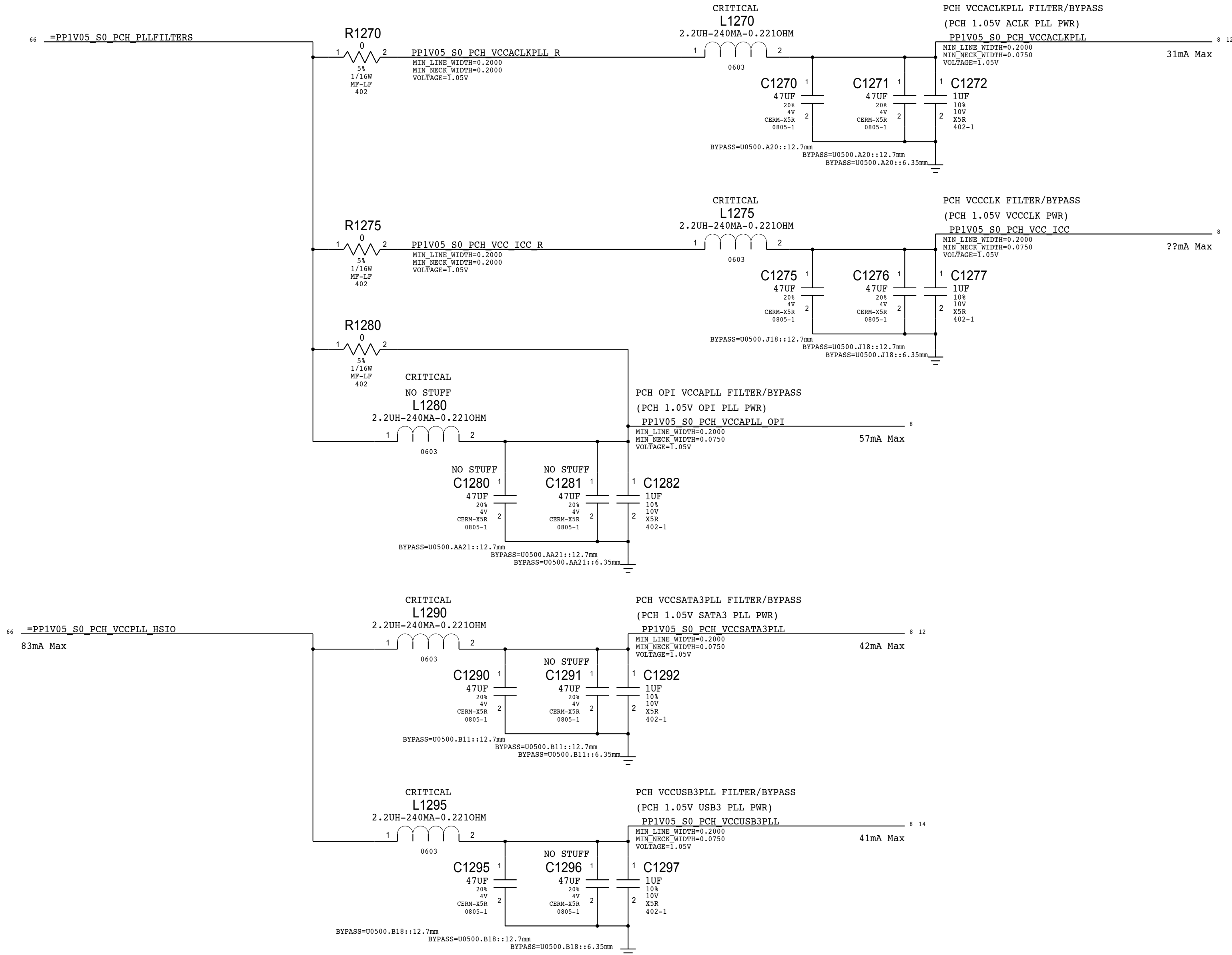
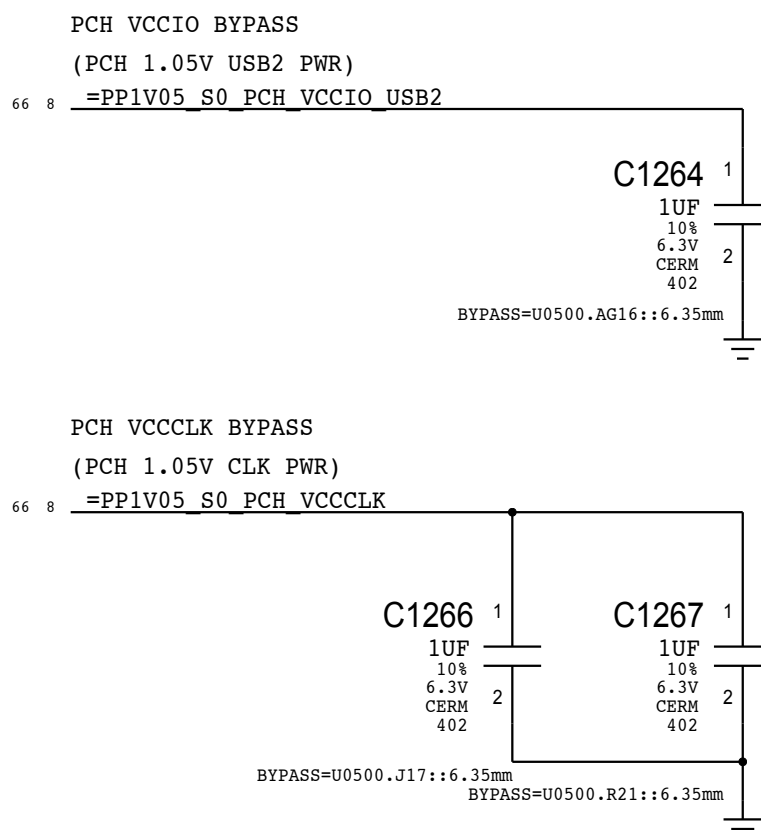
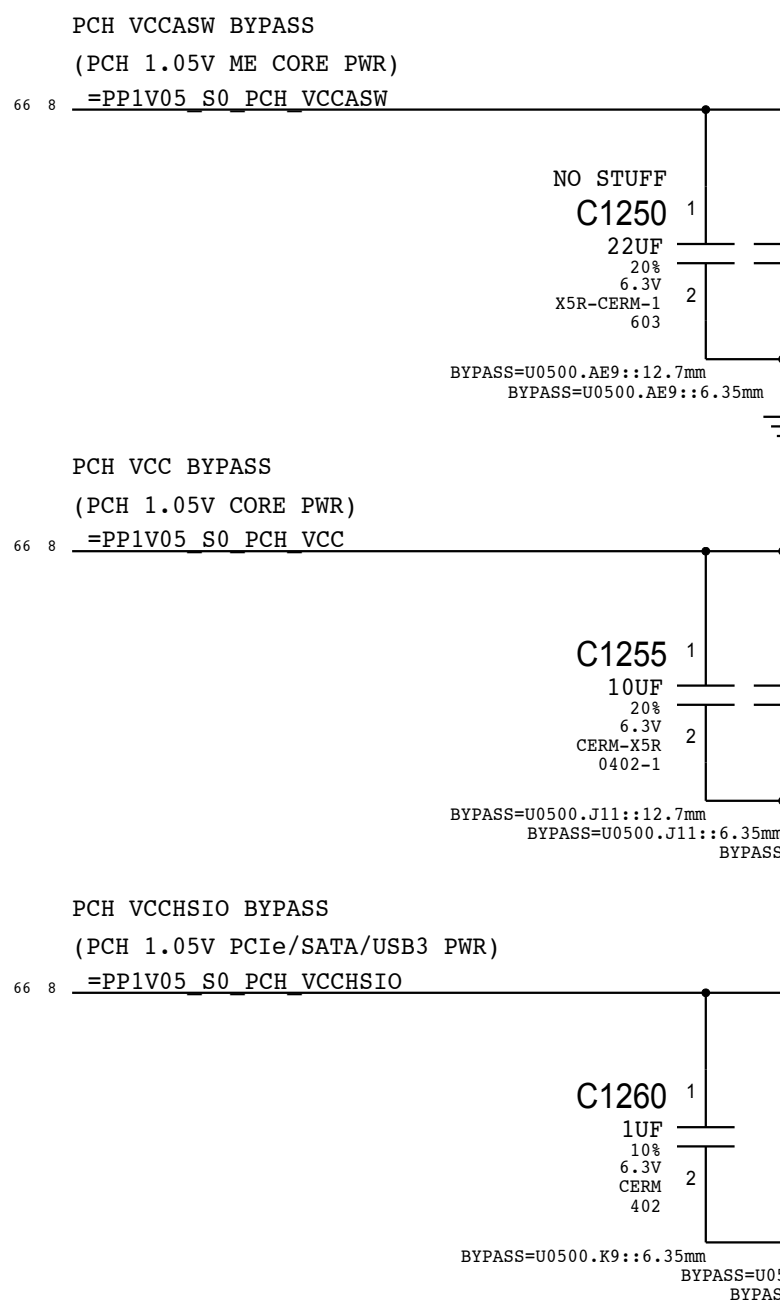
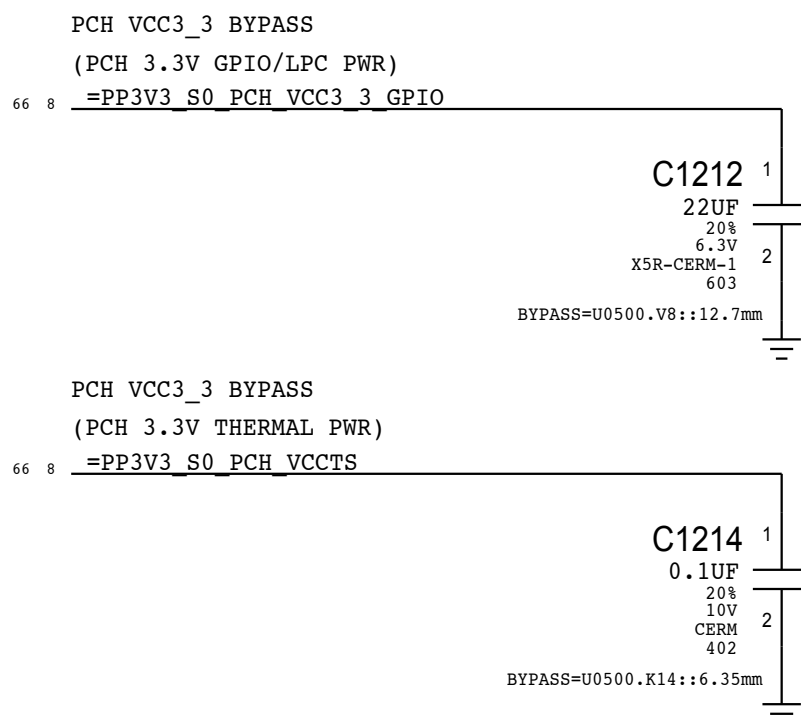
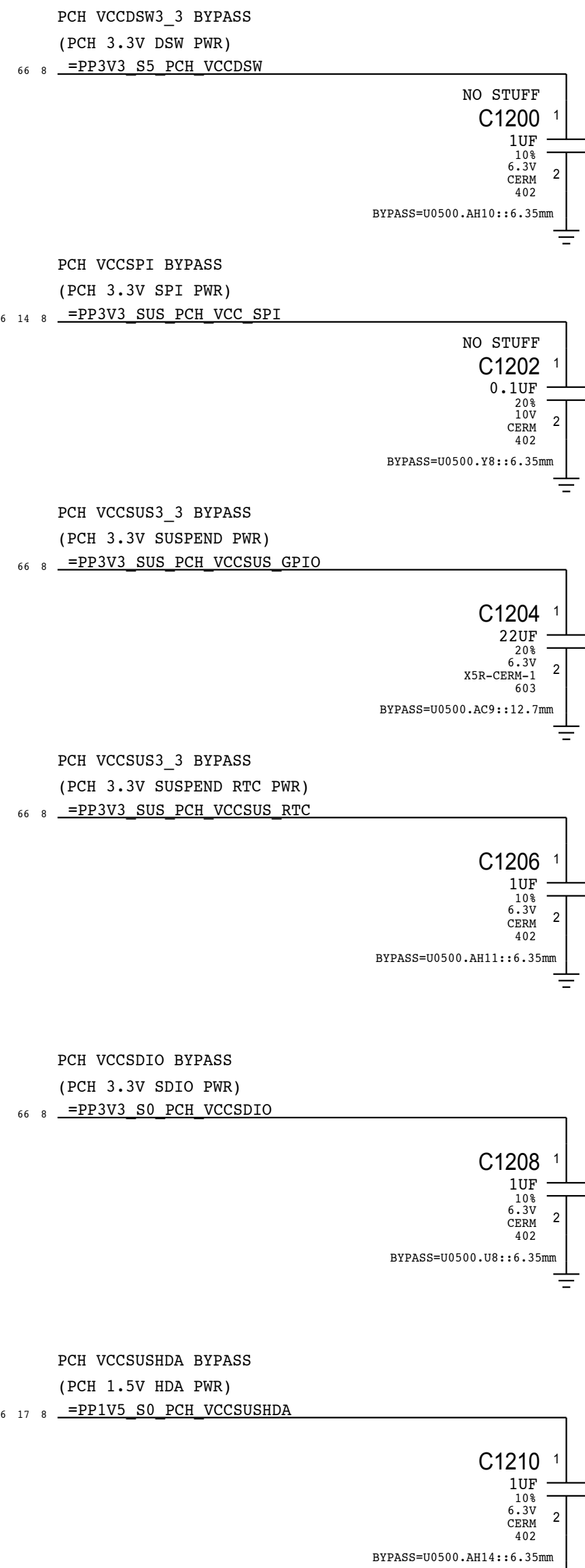
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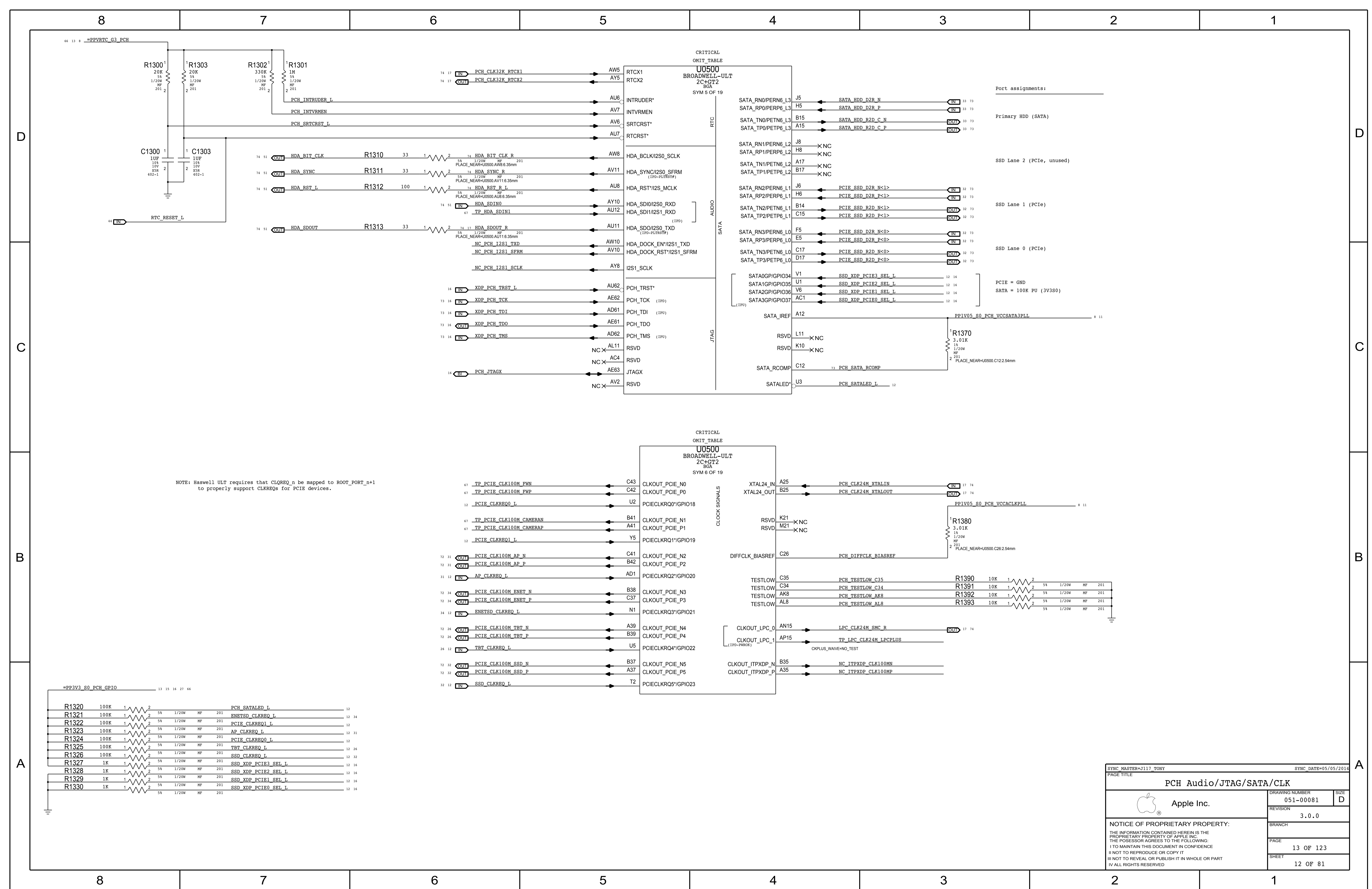
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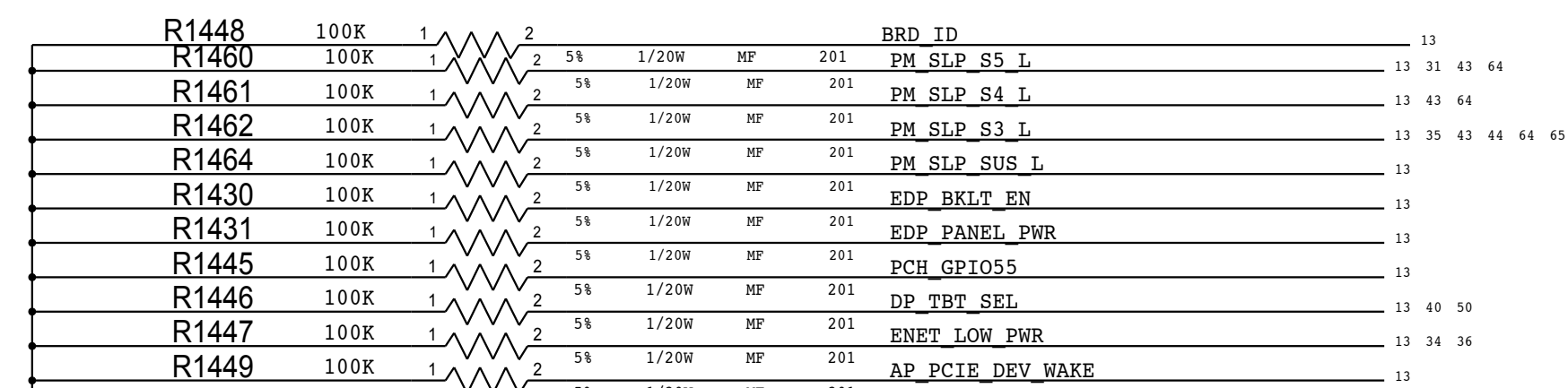
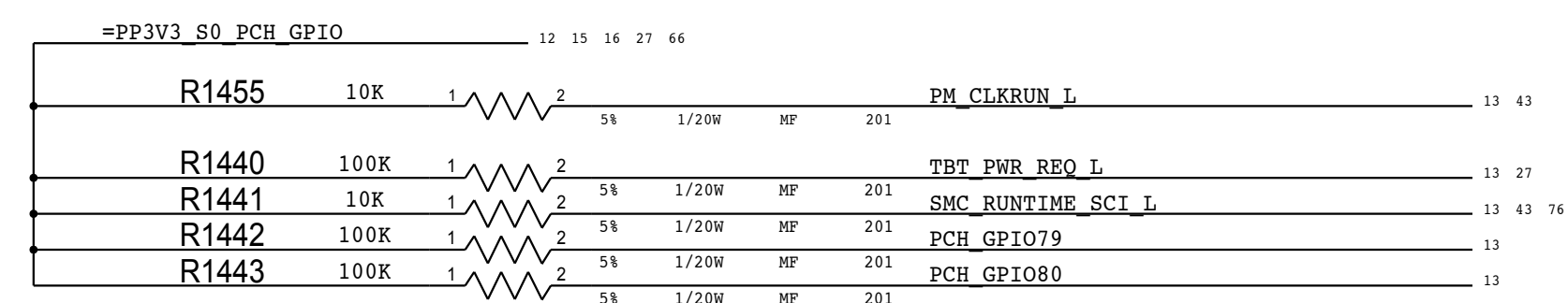
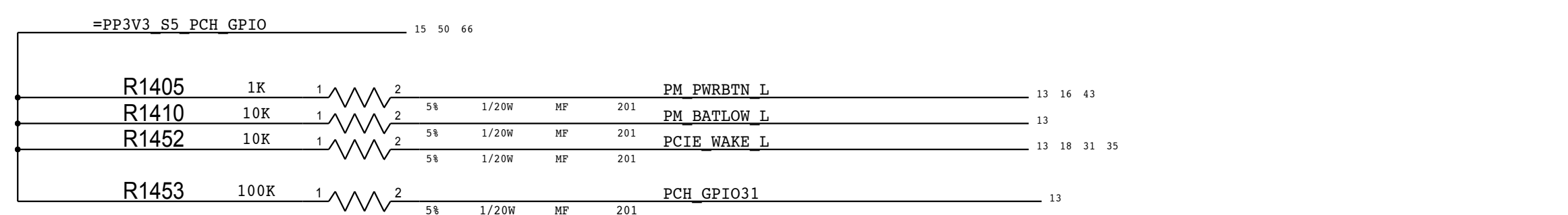
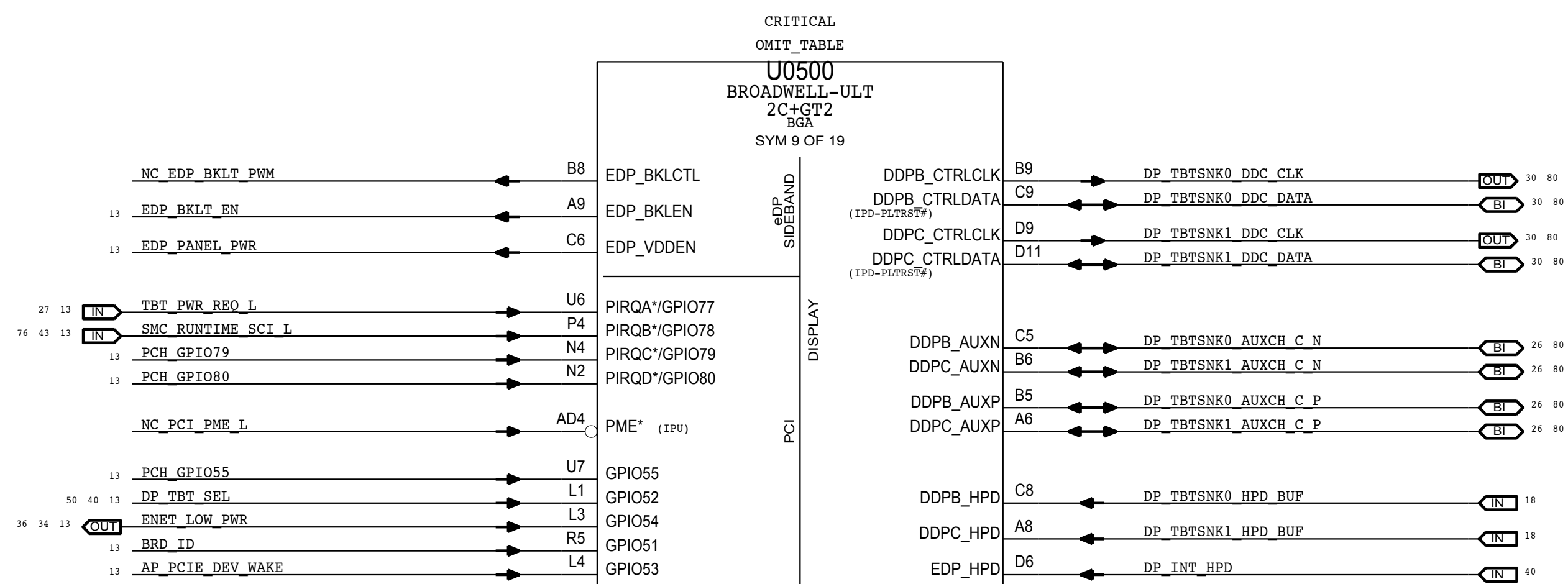
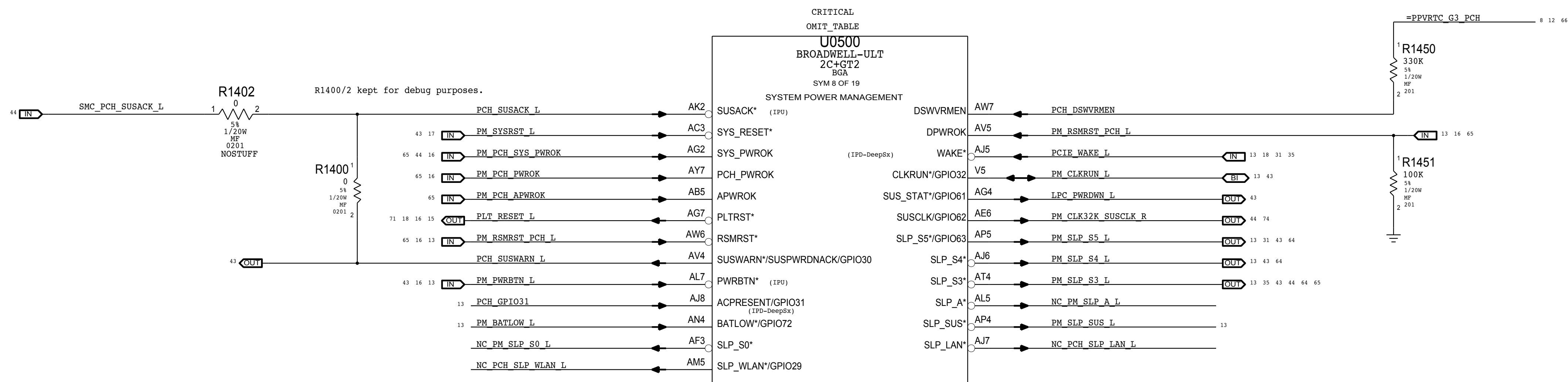
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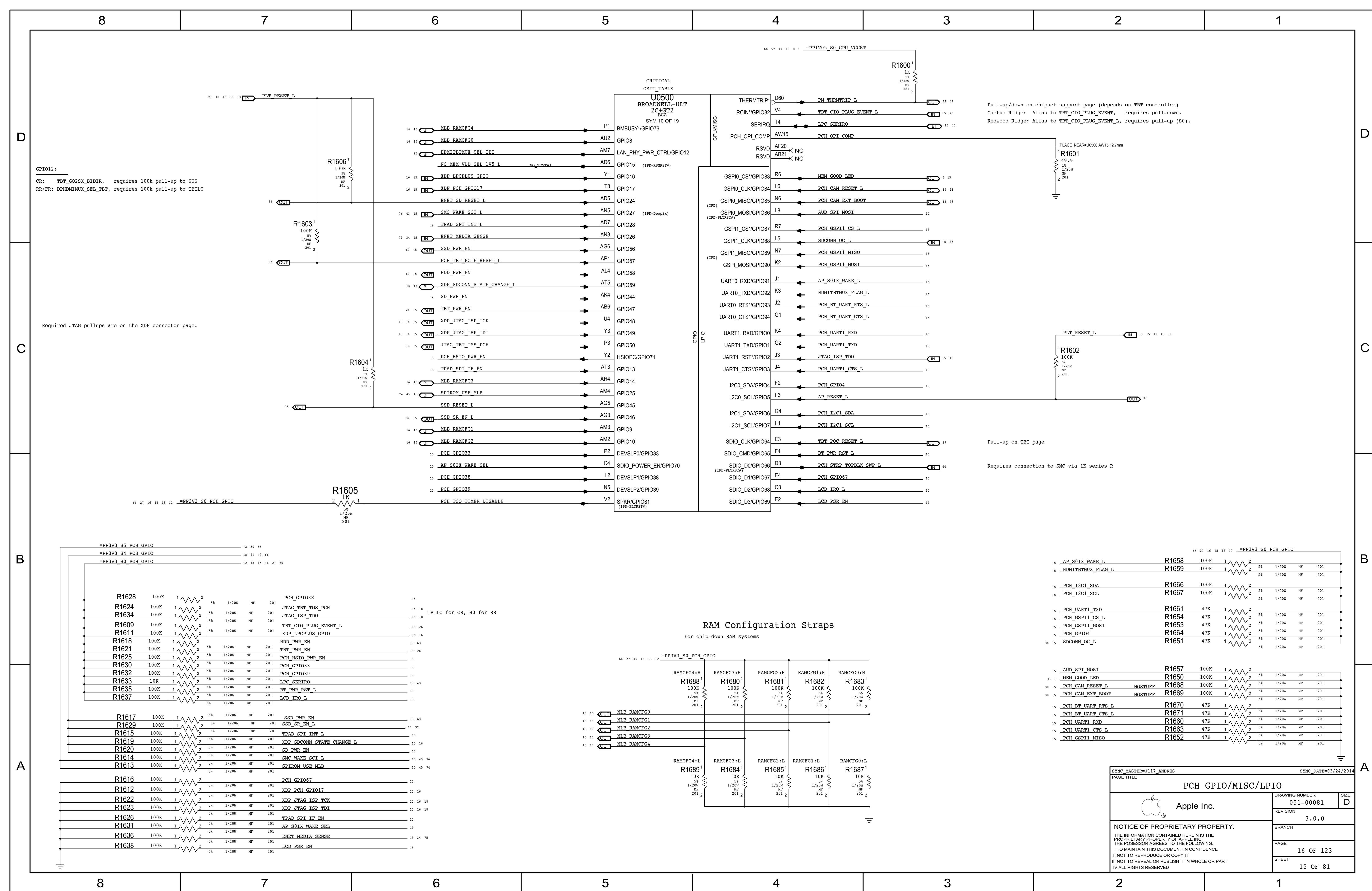
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		PAGE 12 OF 123	
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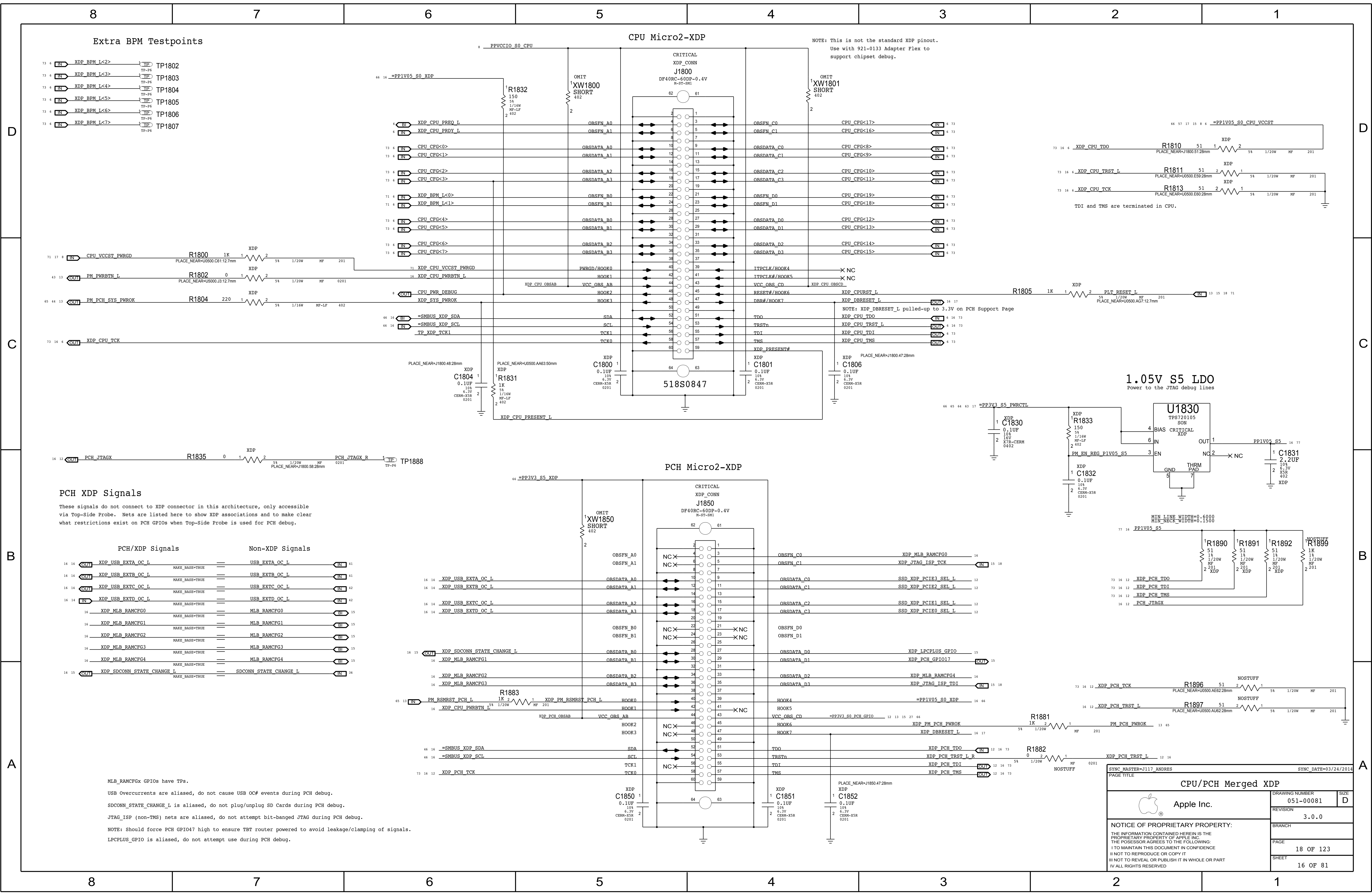






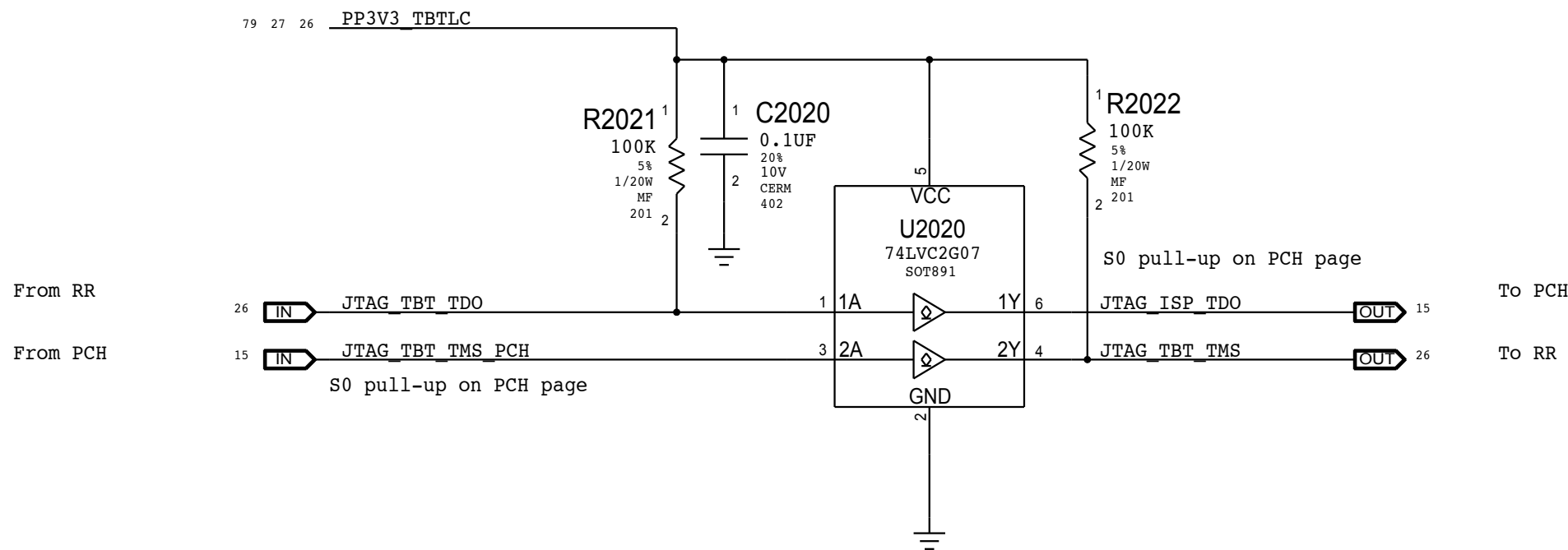
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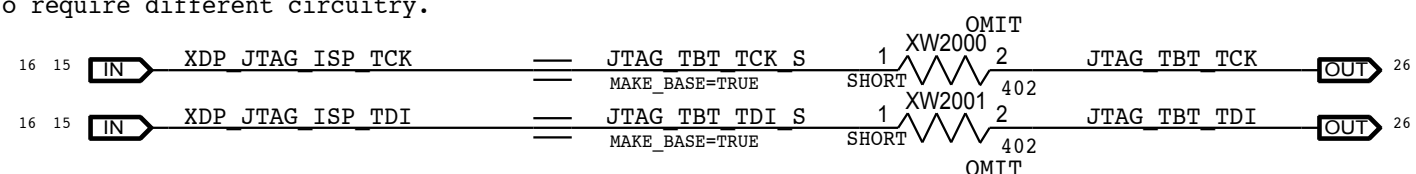
Falcon Ridge JTAG Isolation

TBTLIC can be on when S0 is off, and vice-versa
Isolation ensures no leakage to RR or PCH



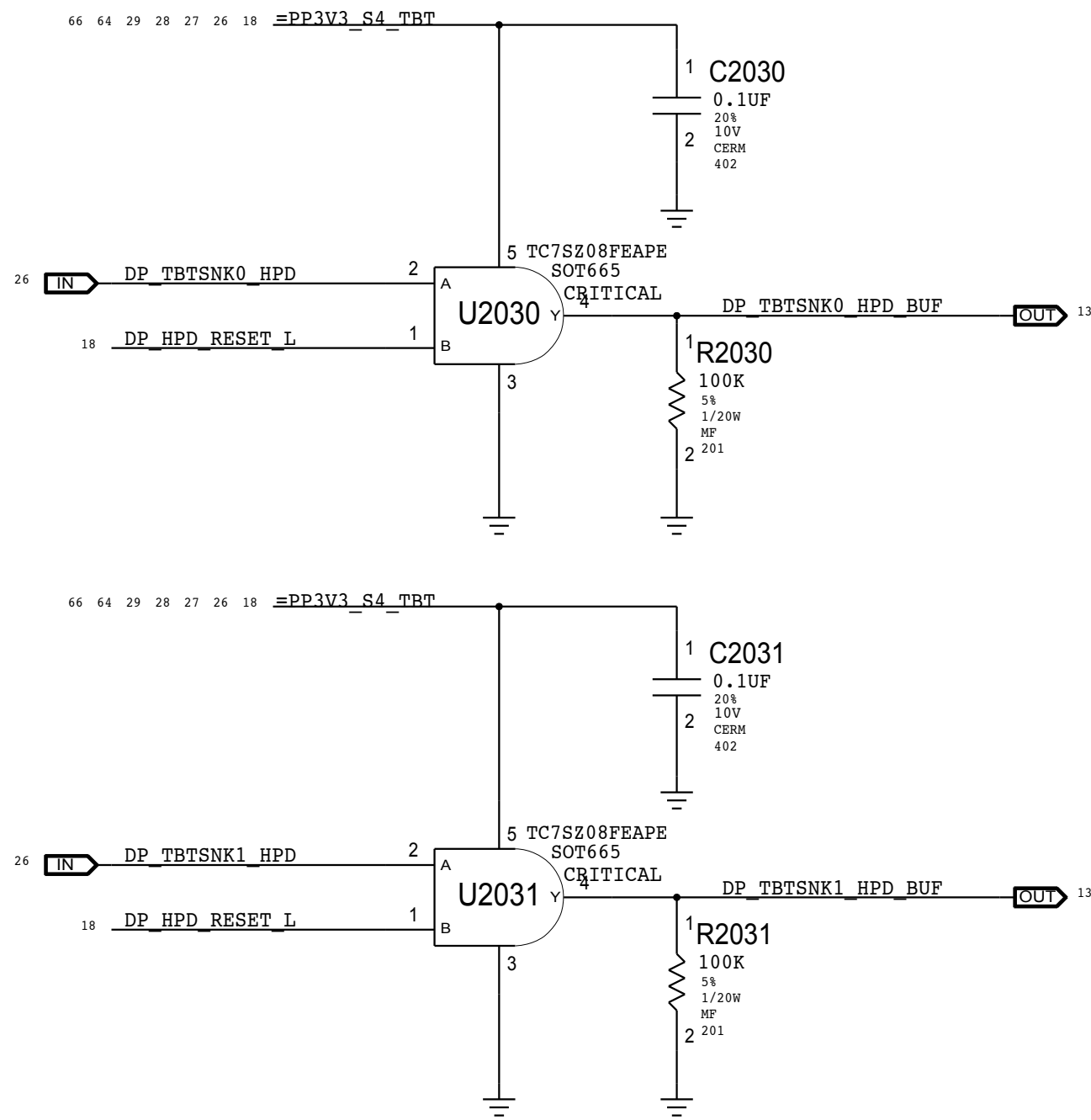
NOTE: Solution shown is for WPT-LP. Other PCH's may require isolation on TCK and TDI as well for PCH glitch-prevention.

NOTE: This reference schematic assumes PCH JTAG GPIOs are only used for Thunderbolt. If other ASIC JTAG signals are wired into these GPIOs different isolation techniques will likely be necessary. Multi-router designs also require different circuitry.

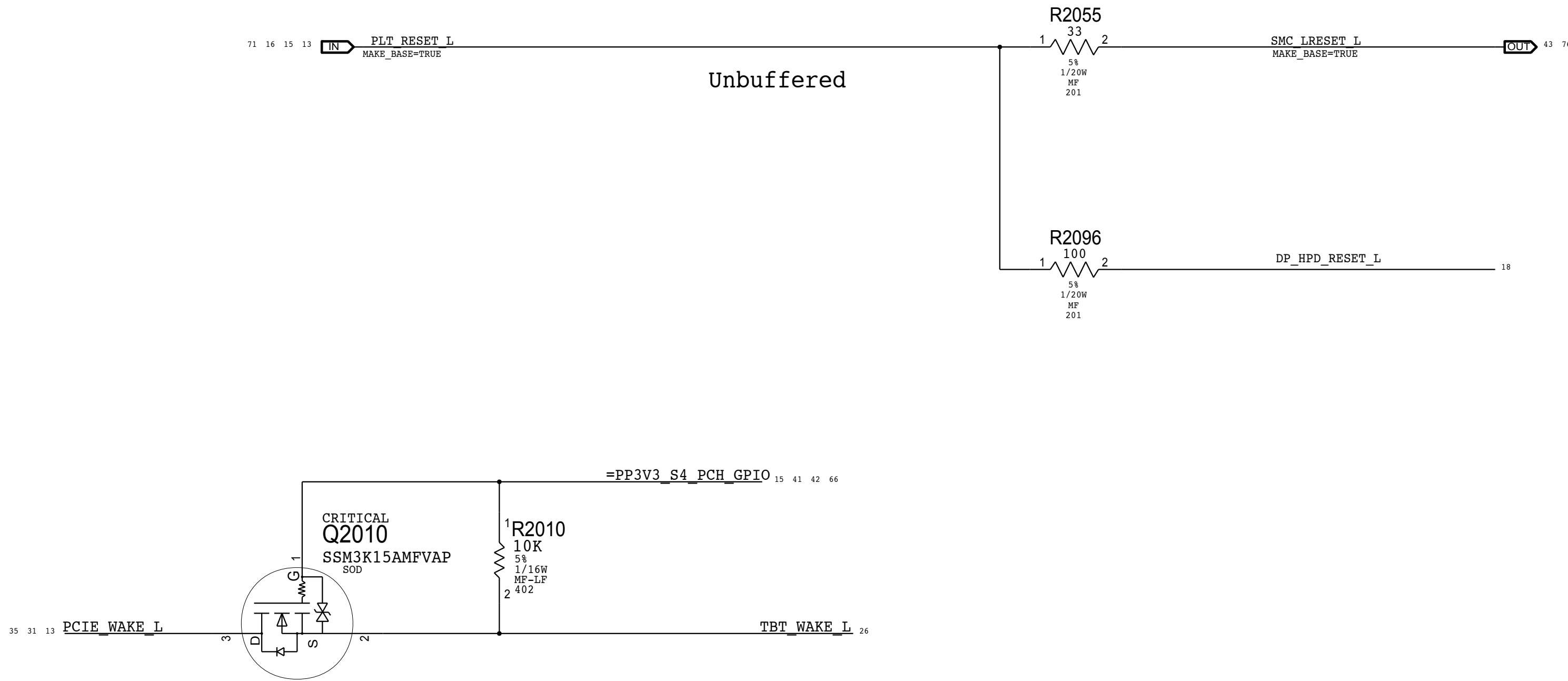


Falcon Ridge HPD Isolation

NOTE: PLT_RESET_L used as the other input to the AND gate so that HPD is only driven high to the PCH in S0.

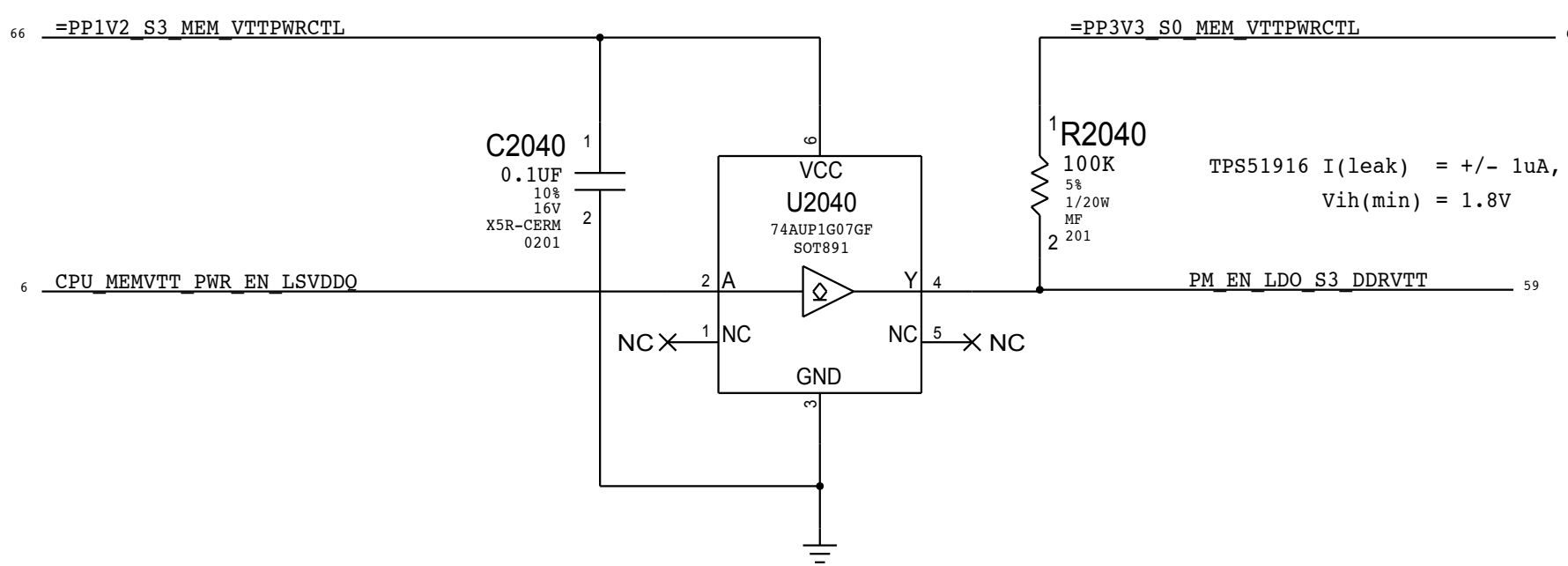


Platform Reset Connections



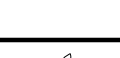
Memory VTT Enable Level-Shifter

CPU output is on VDDQ rail (1.2V), TP851916 has 1.8V Vih(min).



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Memory Bit/Byte Swizzle																
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8		7		6		5		4		3		2		1	
DDR3 Signal Aliases															
Apple Inc.															
051-00081															
3.0.0															
21 OF 123															
19 OF 81															

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 Apple Inc.		DRAWING NUMBER	051-00081
		REVISION	3.0.0
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		21 OF 123	
		SHEET	
		19 OF 81	

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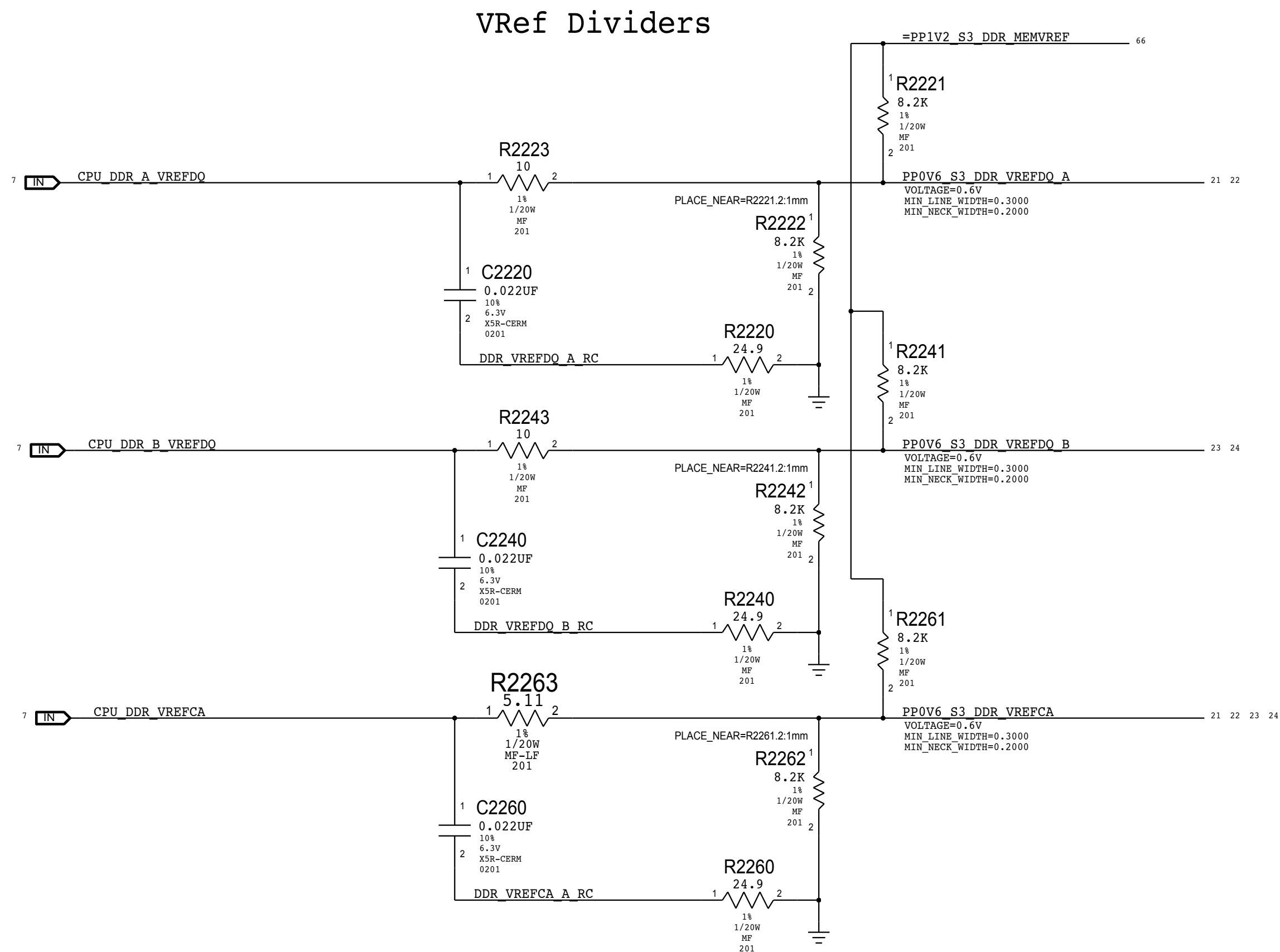
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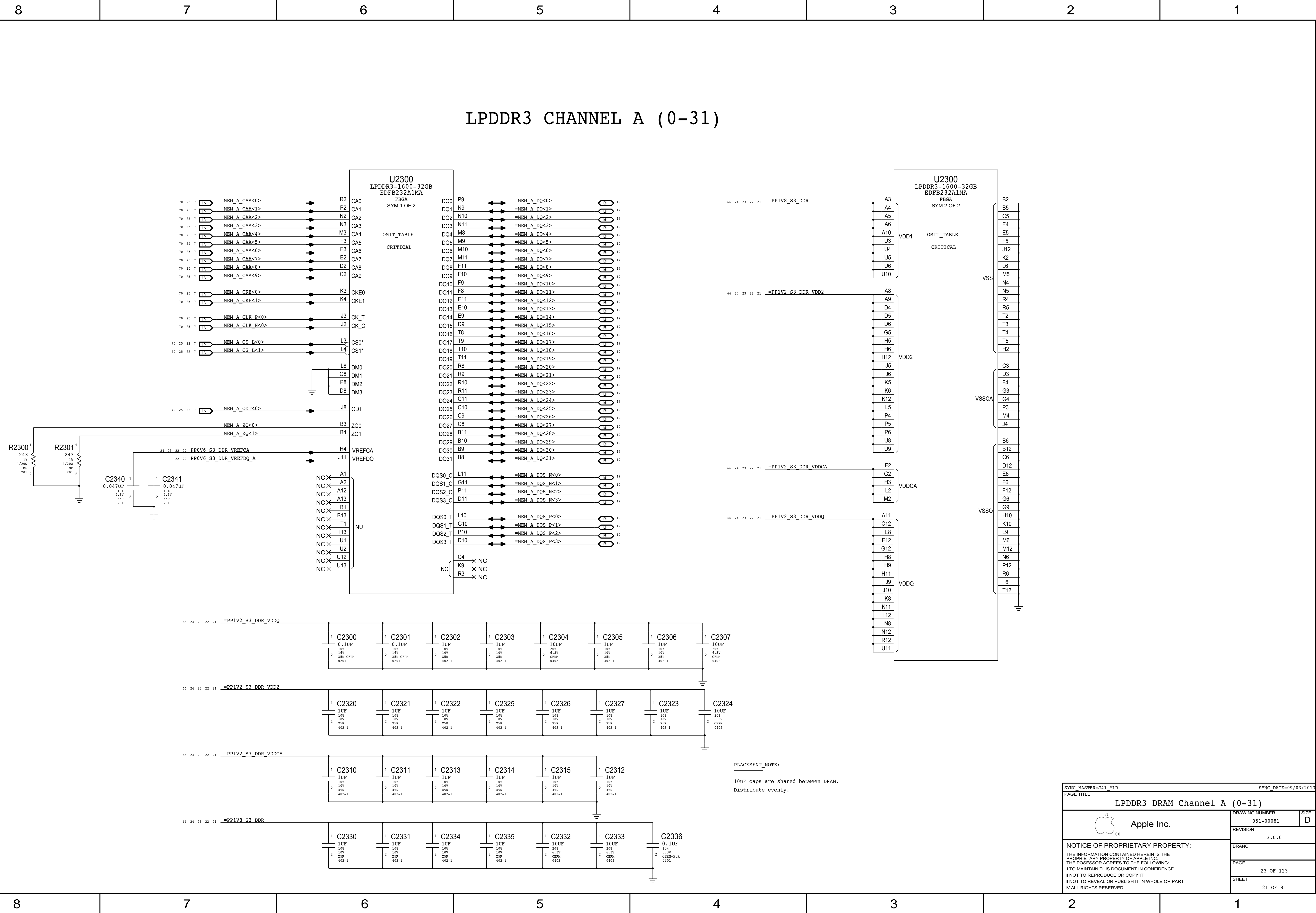
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		PAGE	22 OF 123
		SHEET	20 OF 81



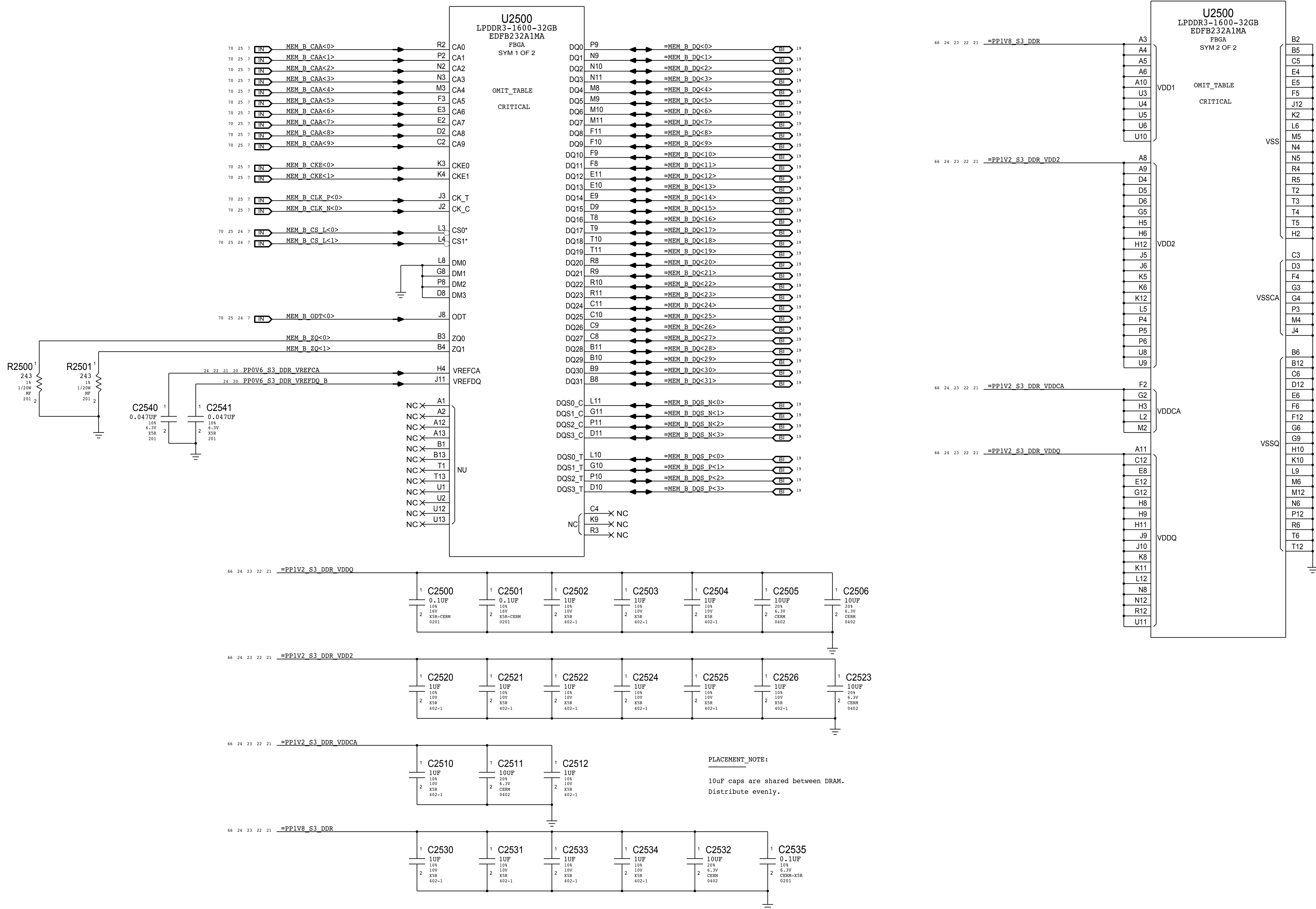
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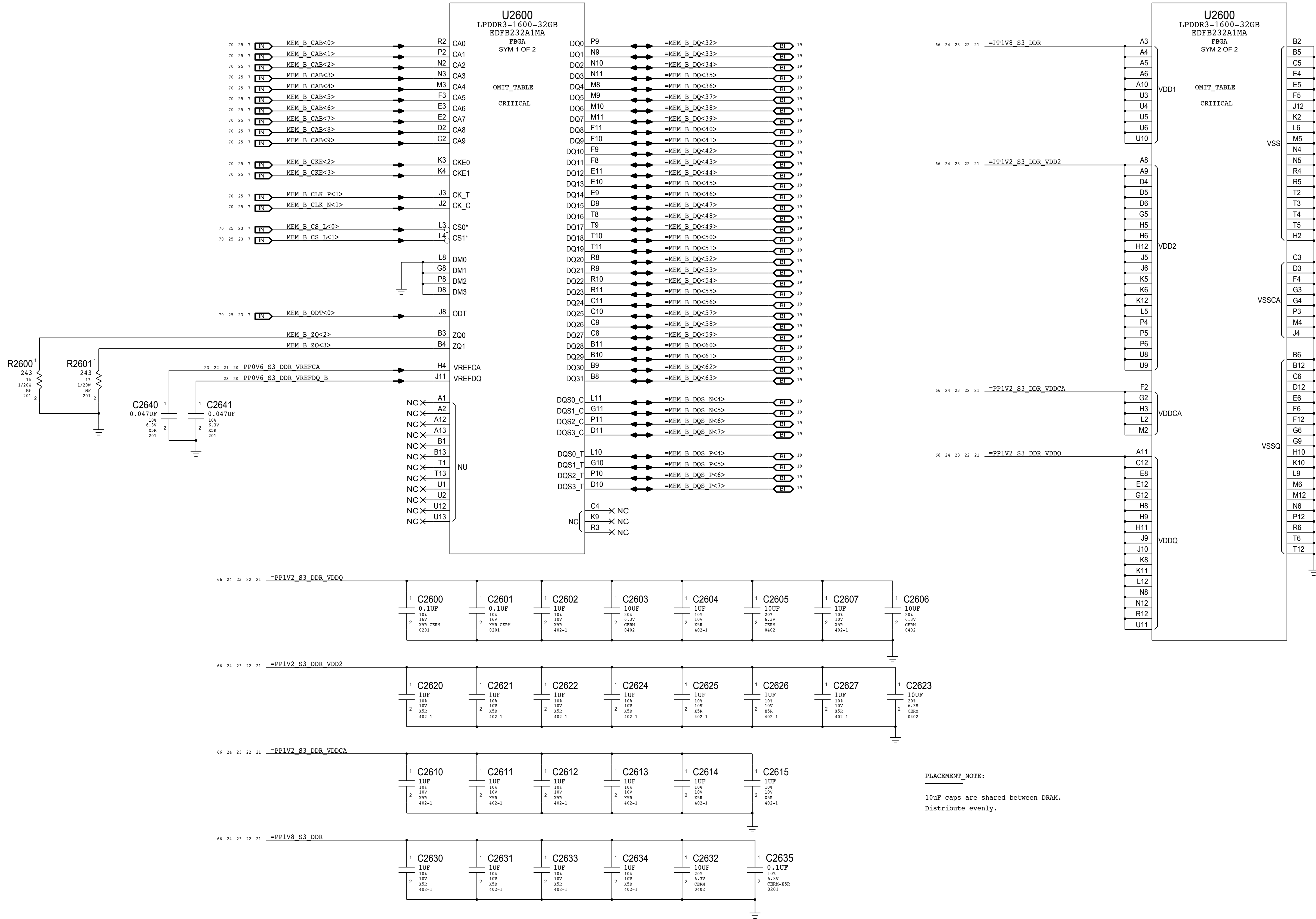
10uF caps are shared between DRAM.
Distribute evenly.

A

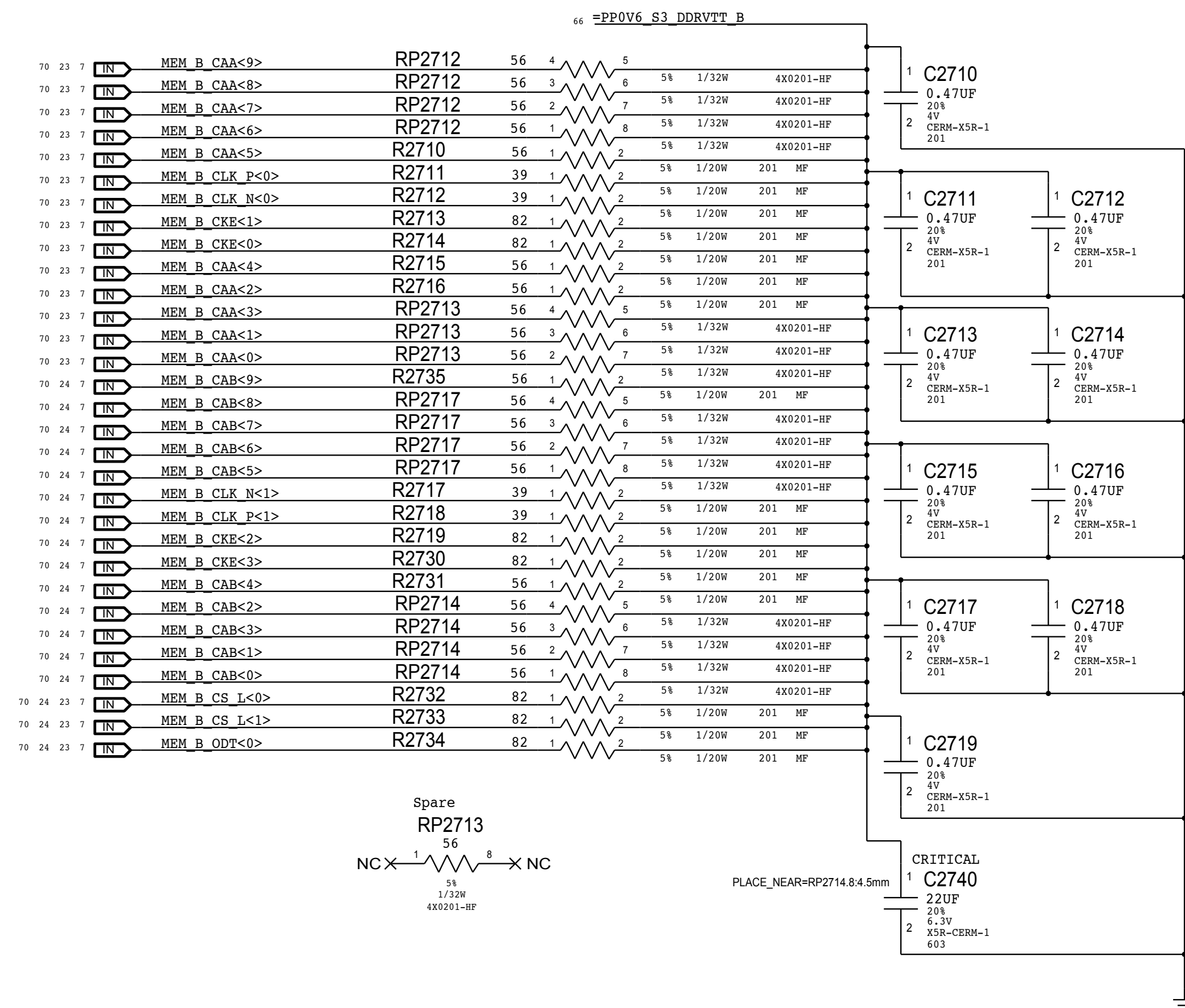
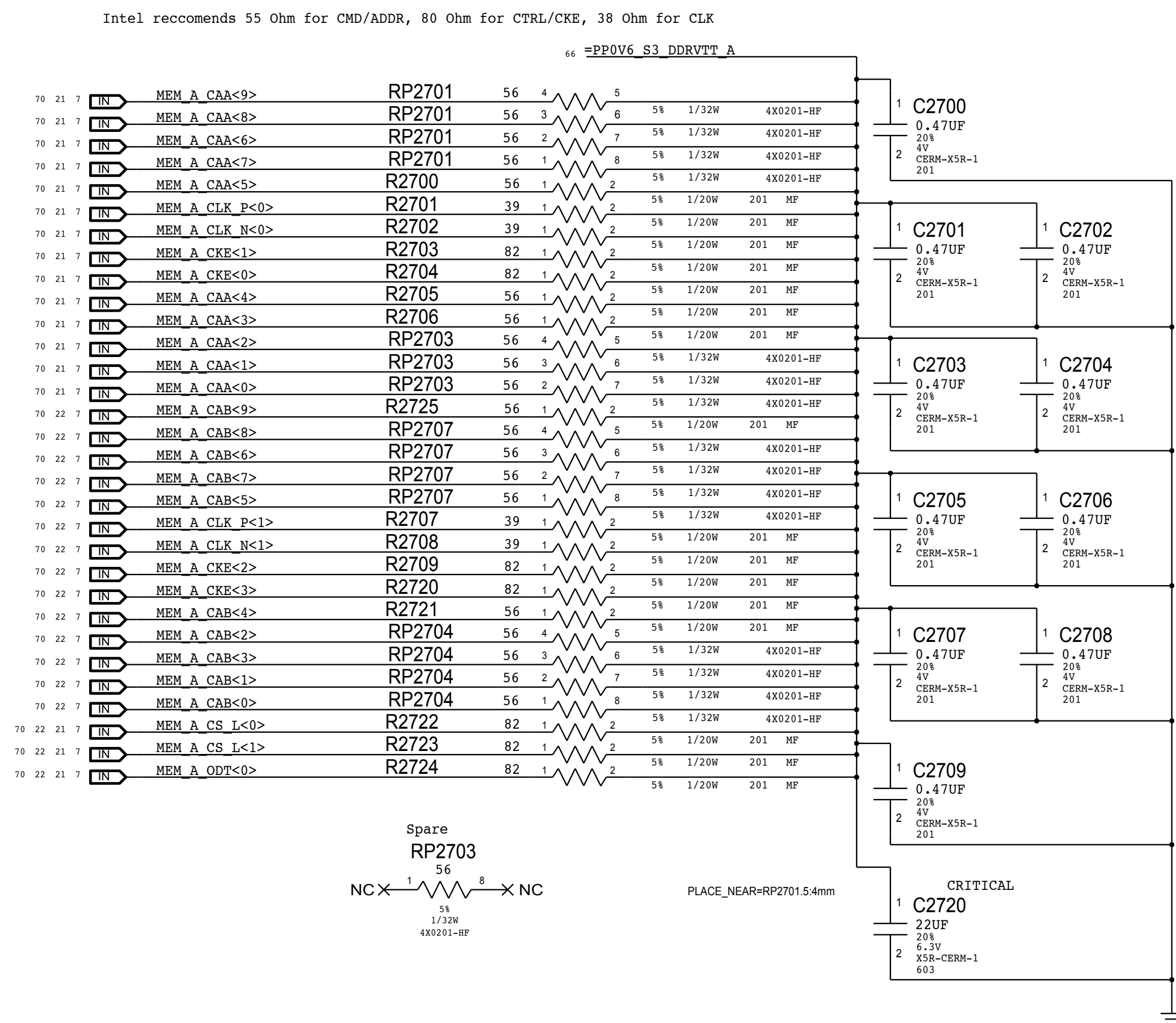
LPDDR3 CHANNEL B (0-31)

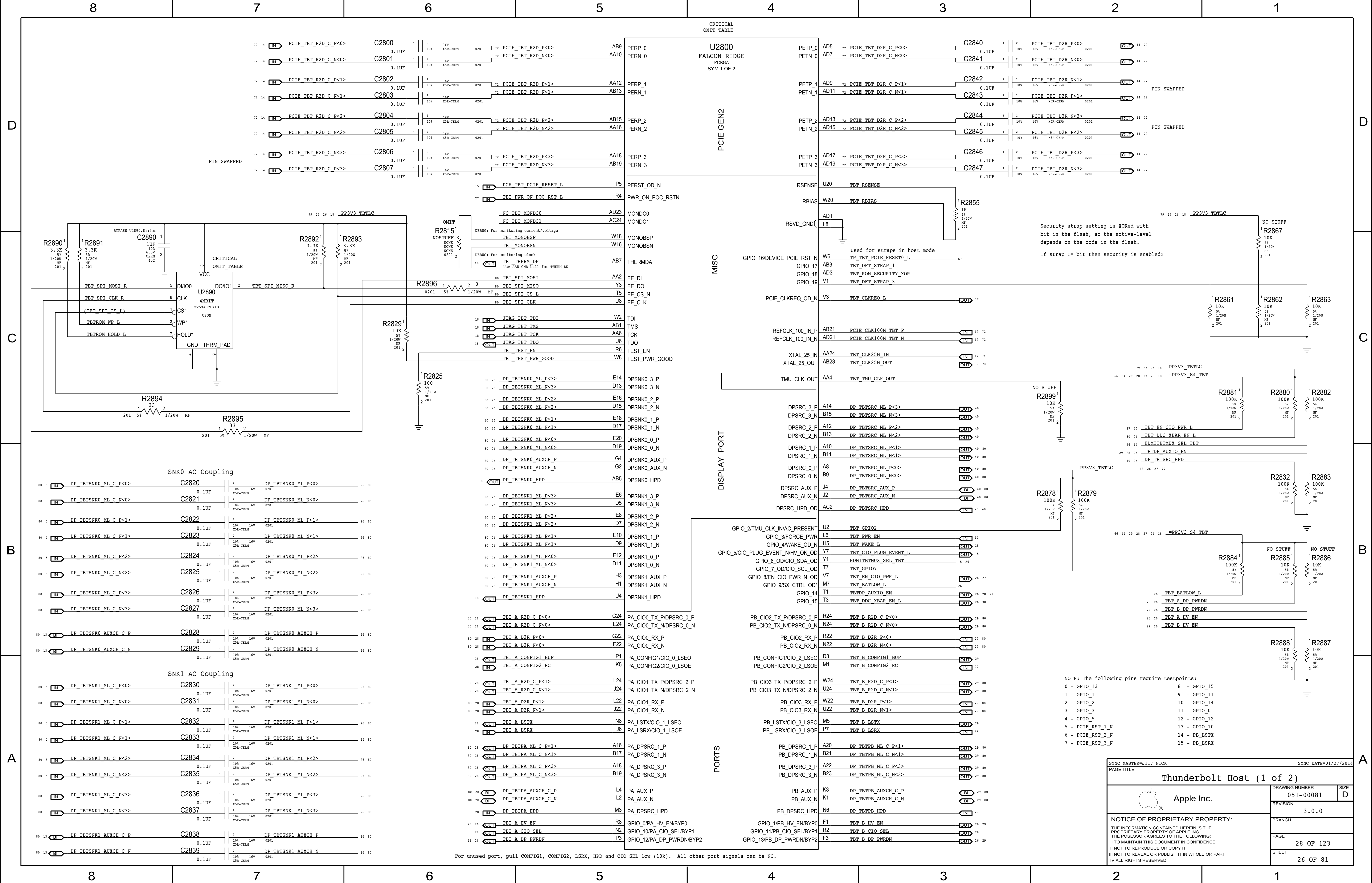


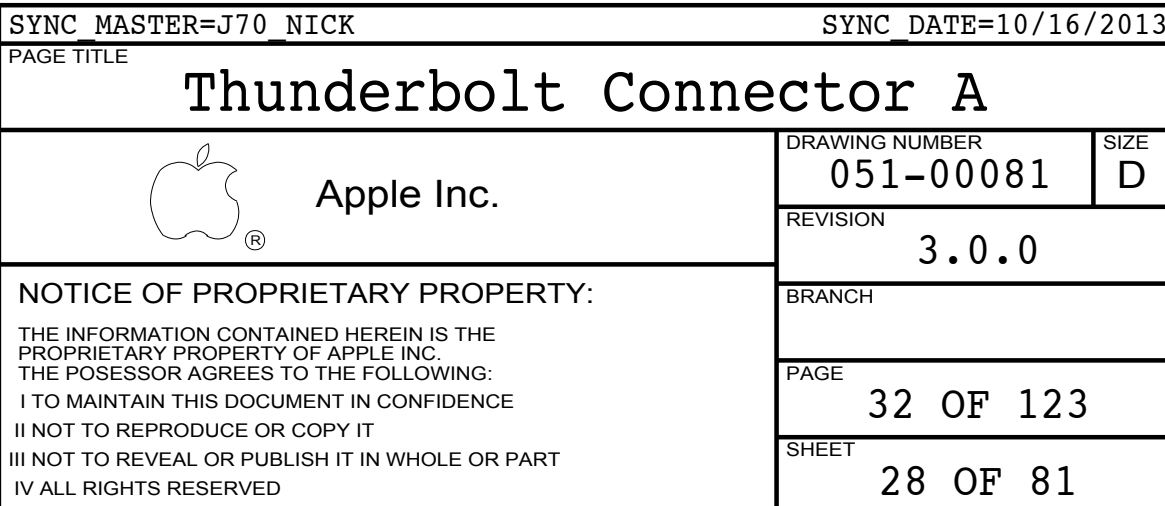
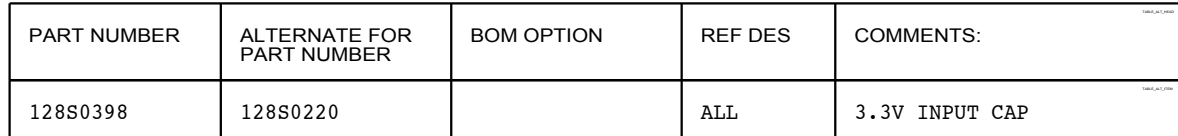
LPDDR3 CHANNEL B (32-63)



PLACEMENT NOTE:
10uF caps are shared between DRAM.
Distribute evenly.







3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

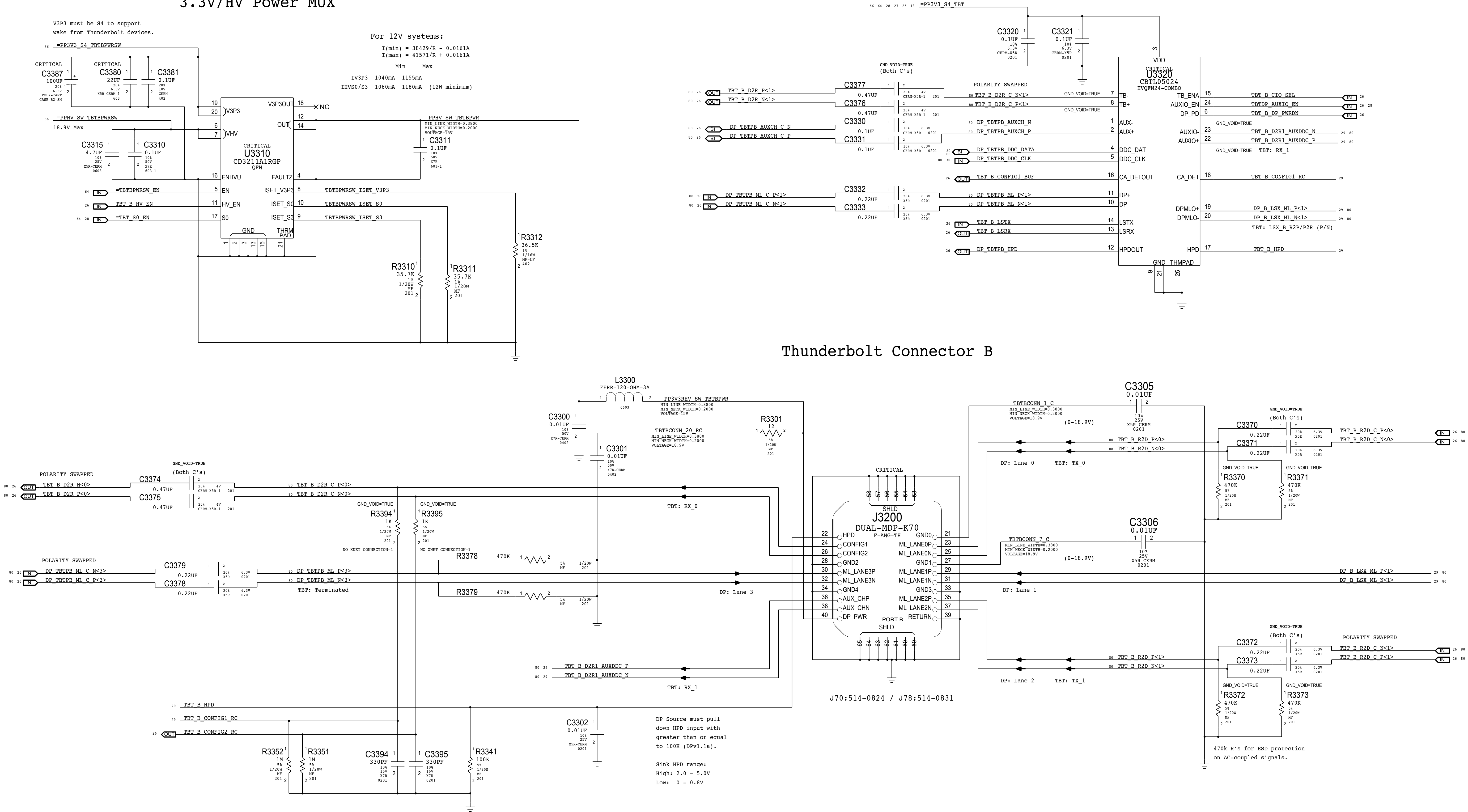
For 12V systems:

$I(\min) = 38429/R - 0.0161A$
 $I(\max) = 41571/R + 0.0161A$

Min Max

IV3P3 1040mA 1155mA

IHVS0/S3 1060mA 1180mA (12W minimum)



Thunderbolt Connector B

J70:514-0824 / J78:514-0831

DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC_MASTER=J70_NICK SYNC_DATE=10/16/2013

PAGE TITLE

Thunderbolt Connector B



Apple Inc.

DRAWING NUMBER 051-00081

SIZE D

REVISION 3.0.0

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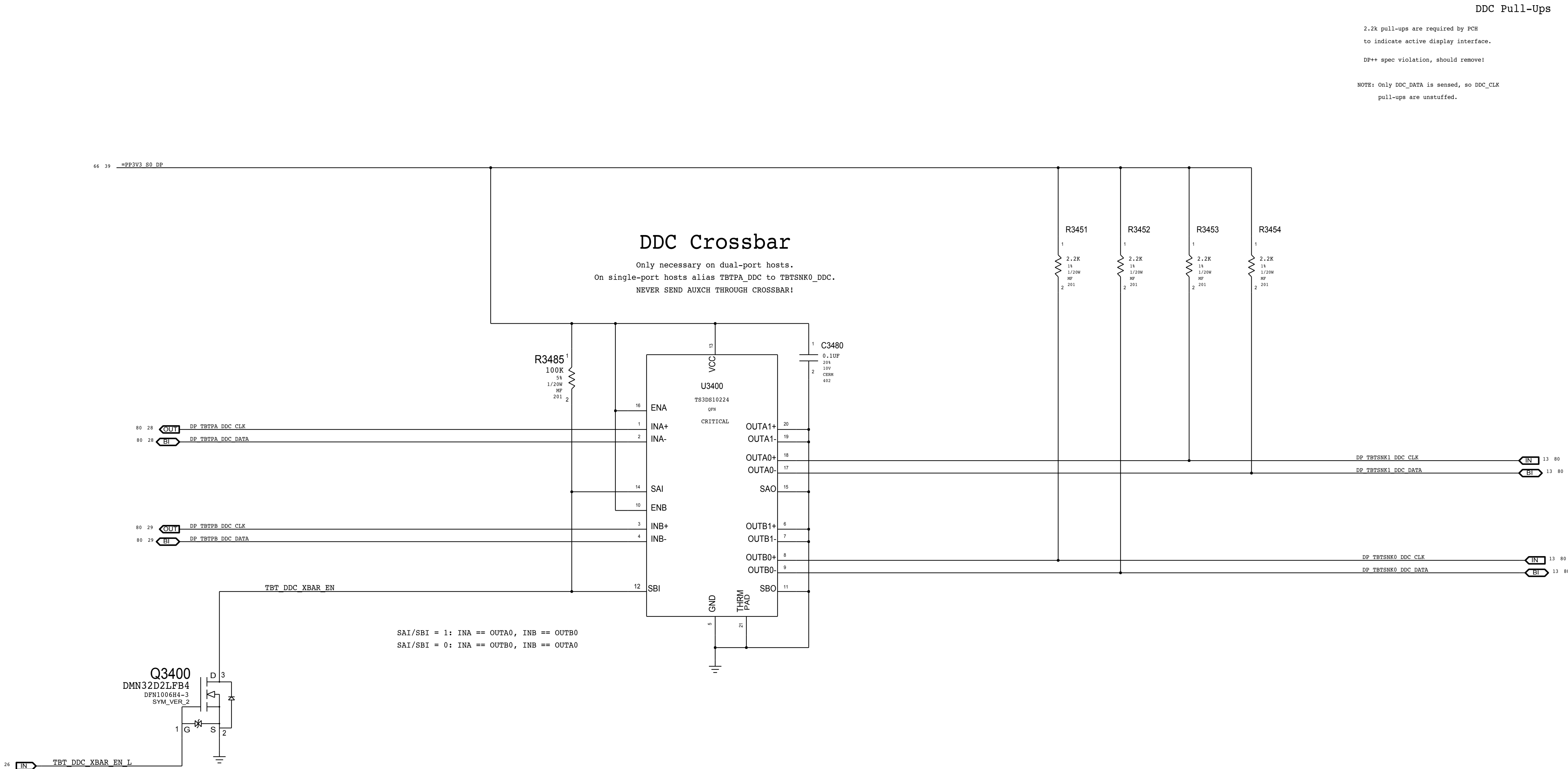
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SYNC_MASTER~J70_TONY

SYNC_DATE~09/13/2013

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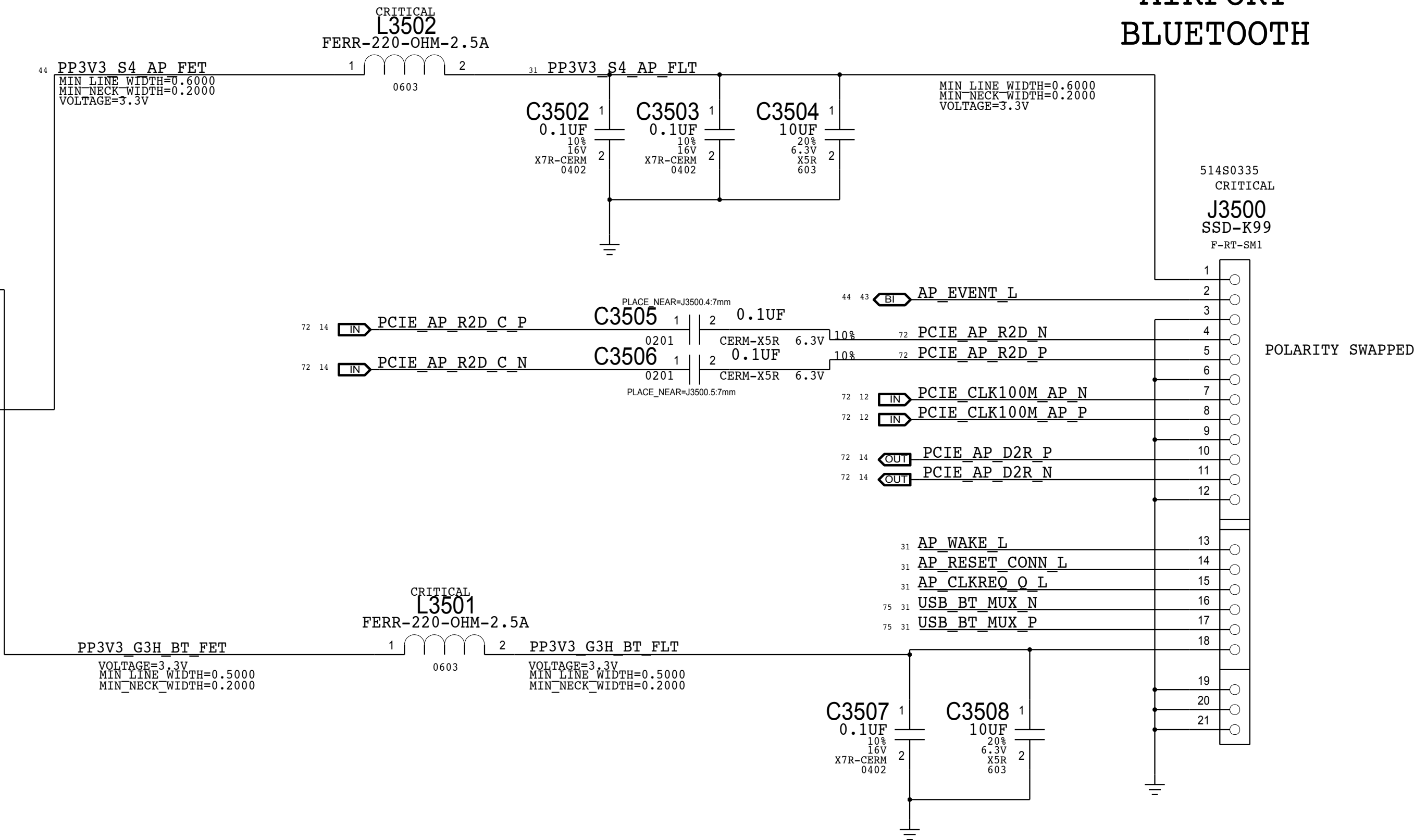
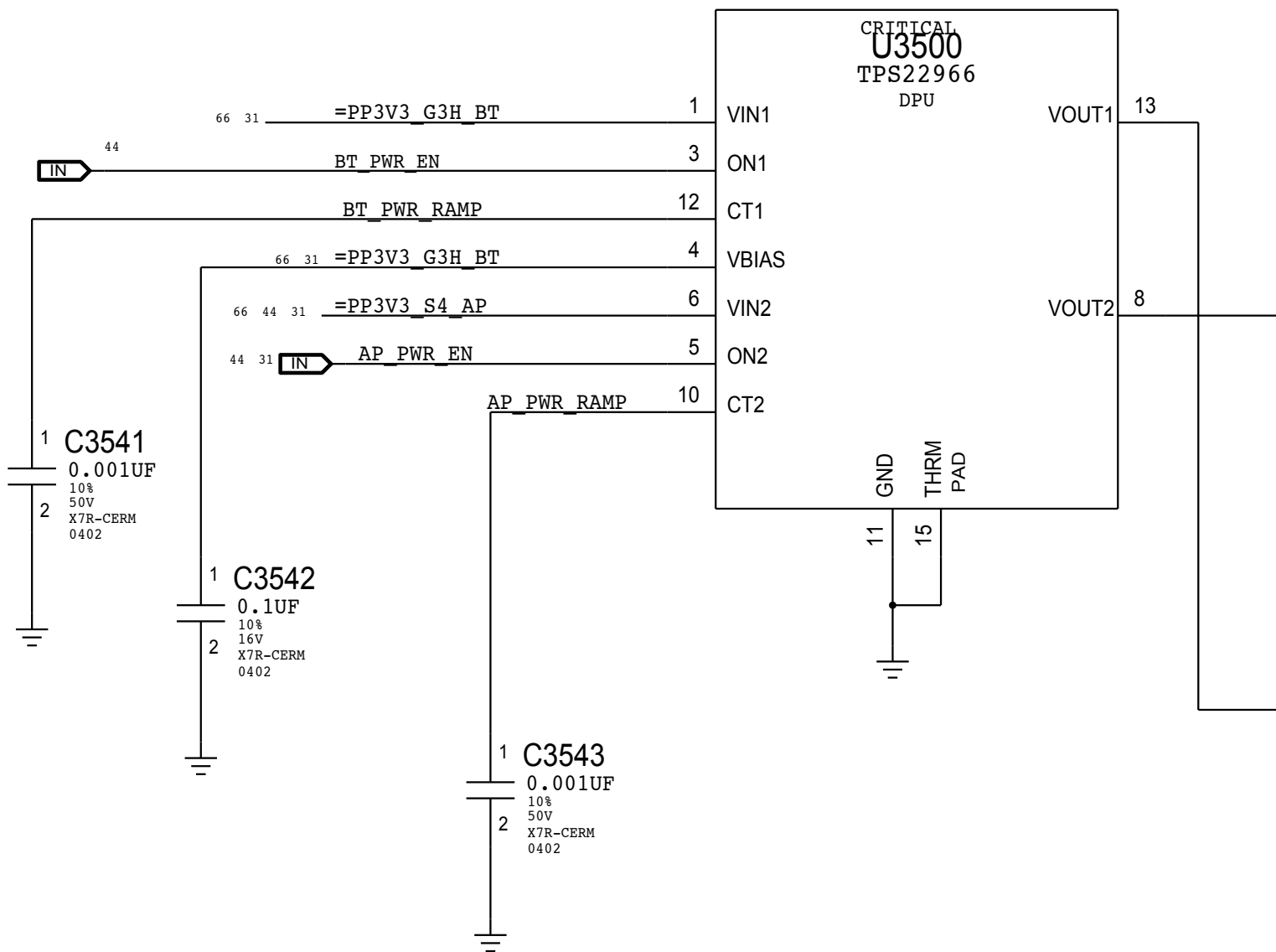
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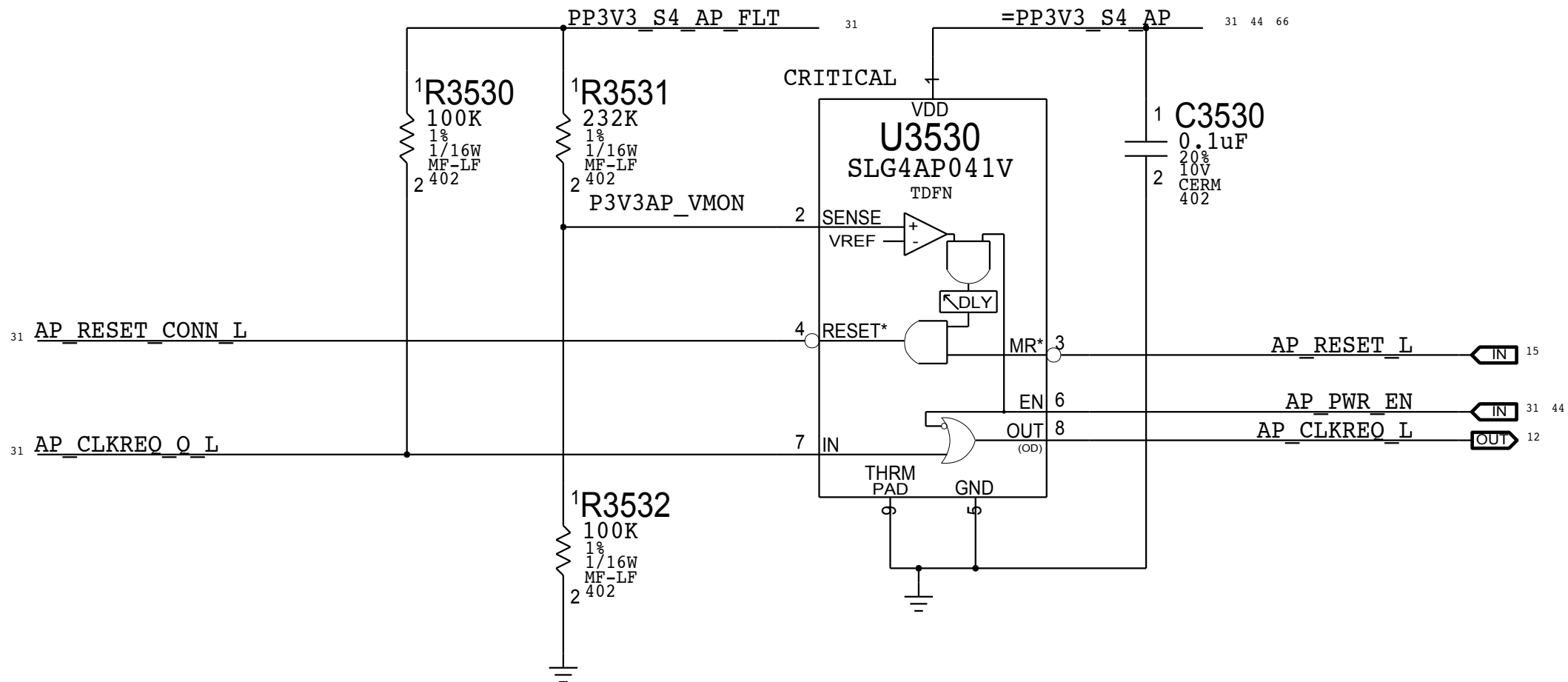
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AP & BT Load Switch

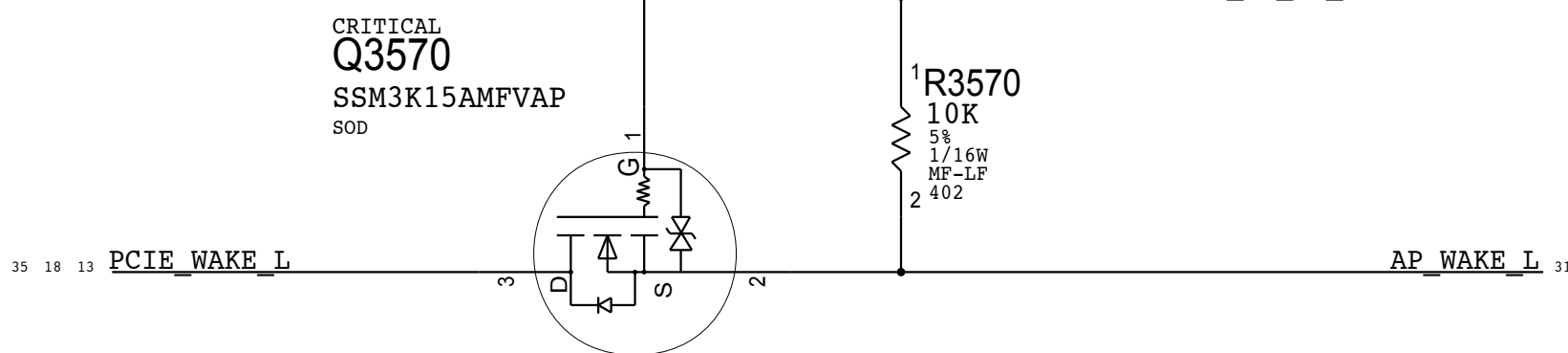
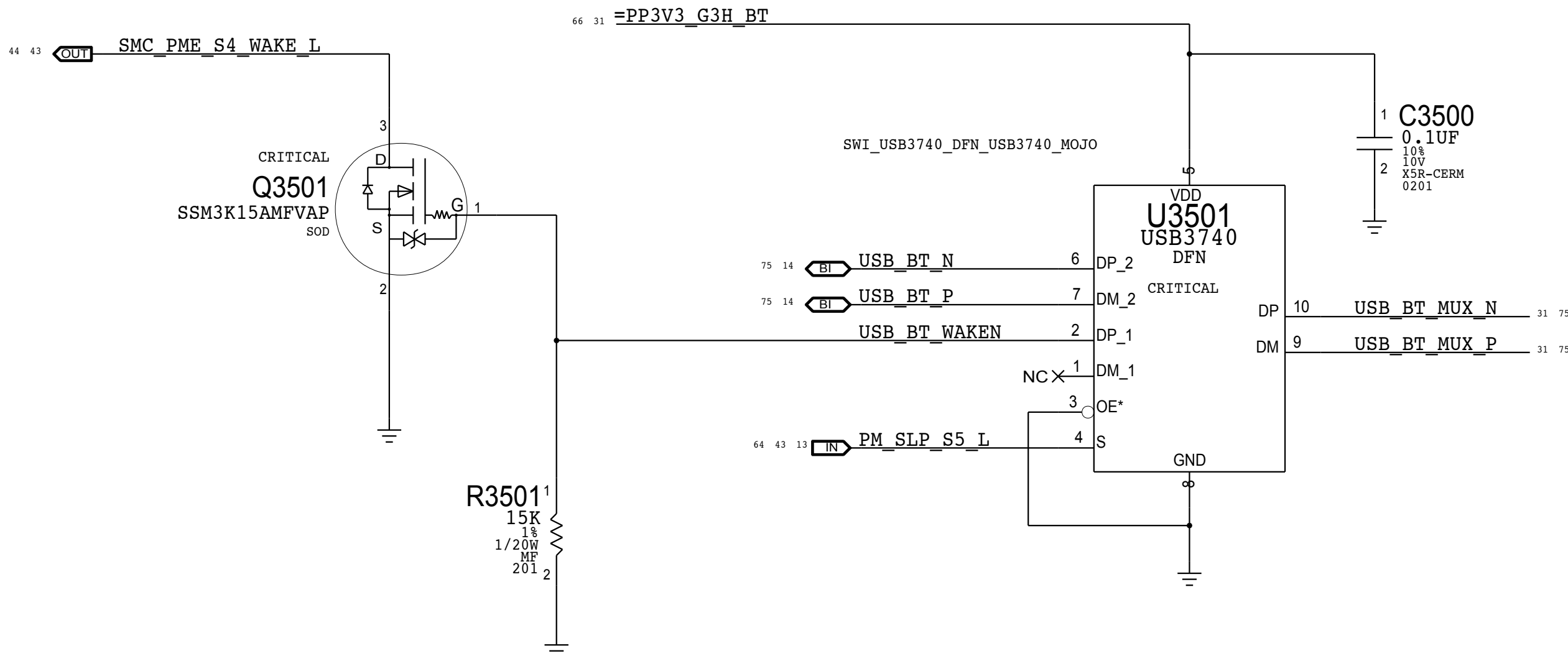
SWITCH	TPS22966
AP SLEW RATE	1185 us
BT SLEW RATE	1185 us
Equation	$0.32 * C_t + 13.7$



Supervisor & CLKREQ# Isolation
Delay = 130 ms +/- 20%



Wake from BT in G3H circuit



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		SHEET	31 OF 81

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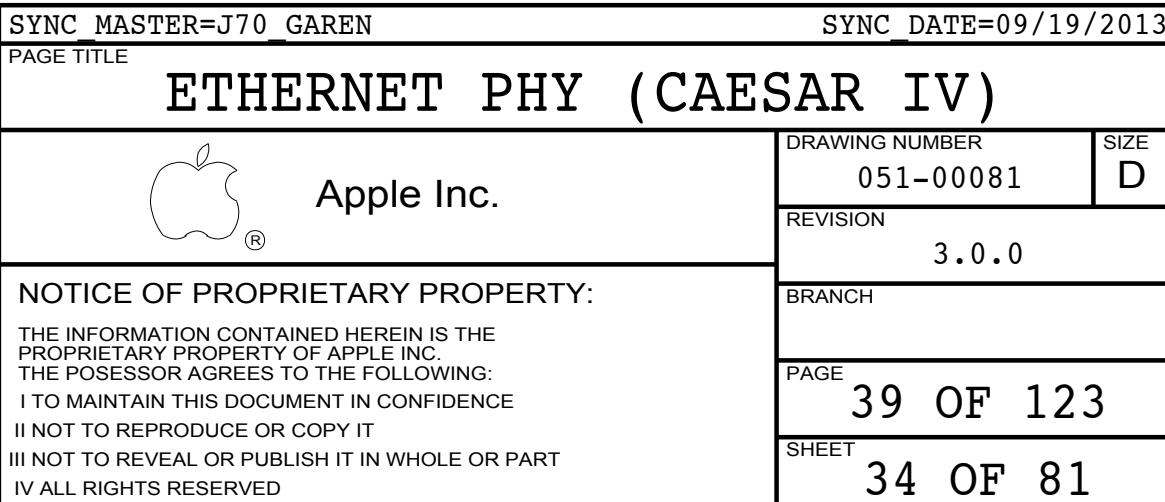
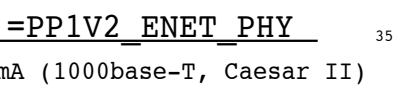


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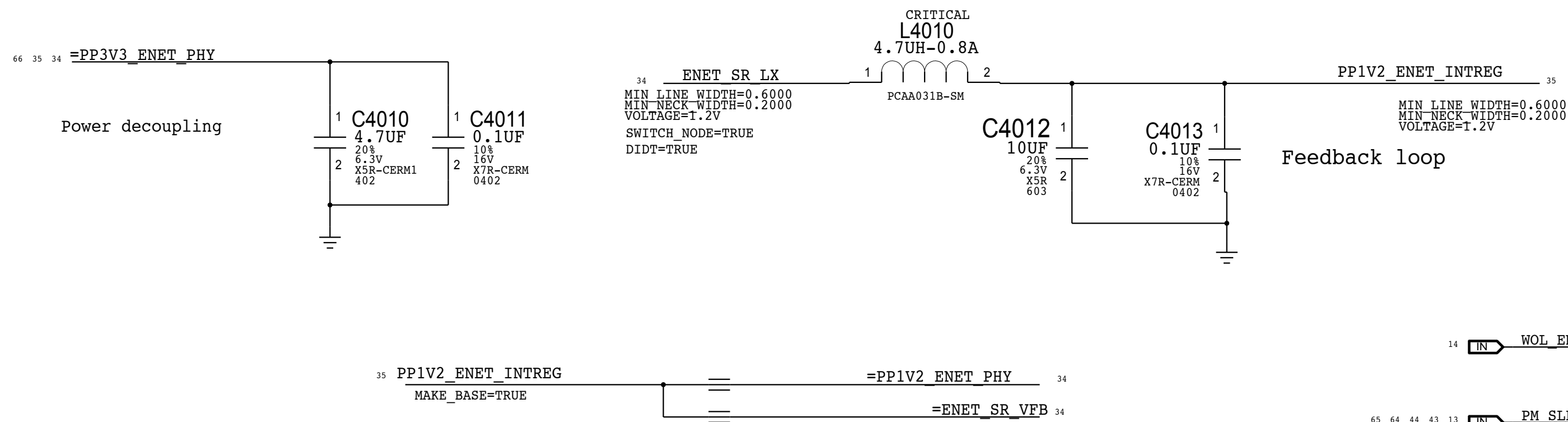
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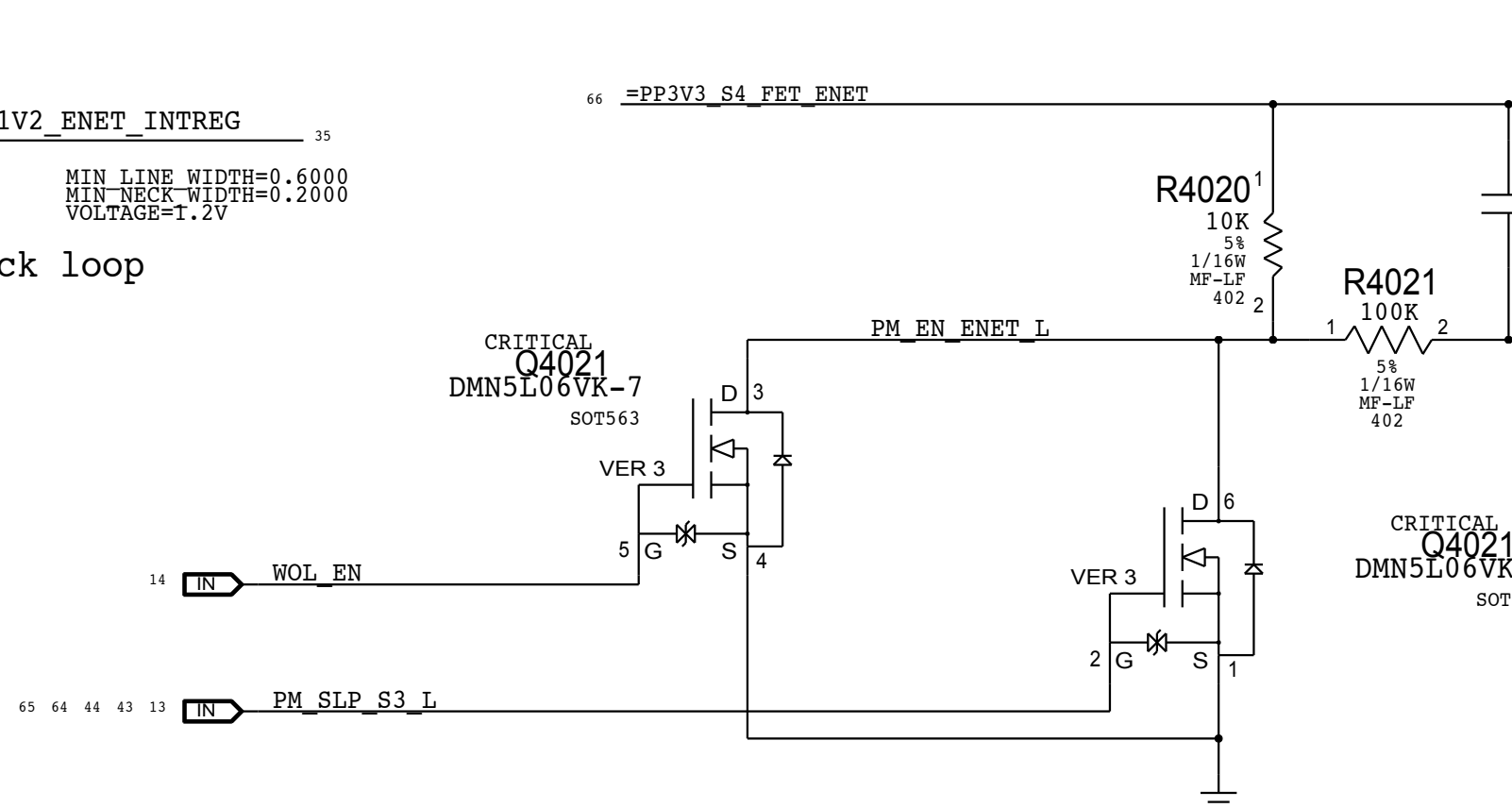
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CAESAR IV 1.2V INT.VR CMPTS

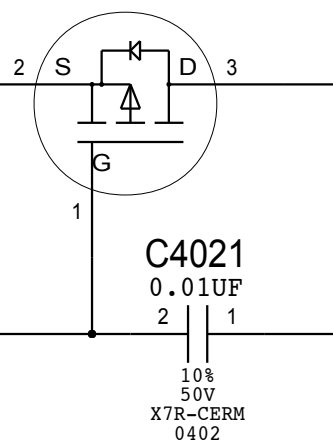


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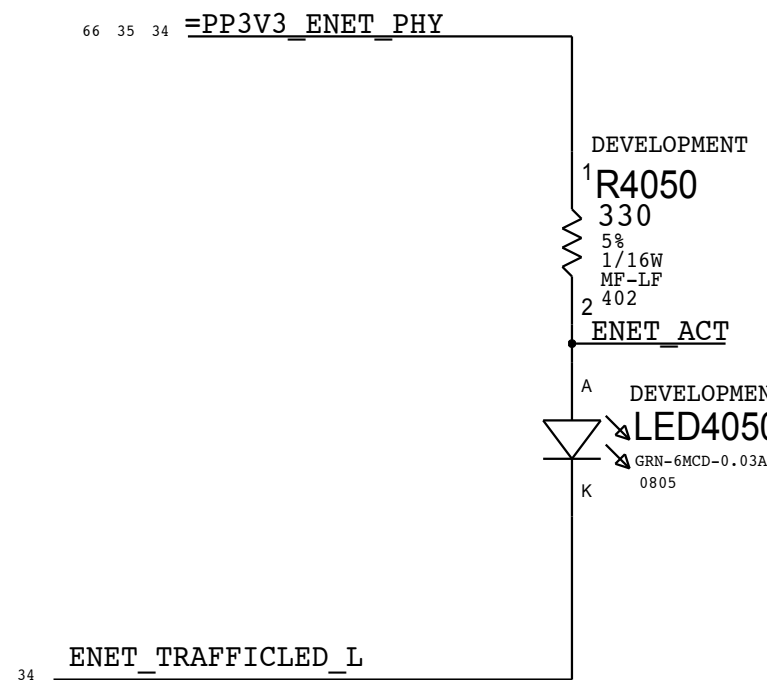


3.3V ENET FET

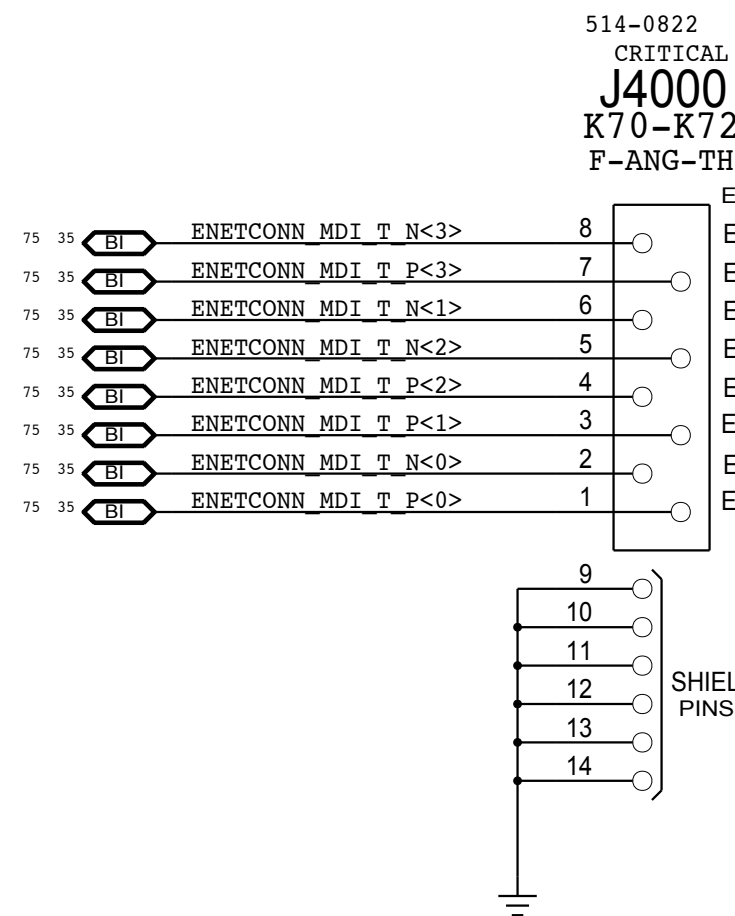
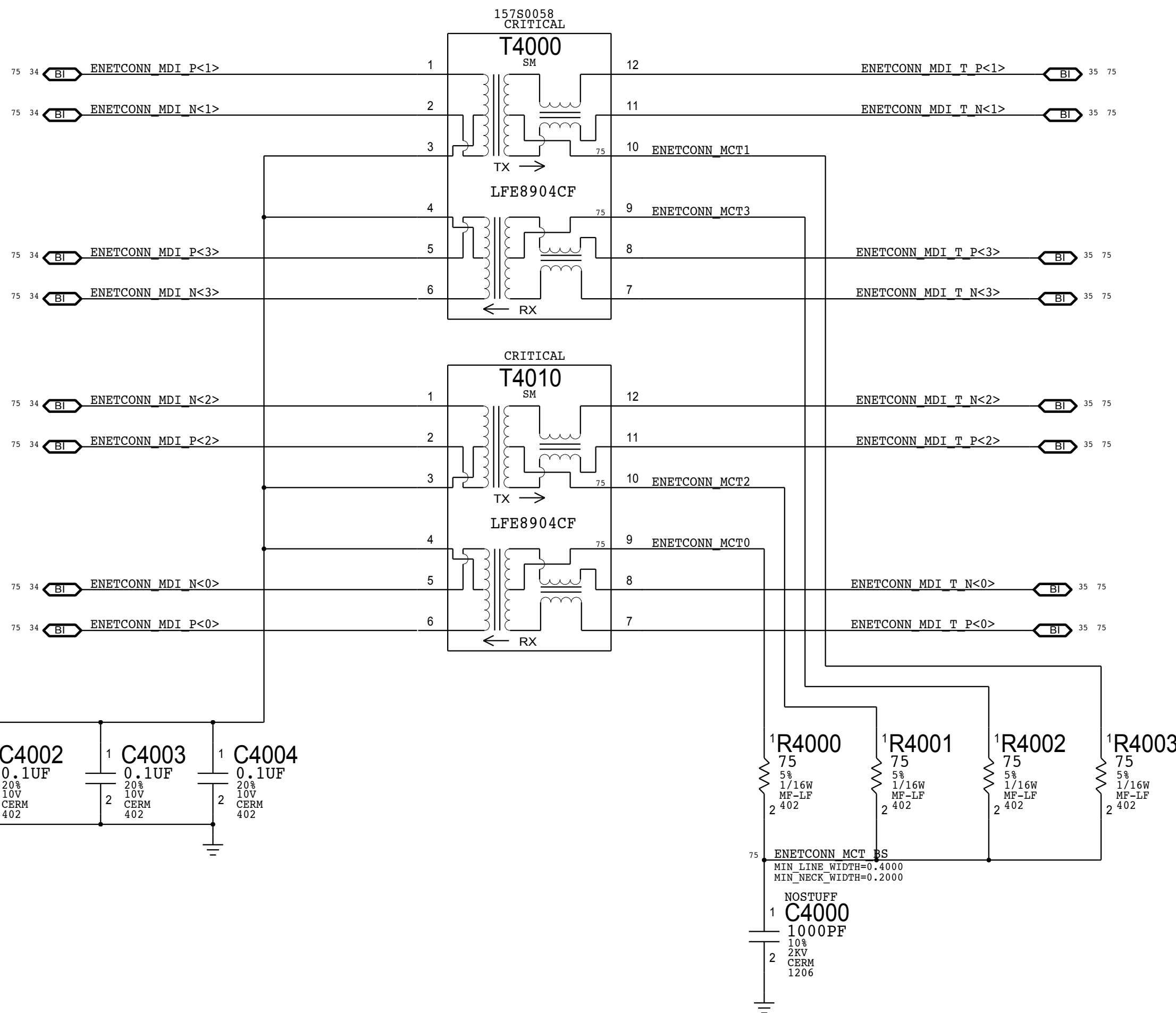
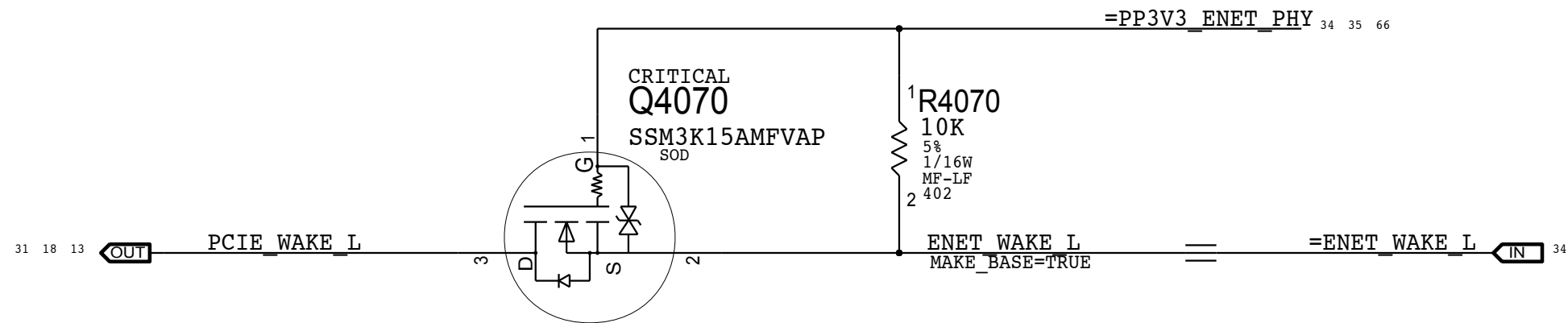
CRITICAL
Q4020
NTR4101P
SOT-23-HF



CAESAR IV ACTIVITY LED



CAESAR IV WAKE# ISOLATION



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		SHEET	35 OF 81

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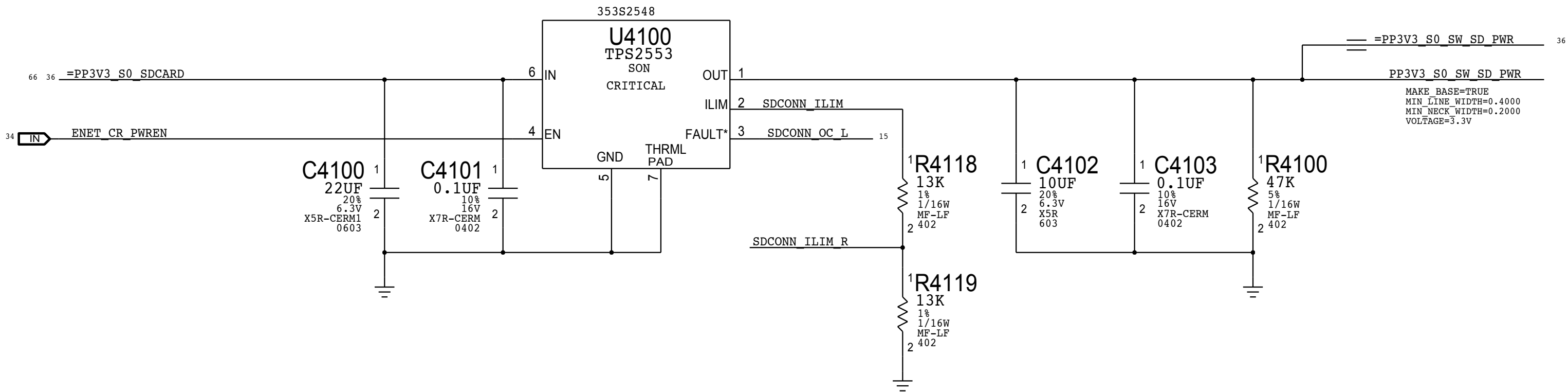
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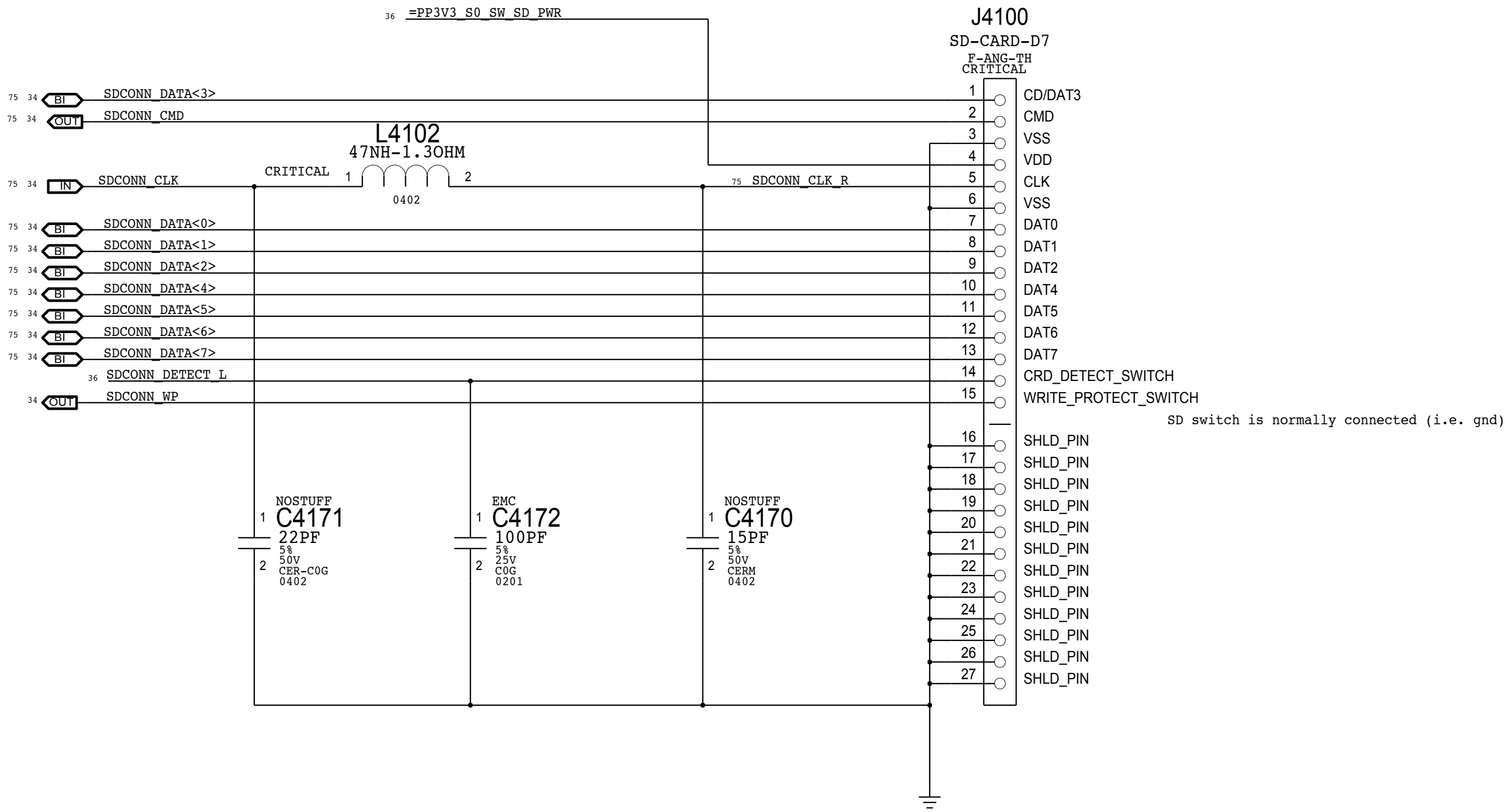
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SD CARD 3.3V OVERCURRENT PROTECTION CHIP

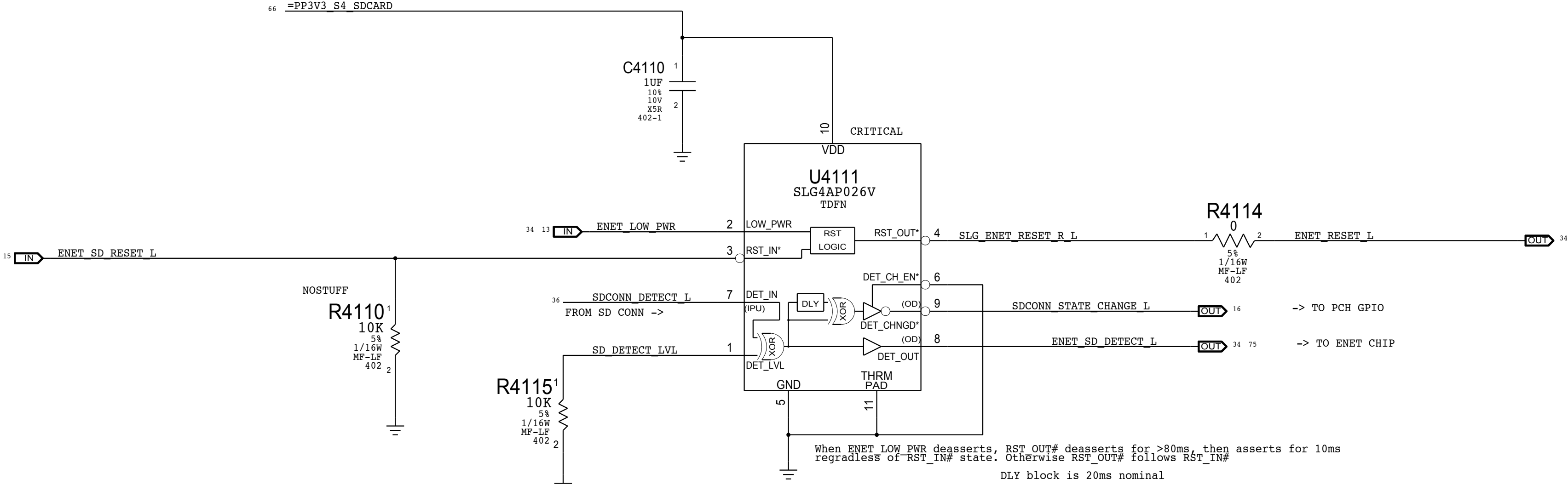


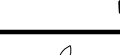
J16:516-0249 / J17:512-0038

SD CARD CONNECTOR



SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



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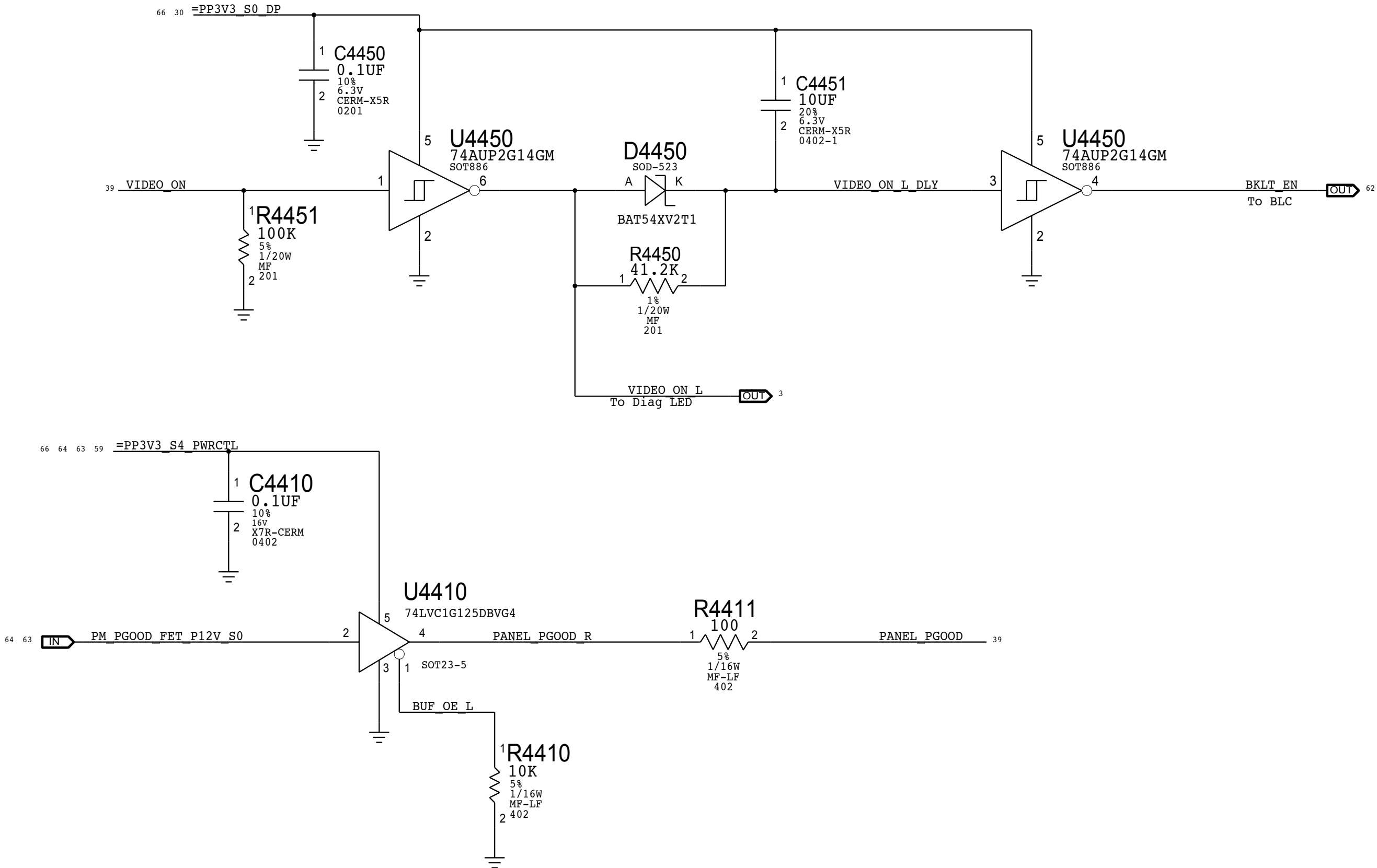
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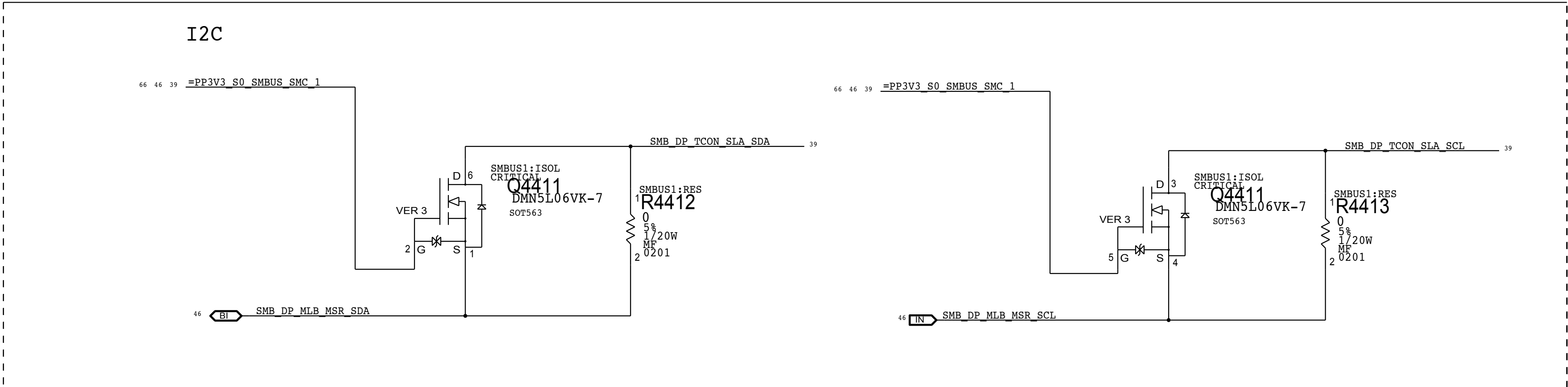
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Backlight Control

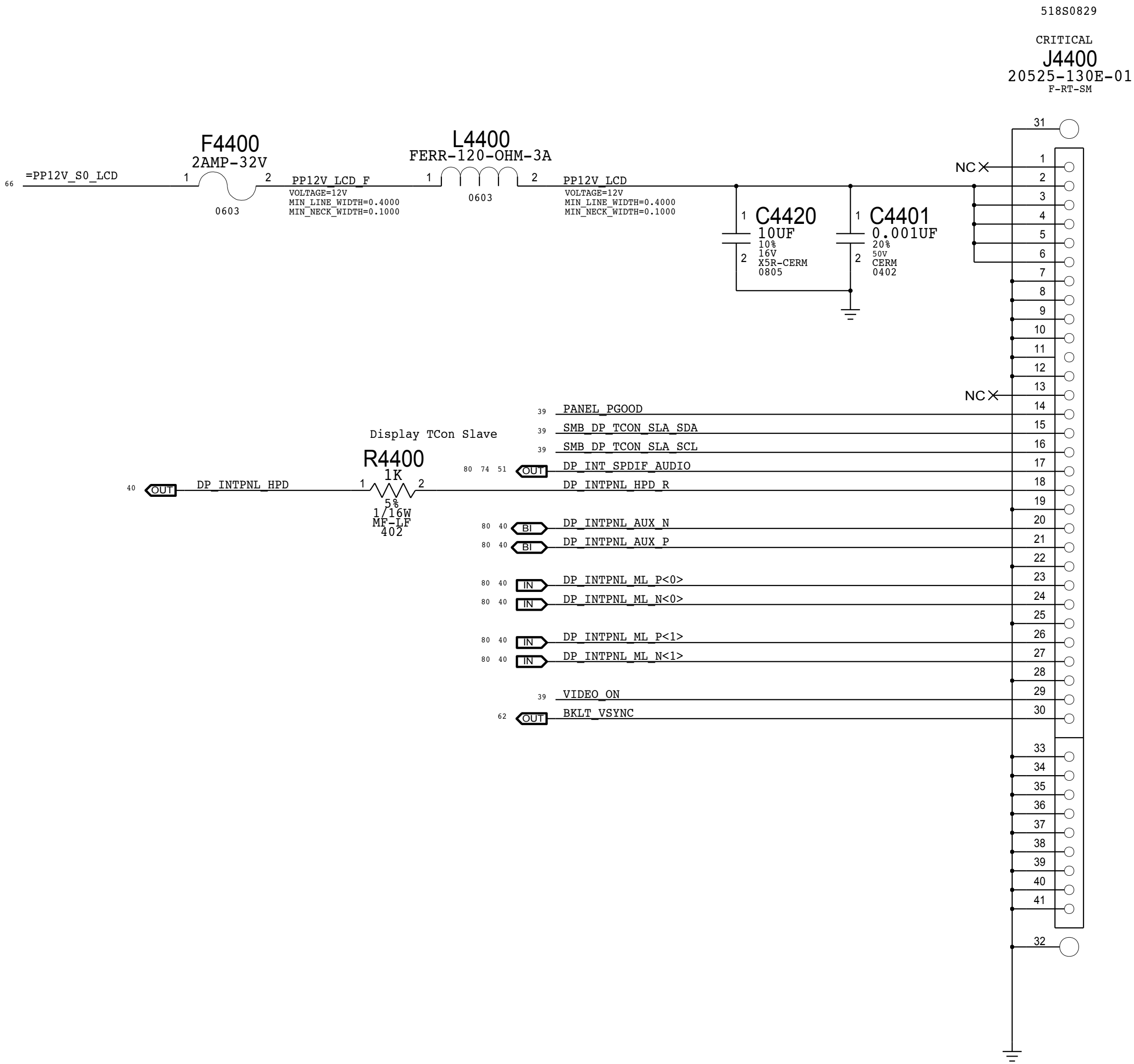
Delay applies only on a L->H transition on VIDEO_ON. This guarantees video is valid before the backlight is enabled.
On a H->L transition, output follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video



SMBus Isolation



Internal DP Connector



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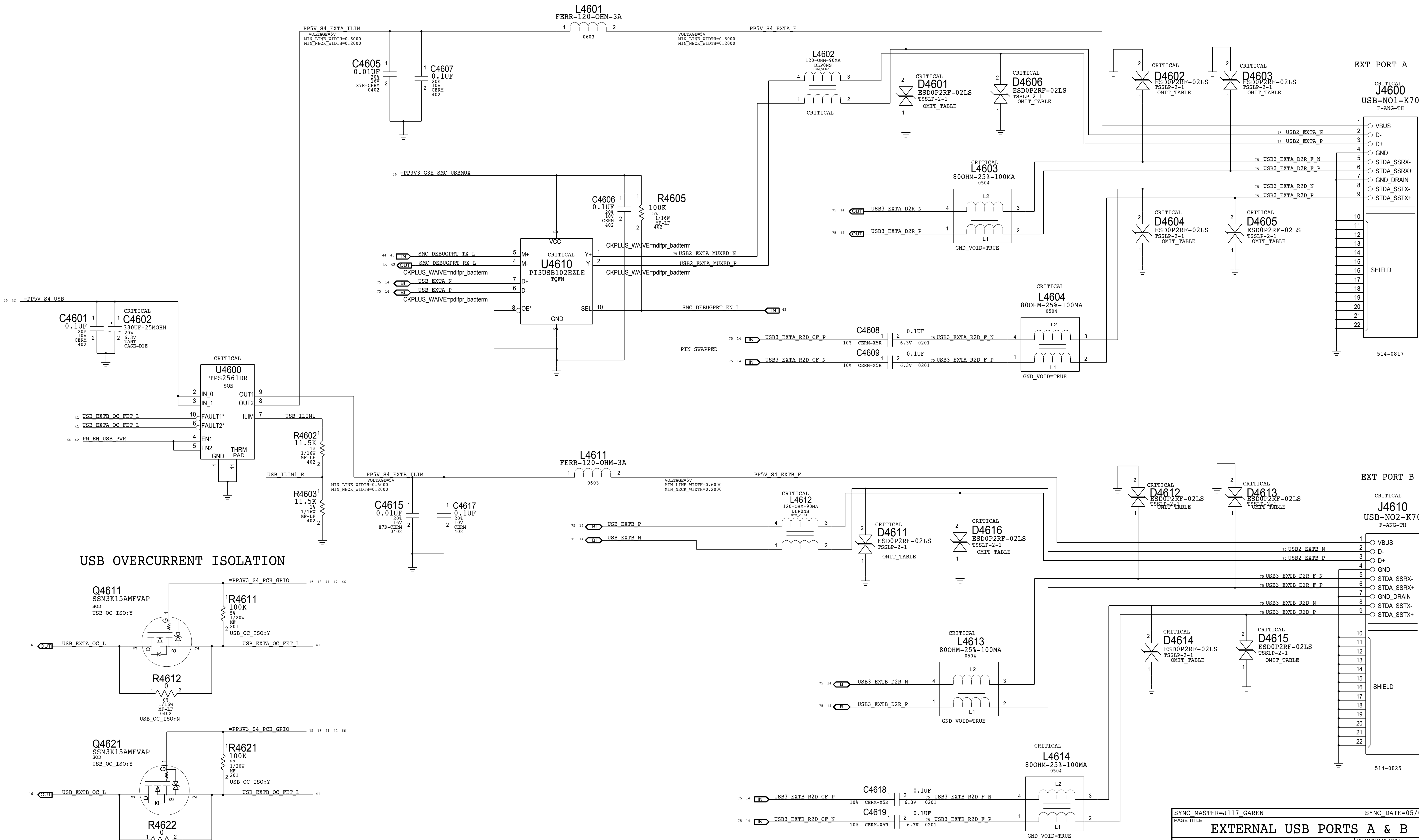
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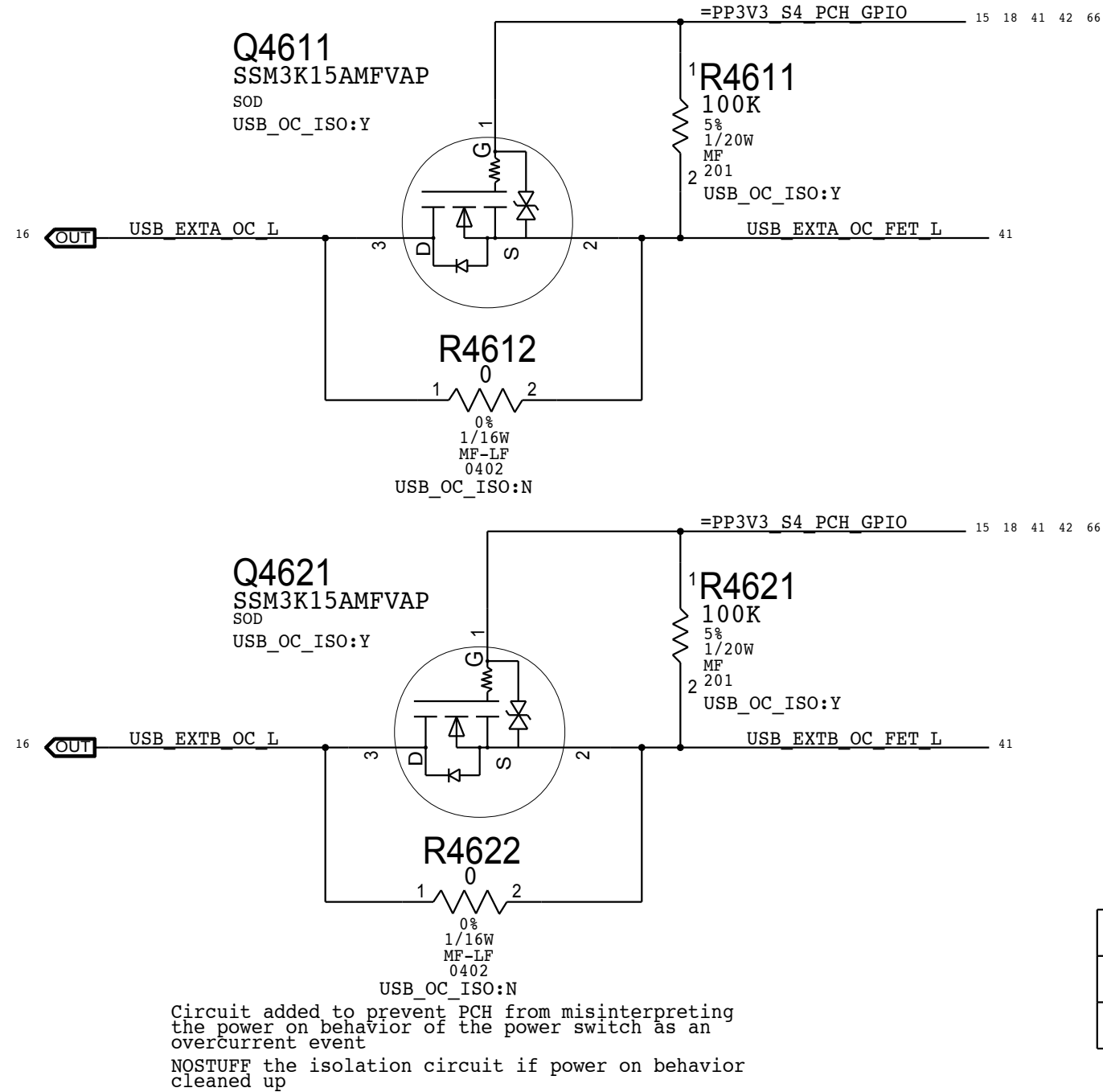
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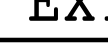


USB OVERCURRENT ISOLATION



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
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37780155	6	USB3 diodes	D4611,D4612,D4613,D4614,D4615,D4616	

P and N pins swapped for cleaner routing
Lane polarity inversion supported by USB3
See Section 6.4.2 of USB3 Spec

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 Apple Inc.		DRAWING NUMBER	051-00081
		REVISION	3.0.0
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		PAGE	46 OF 123
		SHEET	41 OF 81

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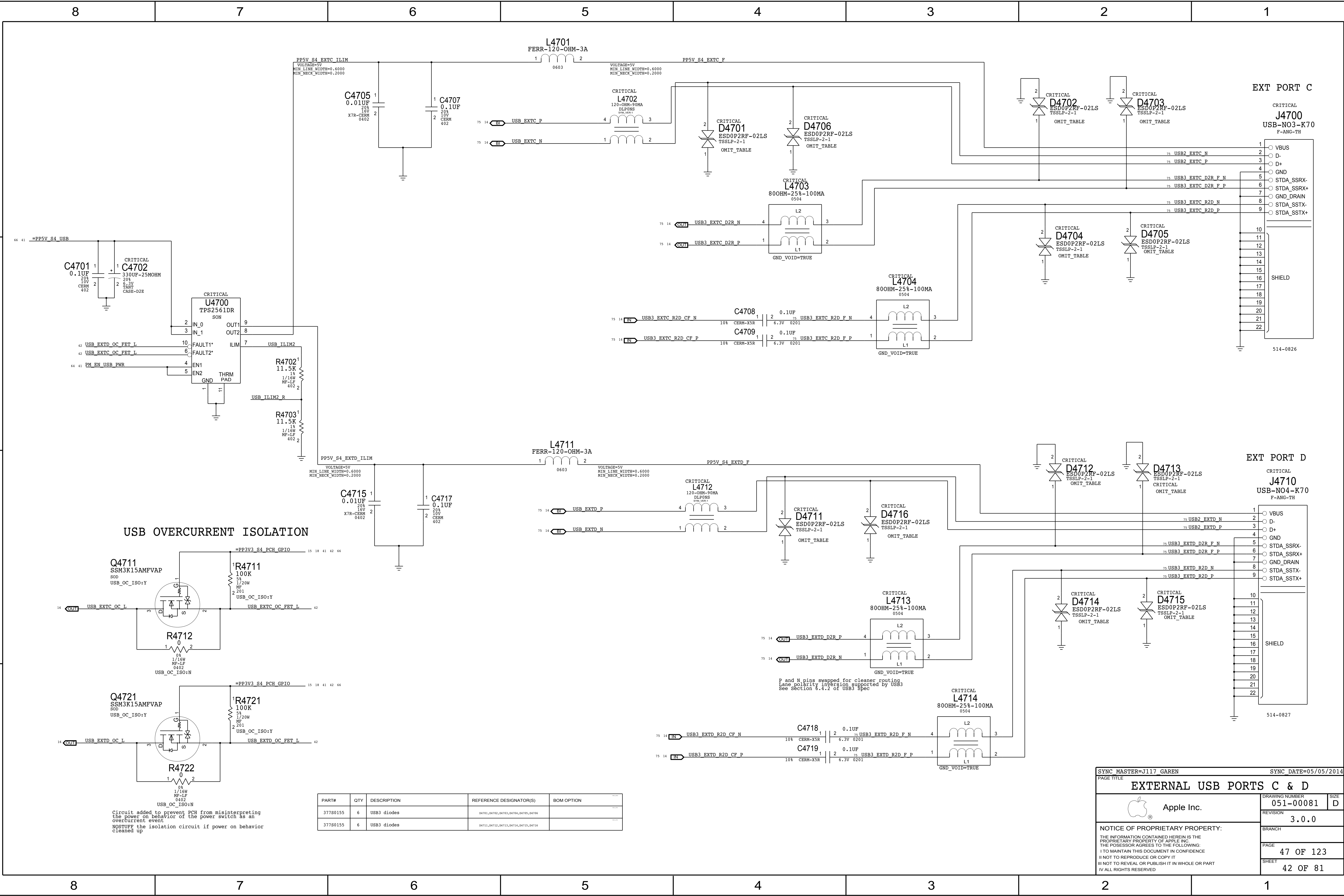
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
37780155	6	USB3 diodes	D4701,D4702,D4703,D4704,D4705,D4706	
37780155	6	USB3 diodes	D4711,D4712,D4713,D4714,D4715,D4716	

SYNC_MASTER=J117_GAREN

SYNC_DATE=05/05/2014

PAGE TITLE

EXTERNAL USB PORTS C & D

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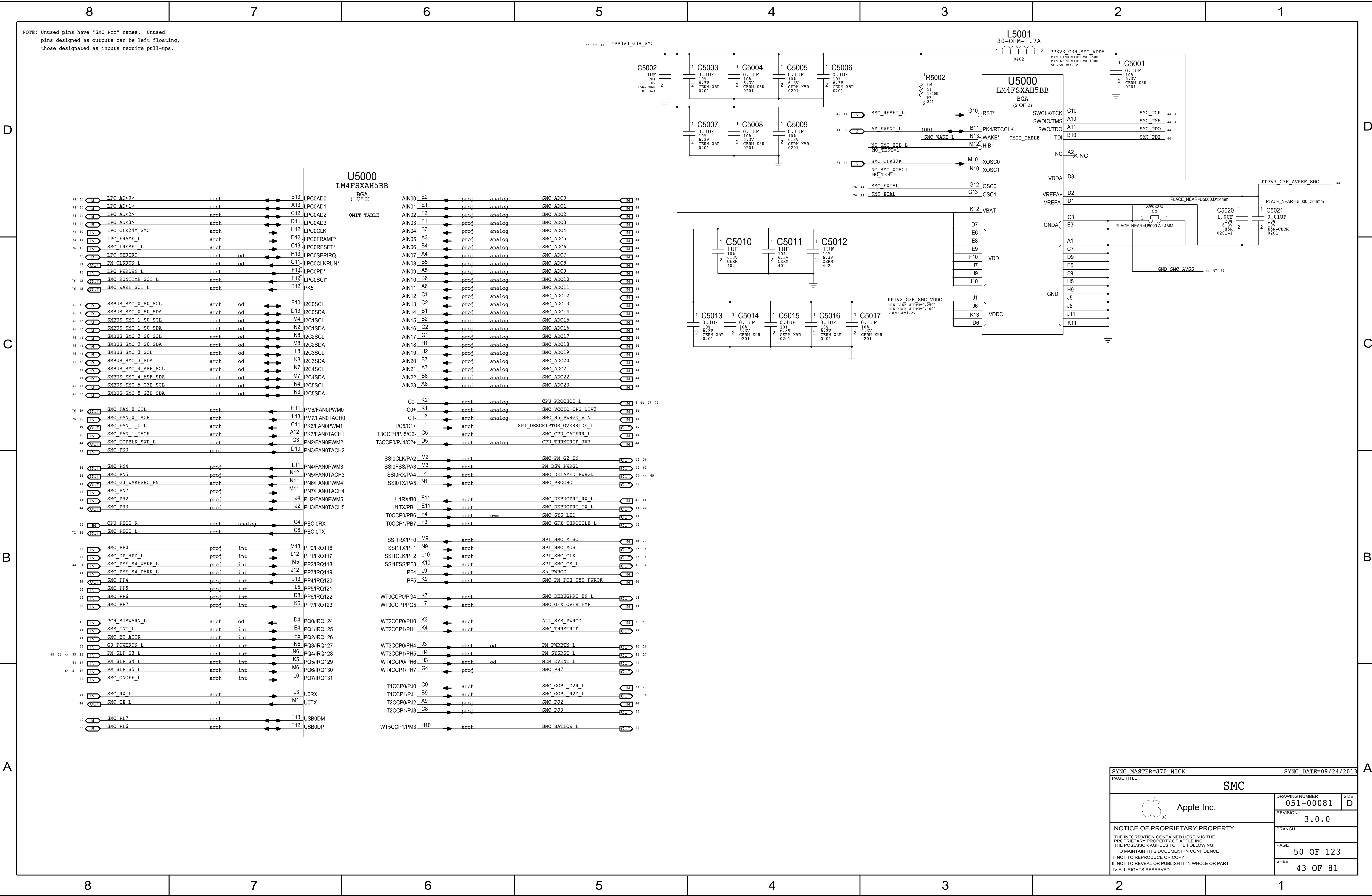
47 OF 123

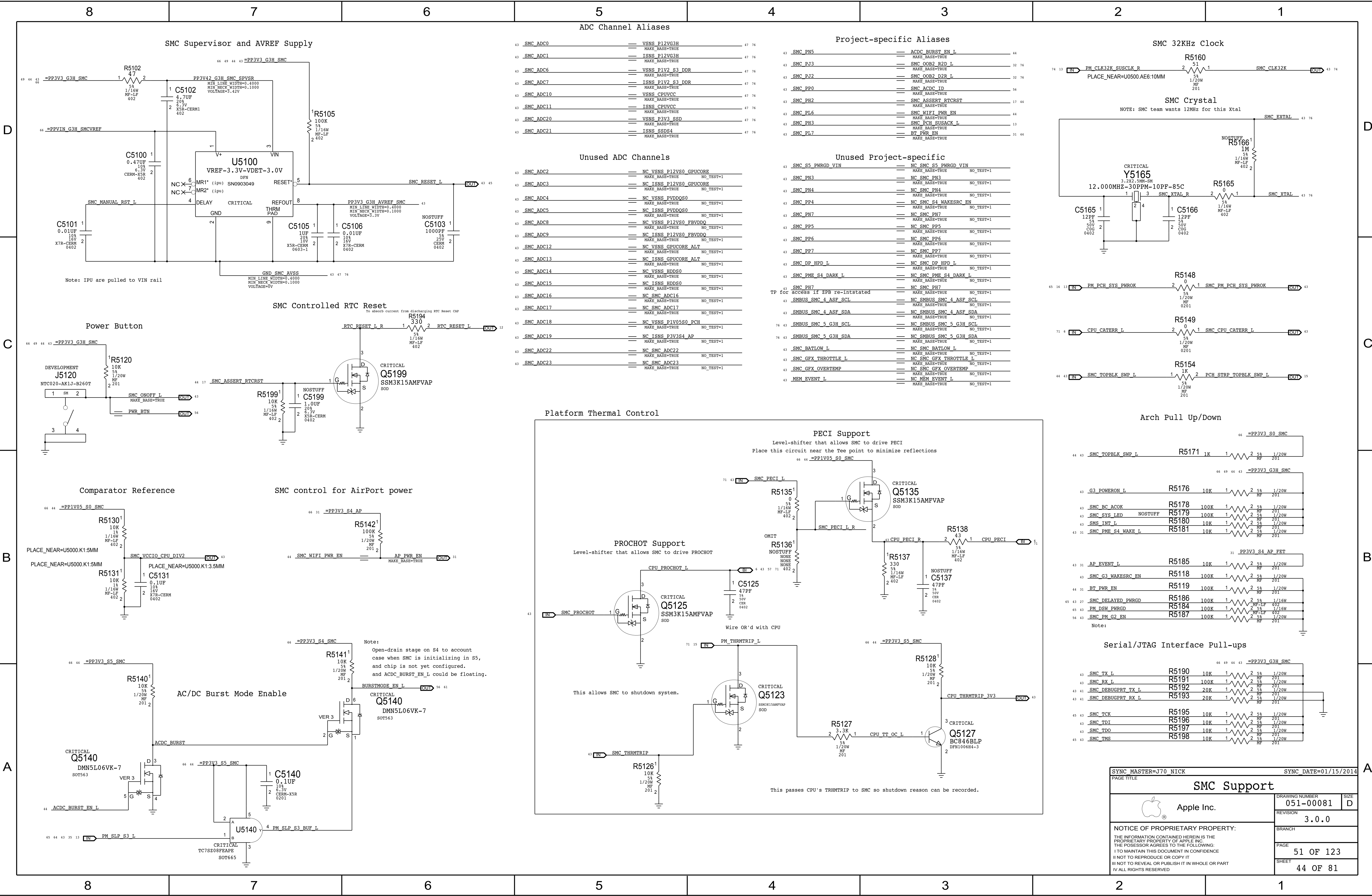
SHEET

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USB OVERCURRENT ISOLATION

Circuit added to prevent PCB from misinterpreting the power on behavior of the power switch as an overcurrent event
NoSTUFF the isolation circuit if power on behavior cleaned up





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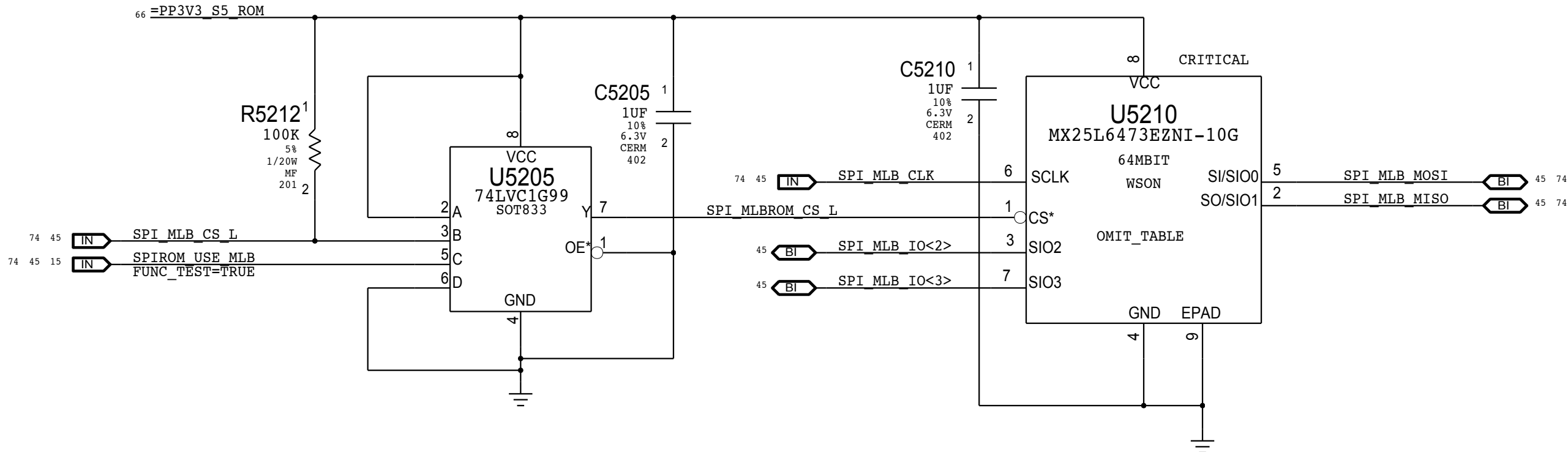
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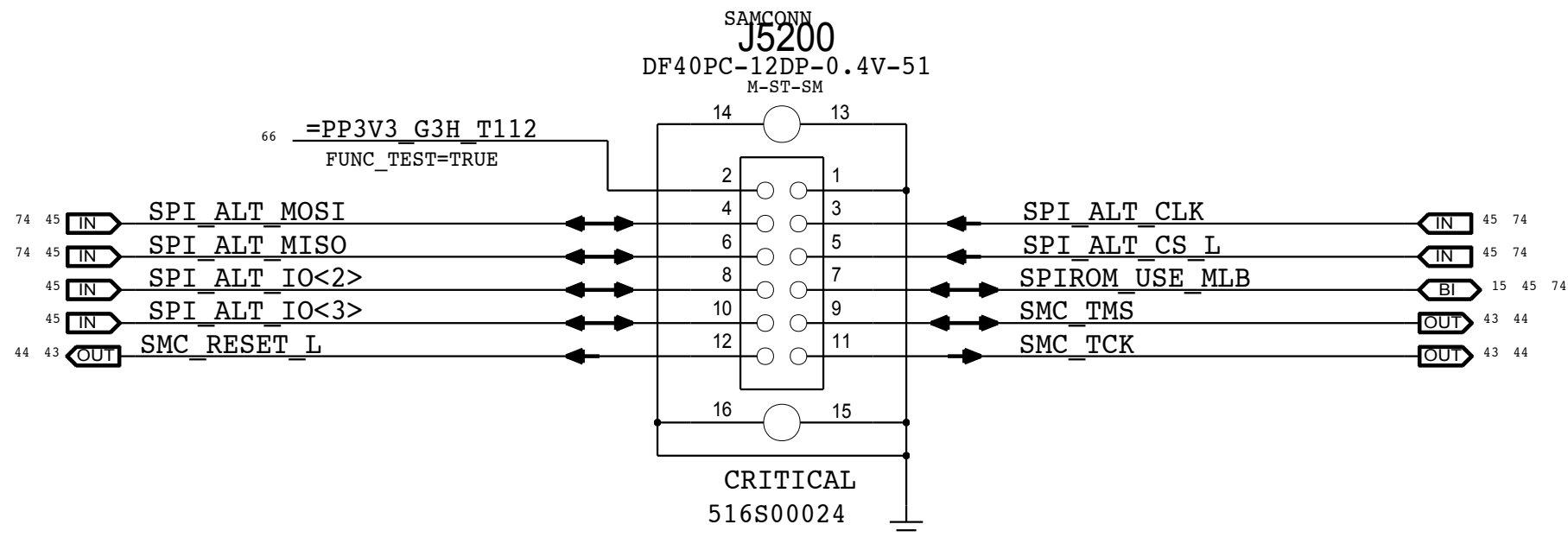
A

SPI BootROM

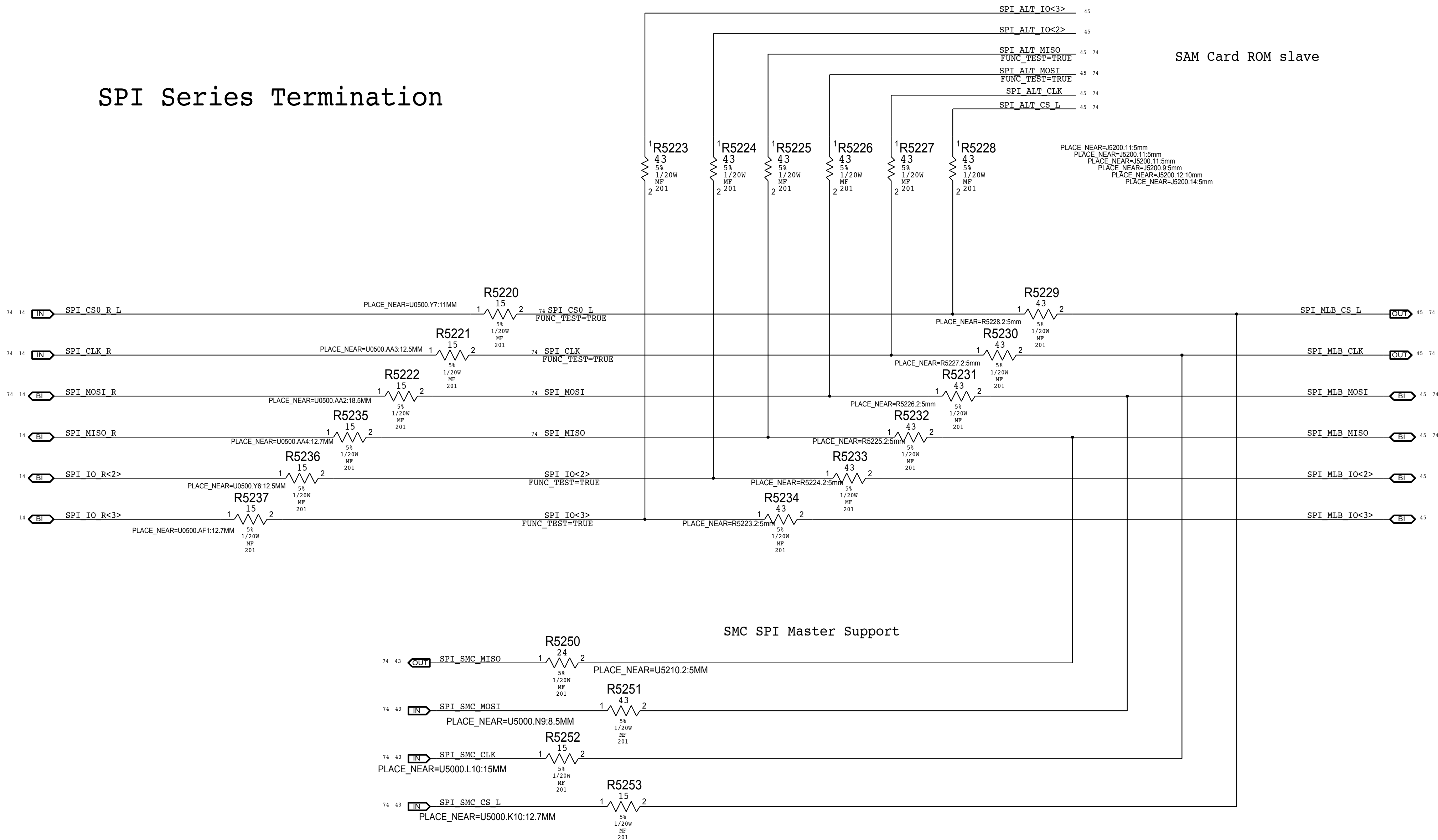
Quad_IO Mode (Mode 0 & 3) supported
SPI Freq: 50MHz for PCH, 20 MHz for SMC



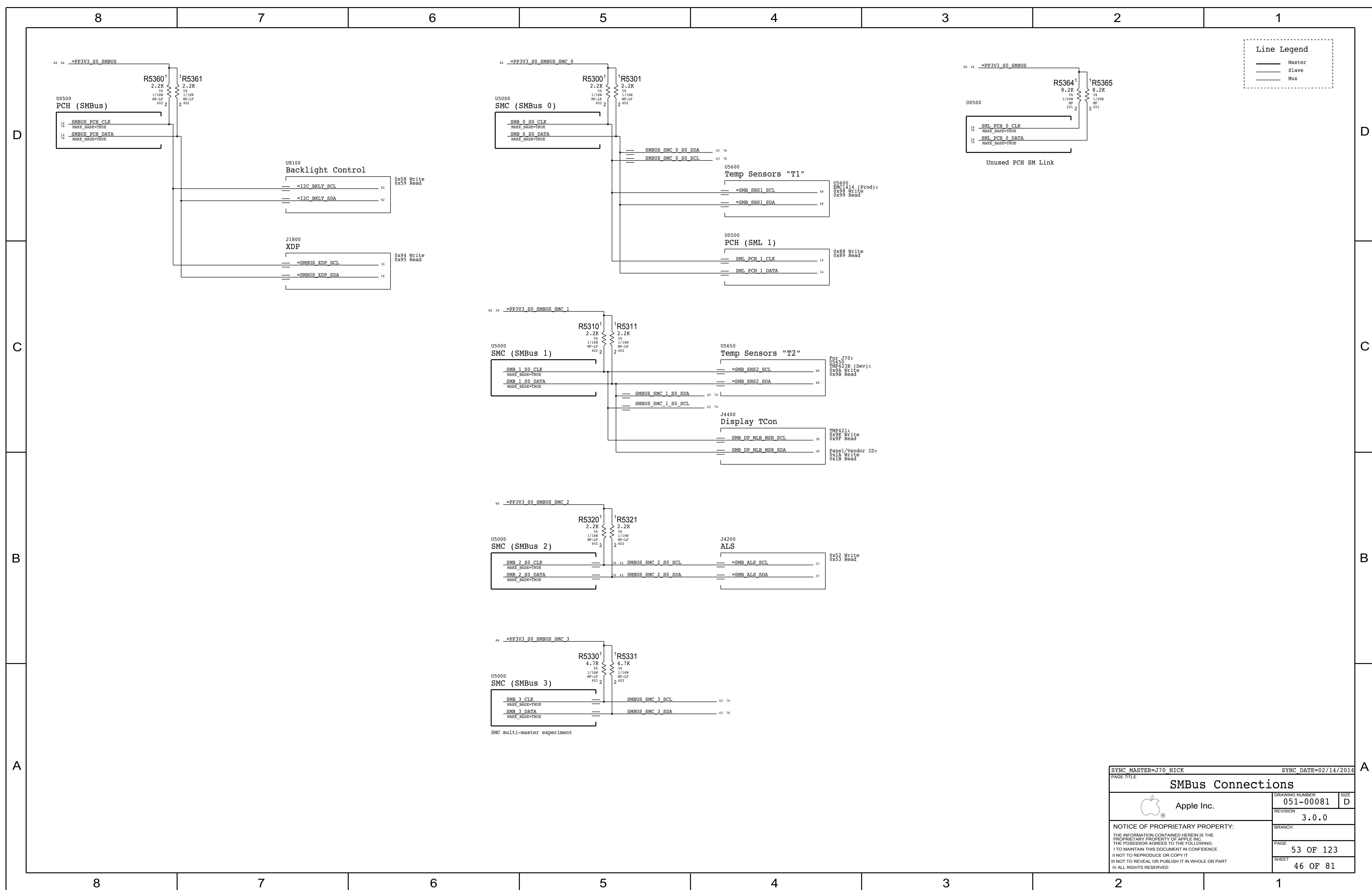
SPI+SWD SAM Connector



SPI Series Termination

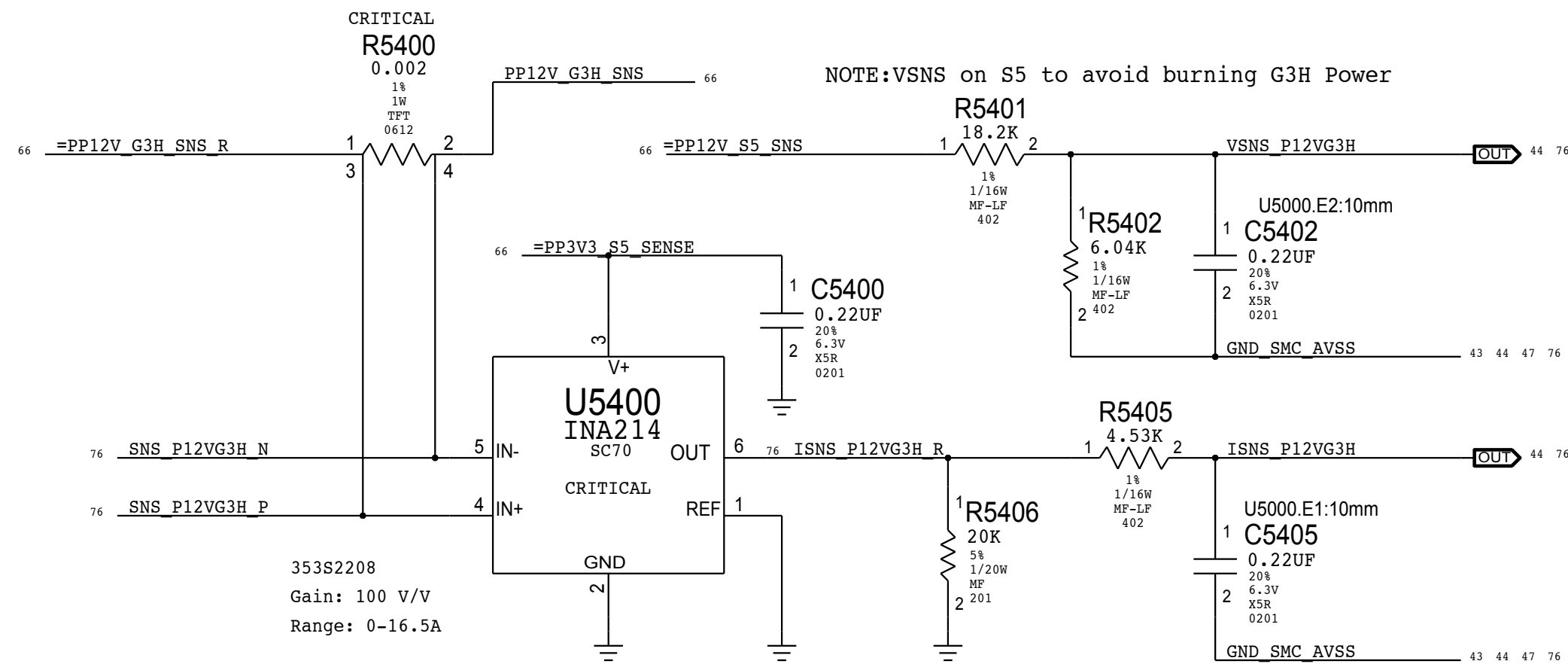


SYNC_MASTER=J117 TONY		SYNC_DATE=05/05/2014	
PAGE TITLE			
SPI and Debug Connector			
	Apple Inc.	DRAWING NUMBER	051-00081
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		PAGE	52 OF 123
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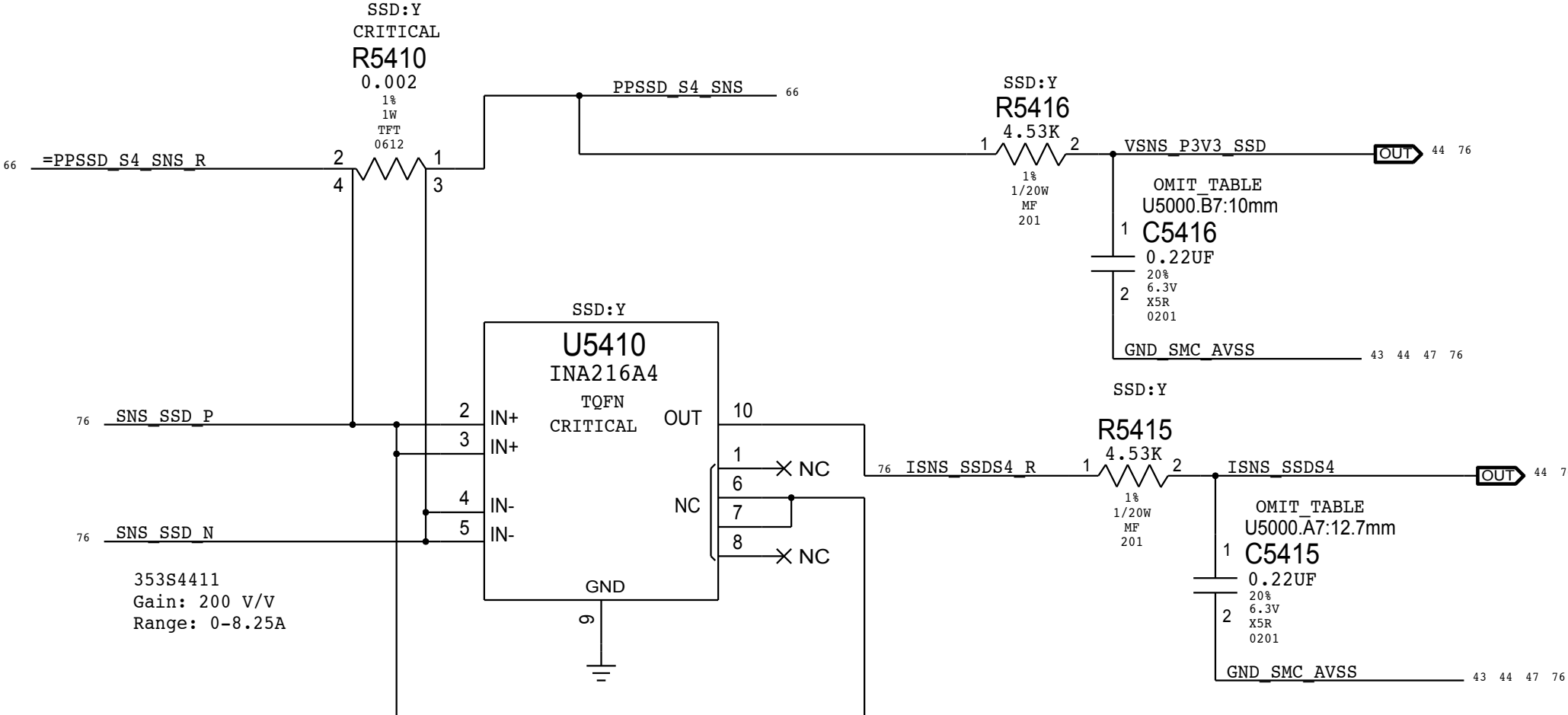
12V G3H (VD2R:ADC0/ID2R:ADC1)

AC/DC lowside sense (System total)



SSD S4 (IH1R:ADC21/VR3R:ADC20)

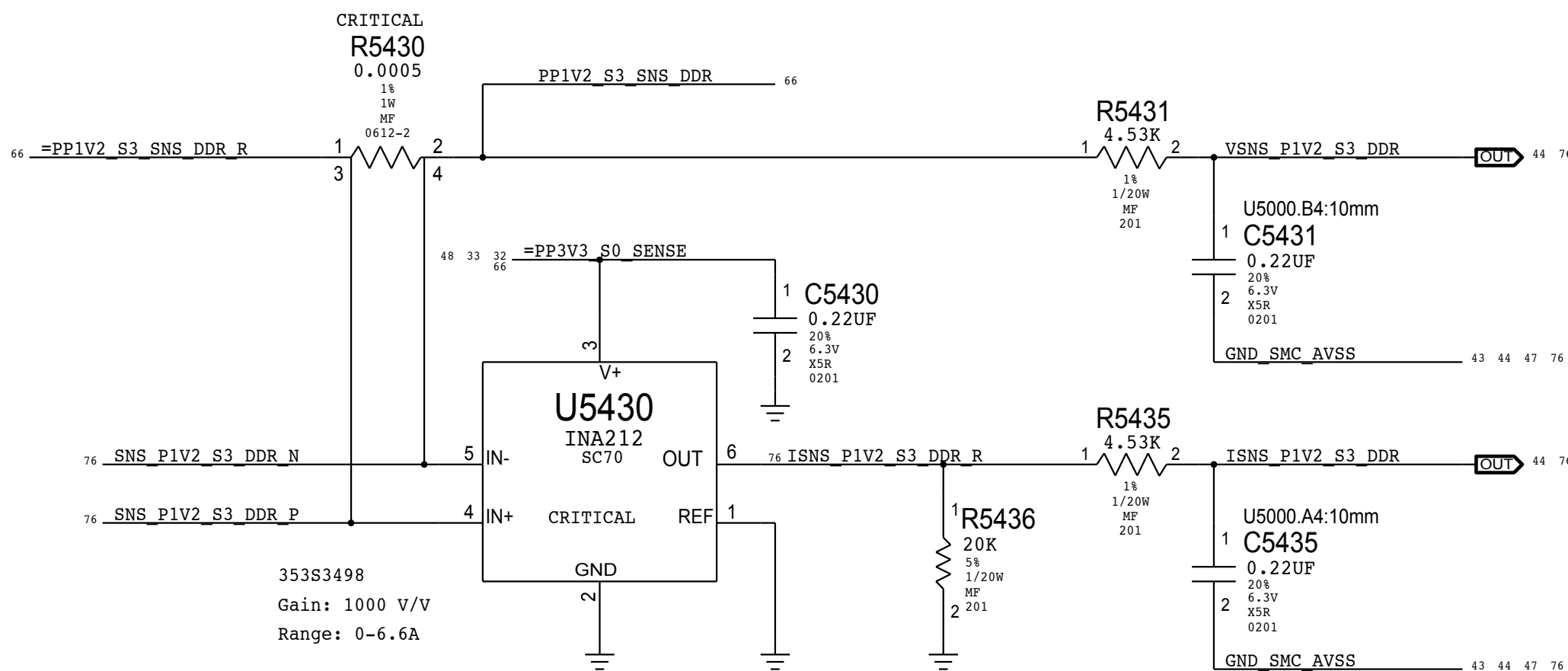
I-sense / V-sense for SSD



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP,0.22UF,201	C5415,C5416	SSD+Y
11780201	2	RES,0 OHM,201	C5415,C5416	SSD+N

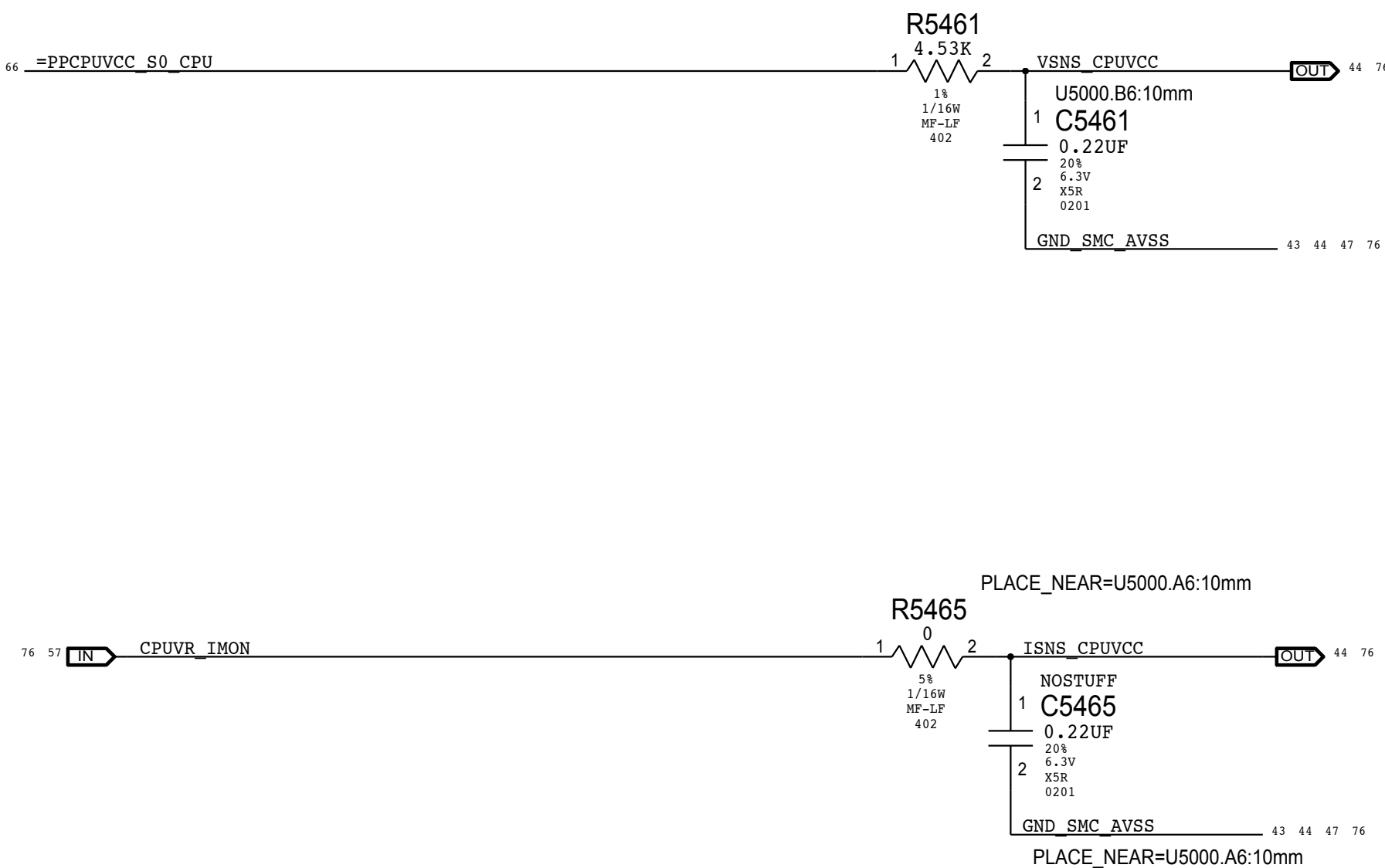
VDDQ S3 (VM0R:ADC6/IM0R:ADC7)

VDDQ lowside sense for DDR



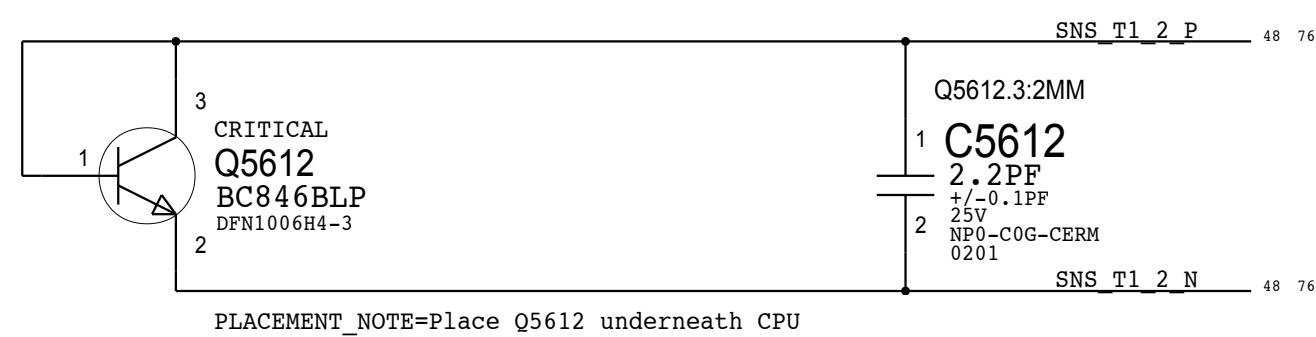
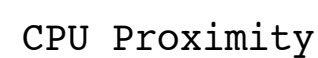
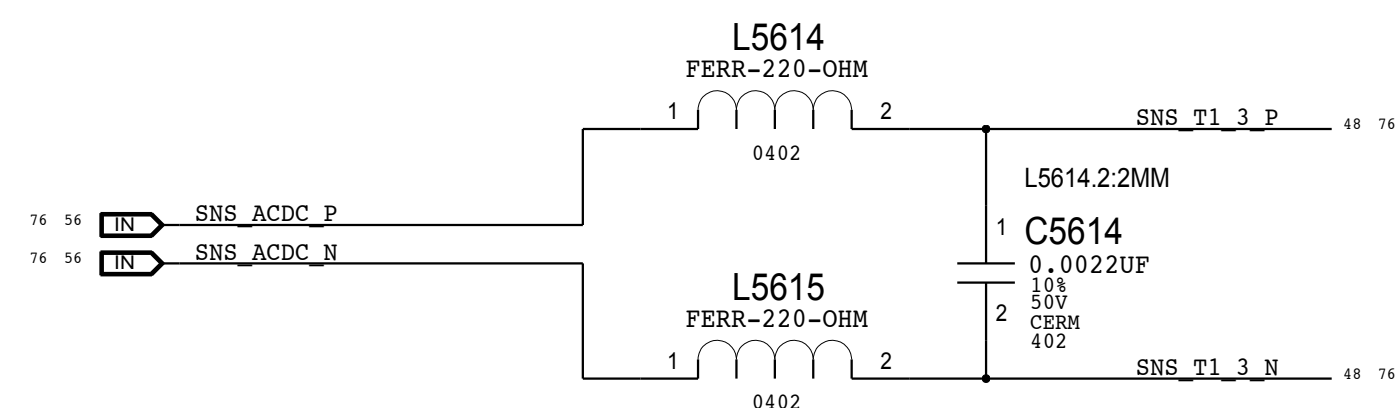
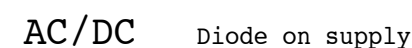
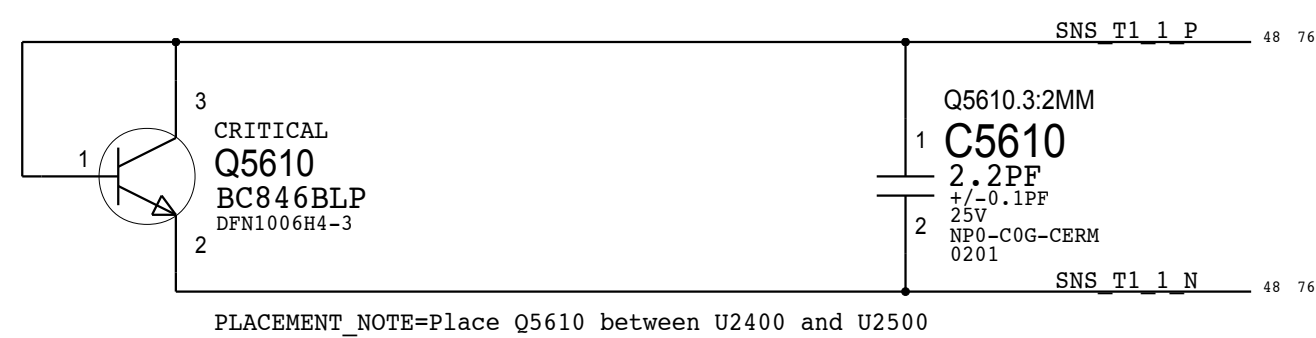
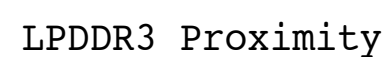
CPU Core (VC0C:ADC10/IC0C:ADC11)

Voltage sense and IMON amp (VC0C, IC0C)

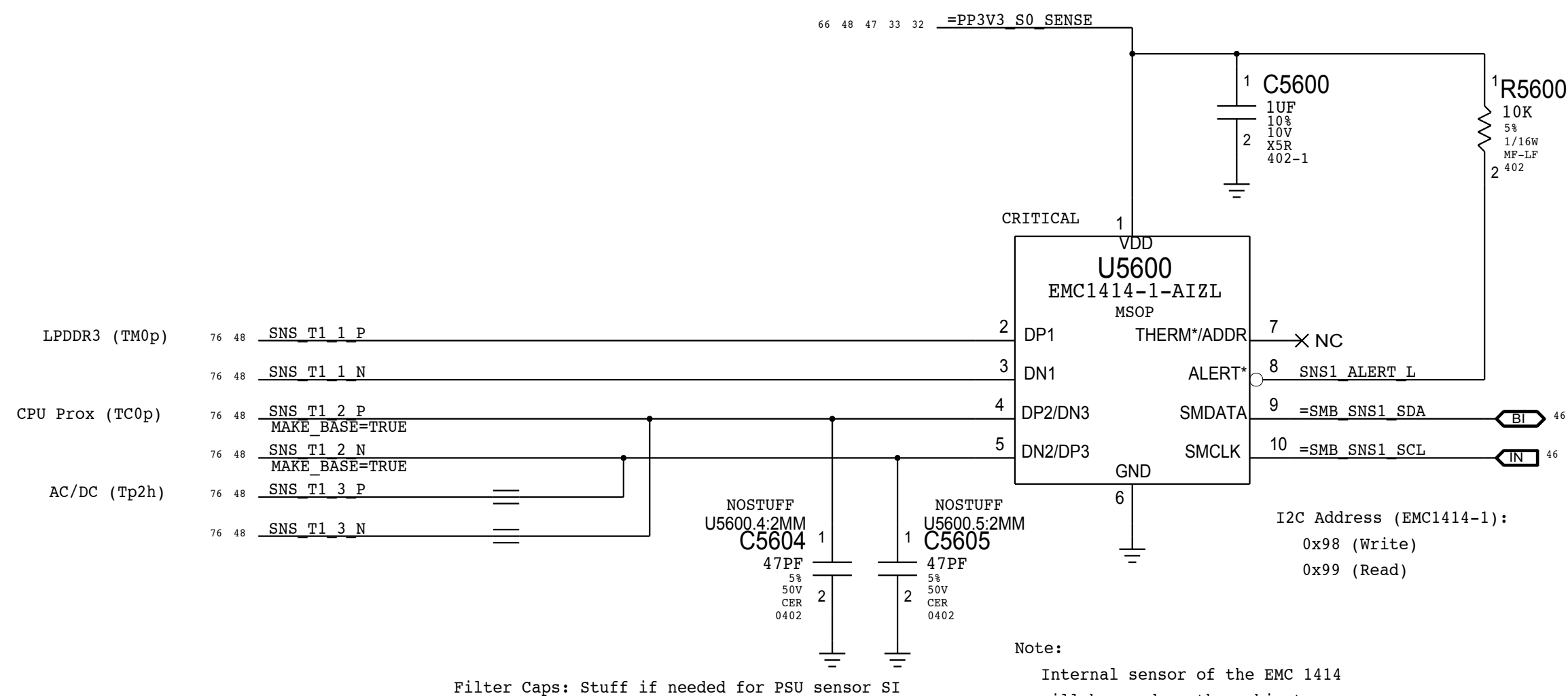


SYNC_MASTER=J70 NICK		SYNC_DATE=11/05/2013	
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I and V Sense			
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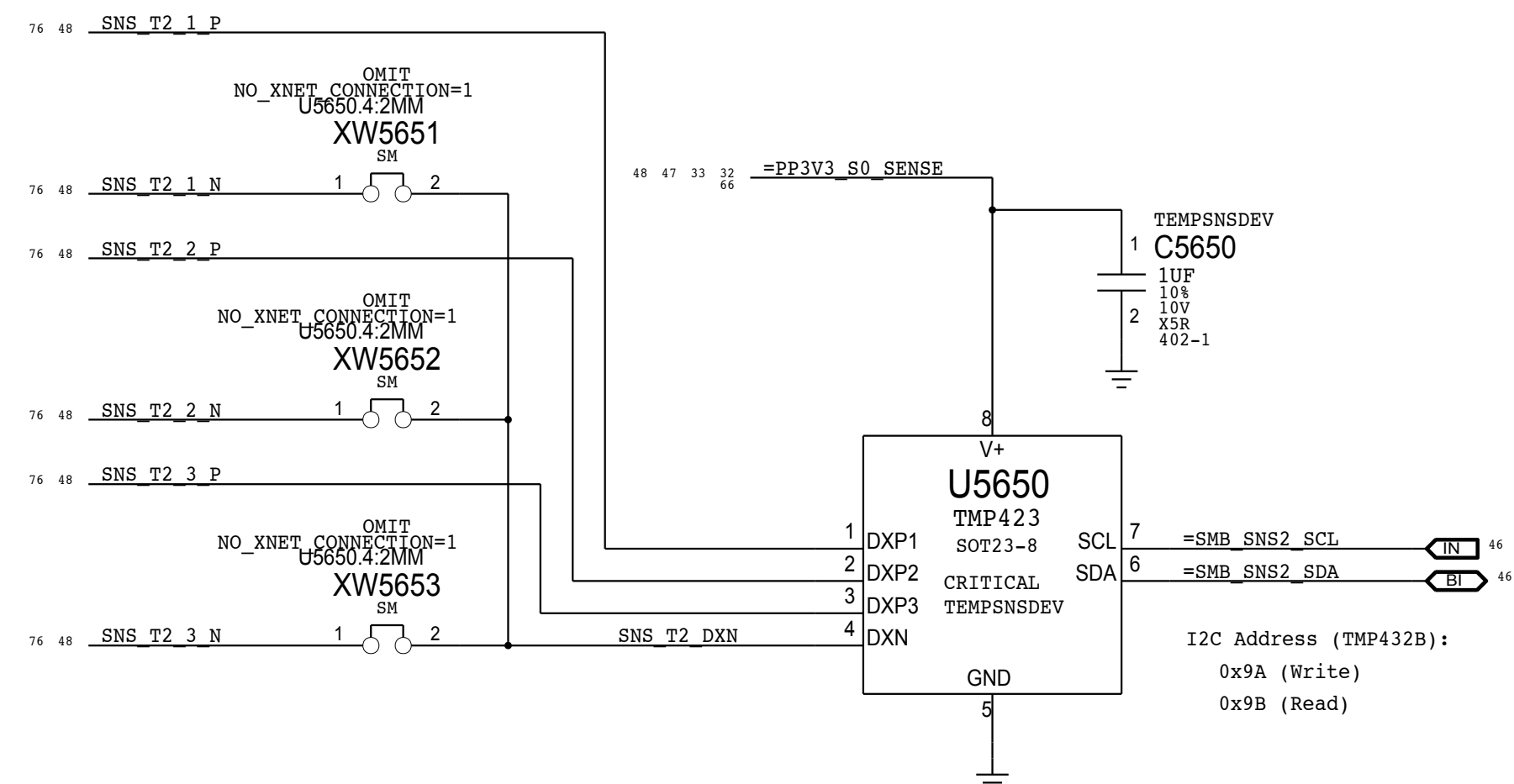
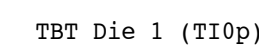
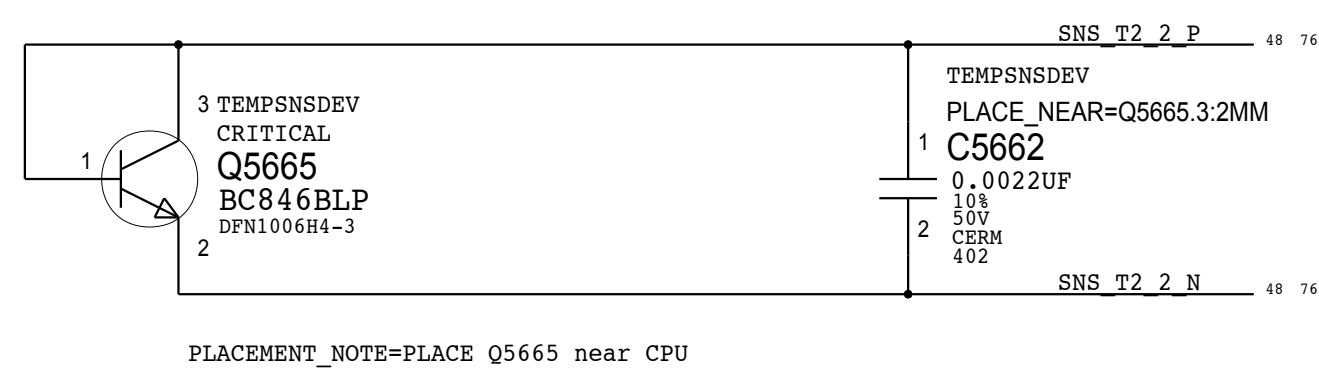
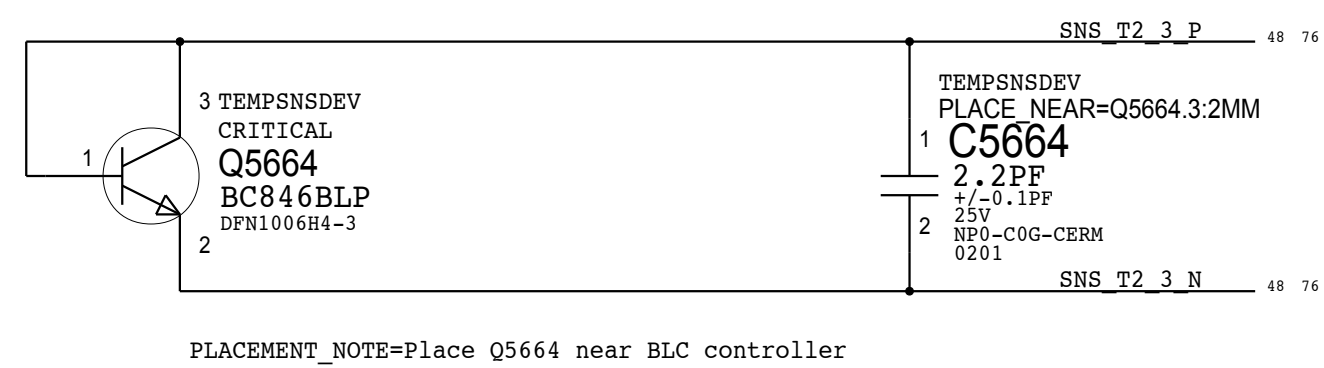
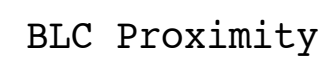
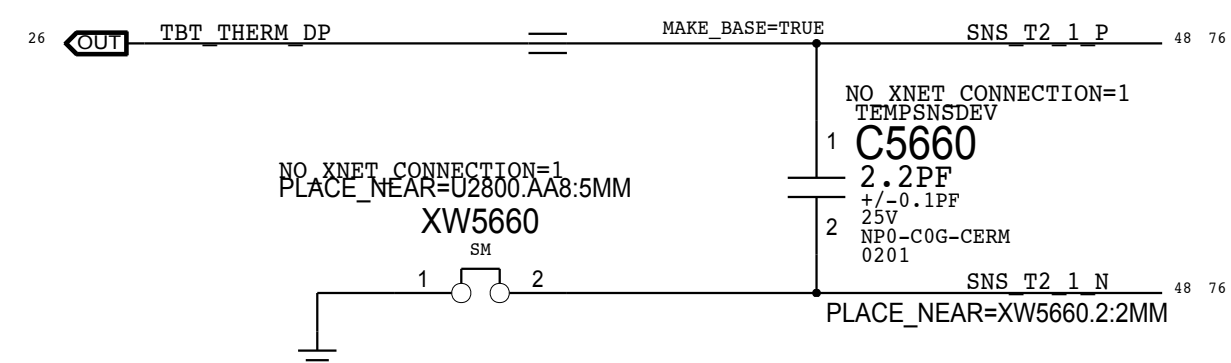
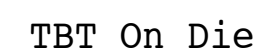
Temperature Sensor T1



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode



Temperature Sensor T2



SYNC MASTER=J70 NICK		SYNC DATE=11/05/2013	
PAGE TITLE			
Temperature Sensors			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00081		D
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PAGE		56 OF 123	
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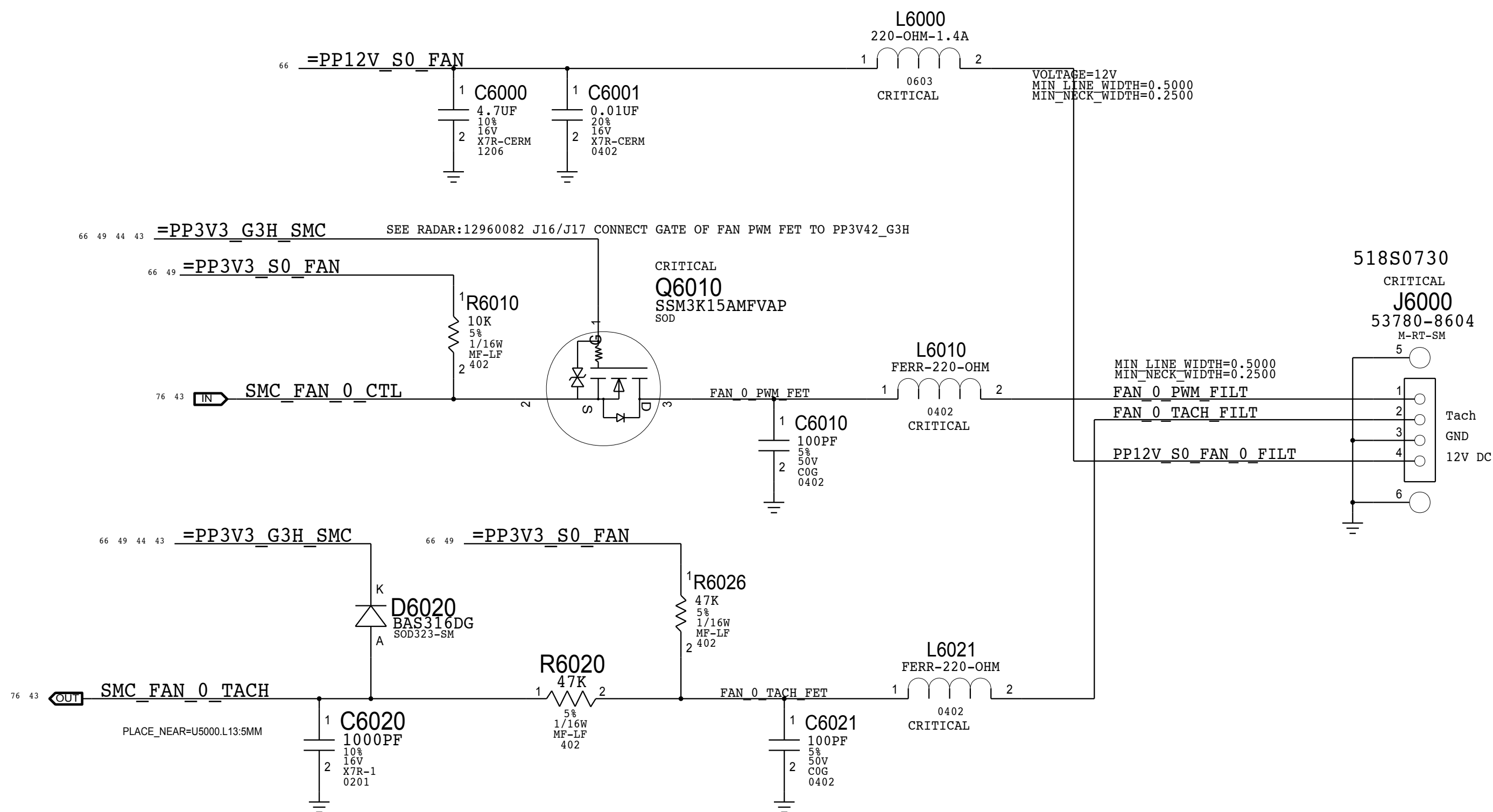
SMC Fan 0 (System)

Note:

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

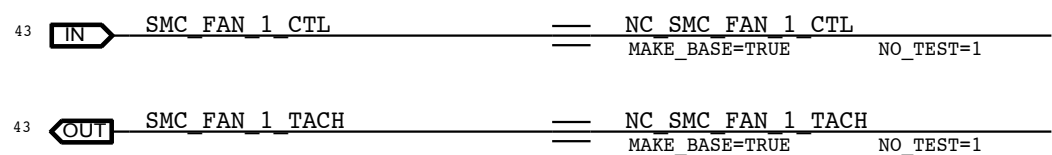
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

SMC Fan 1 (Unused)



SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
PAGE TITLE			
System Fan			
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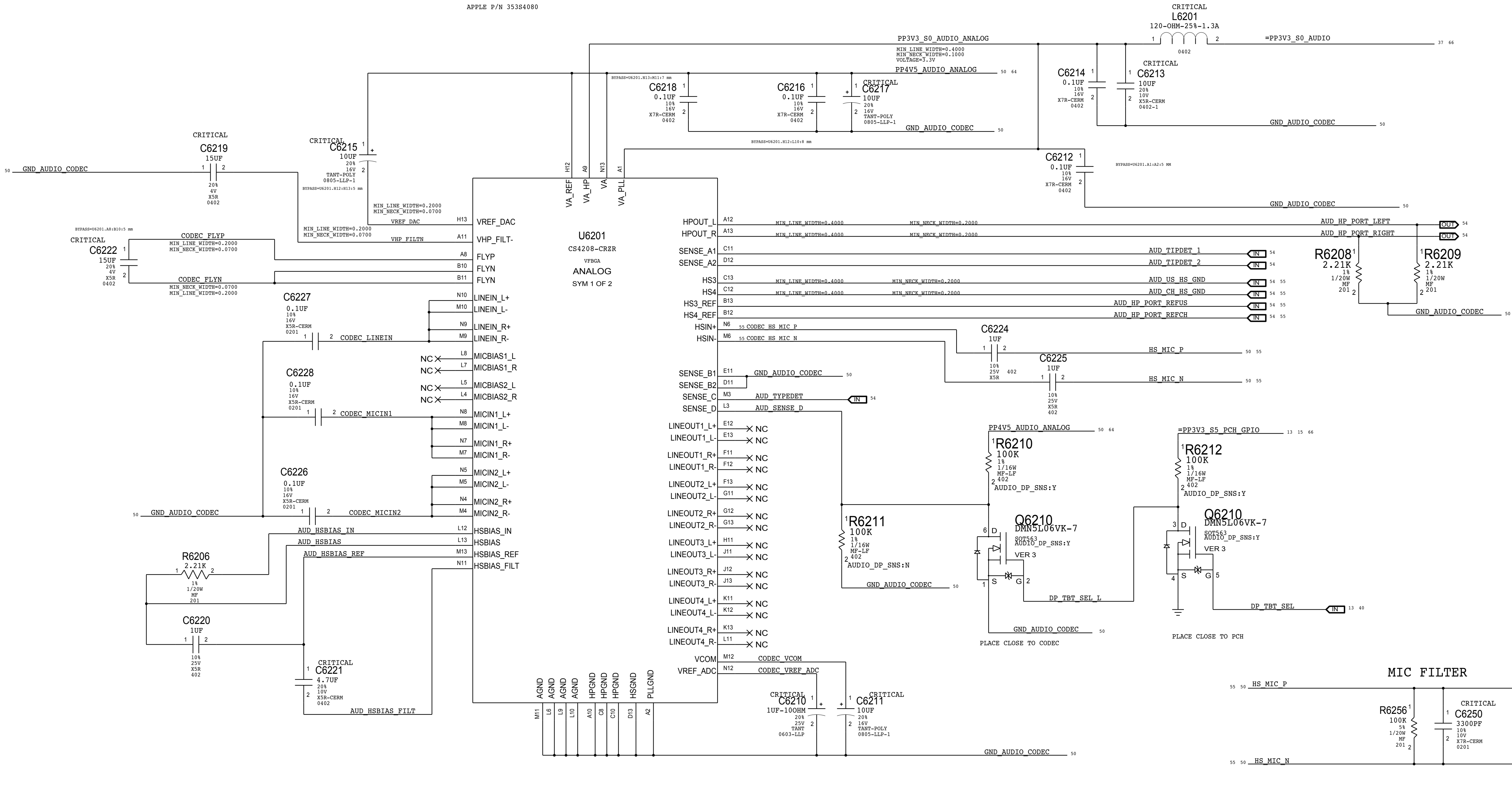
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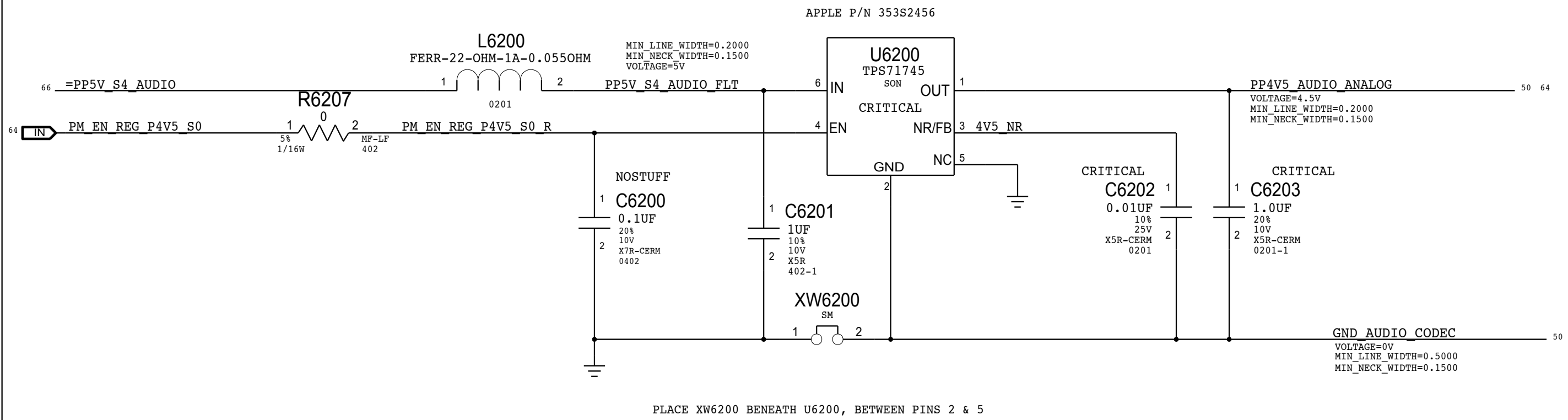
A

AUDIO CODEC, ANALOG BLOCKS

APPLE P/N 353S4080

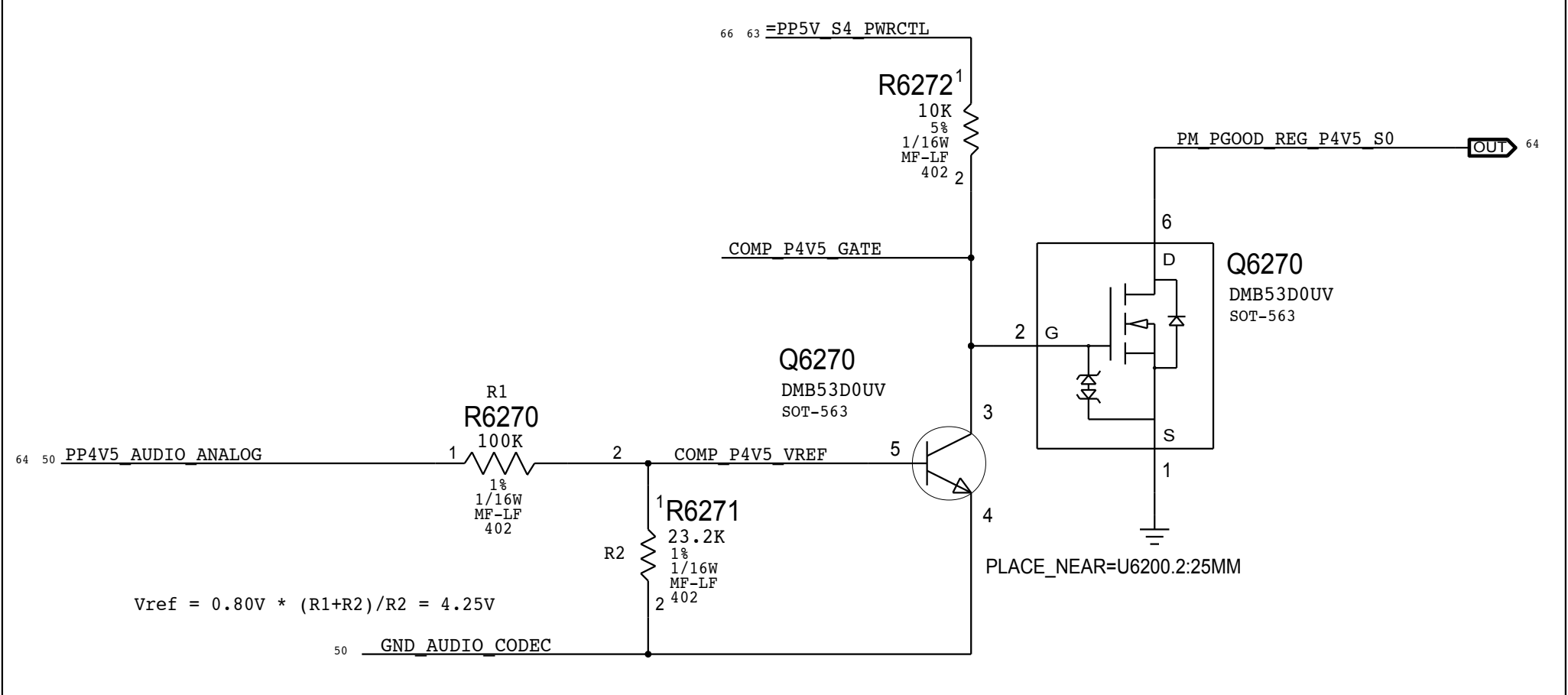


4.5V POWER SUPPLY FOR CODEC



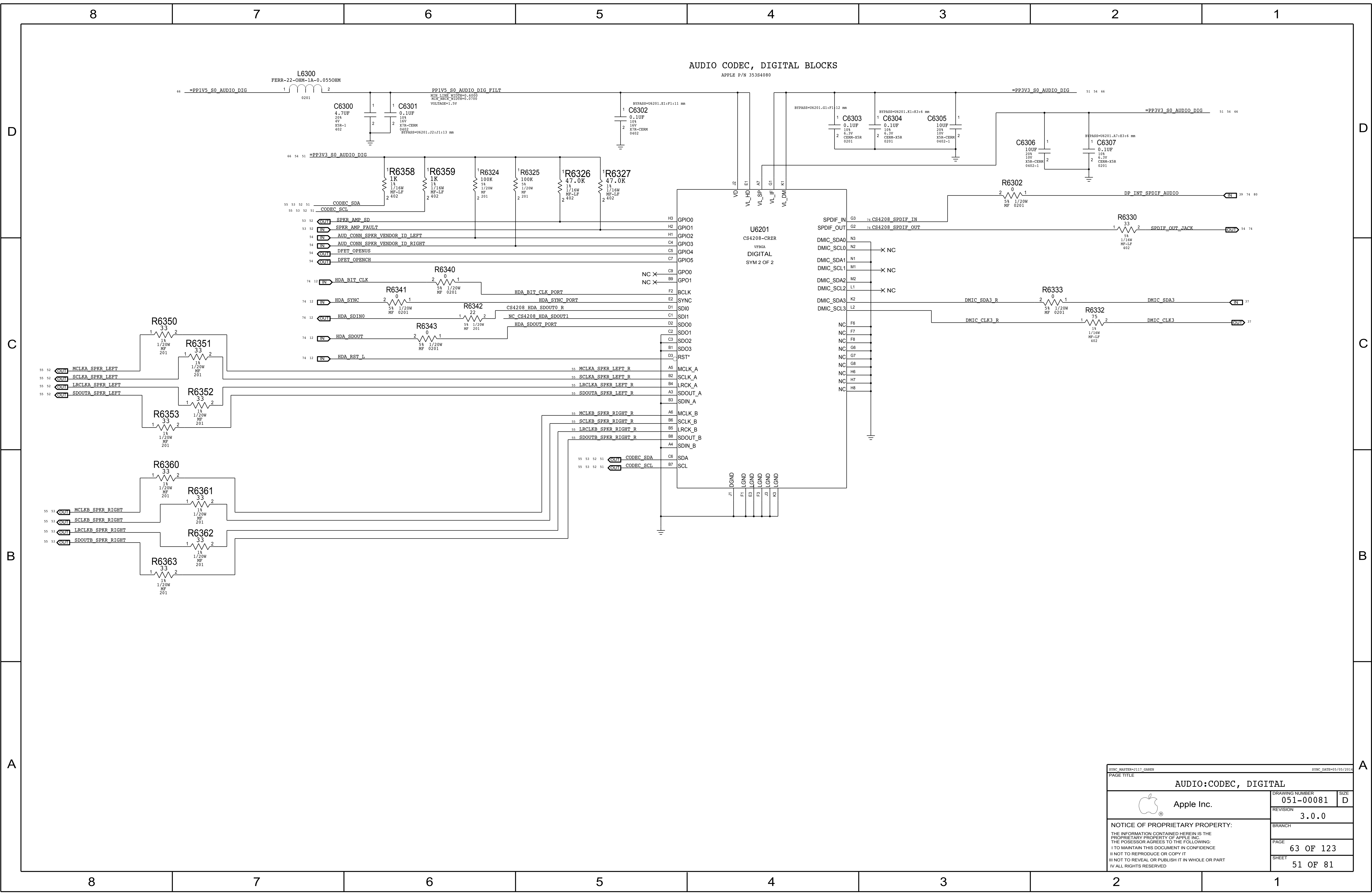
PLACE XW6200 BENEATH U6200, BETWEEN PINS 2 & 5

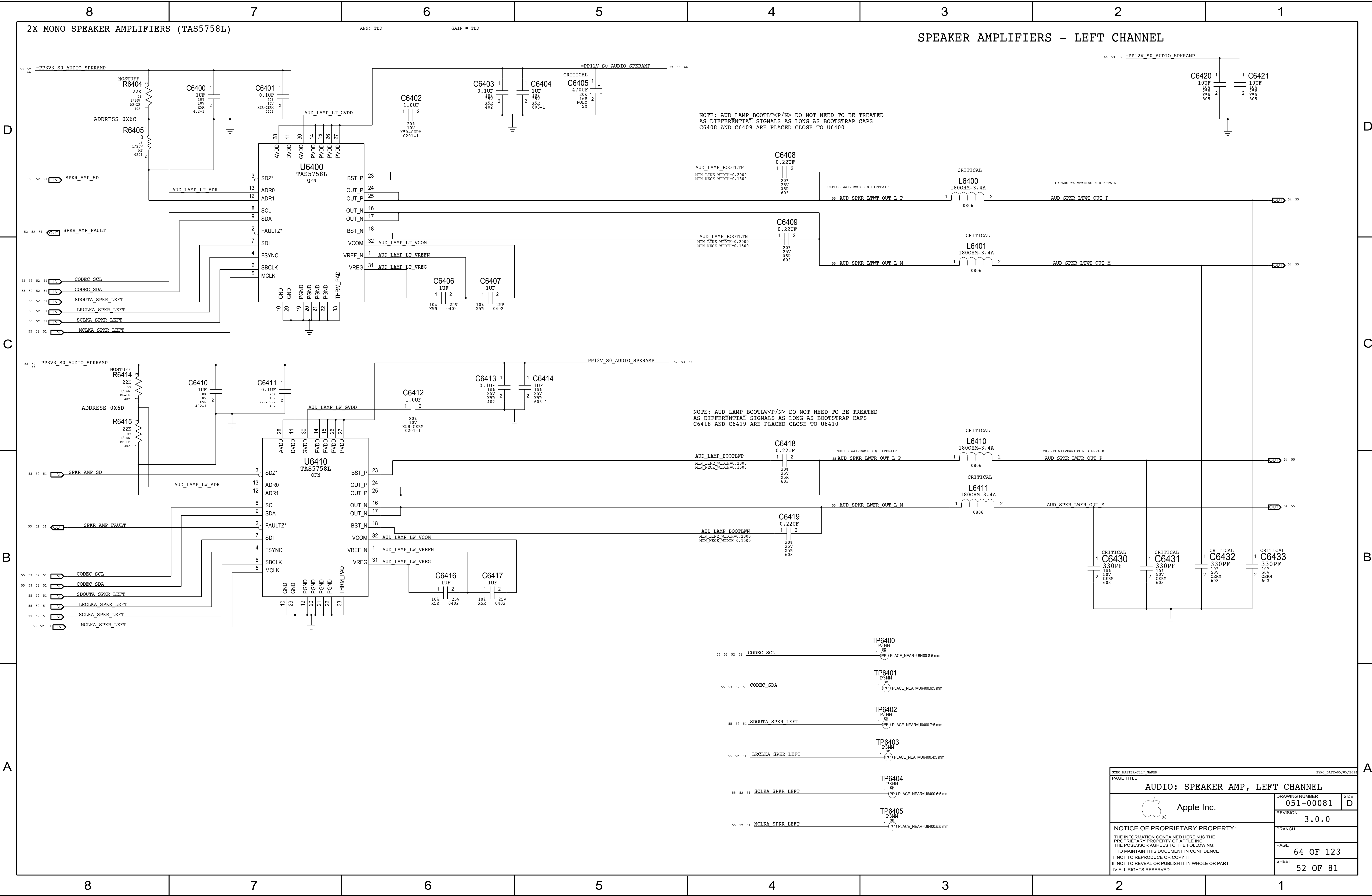
4.5V COMPARATOR

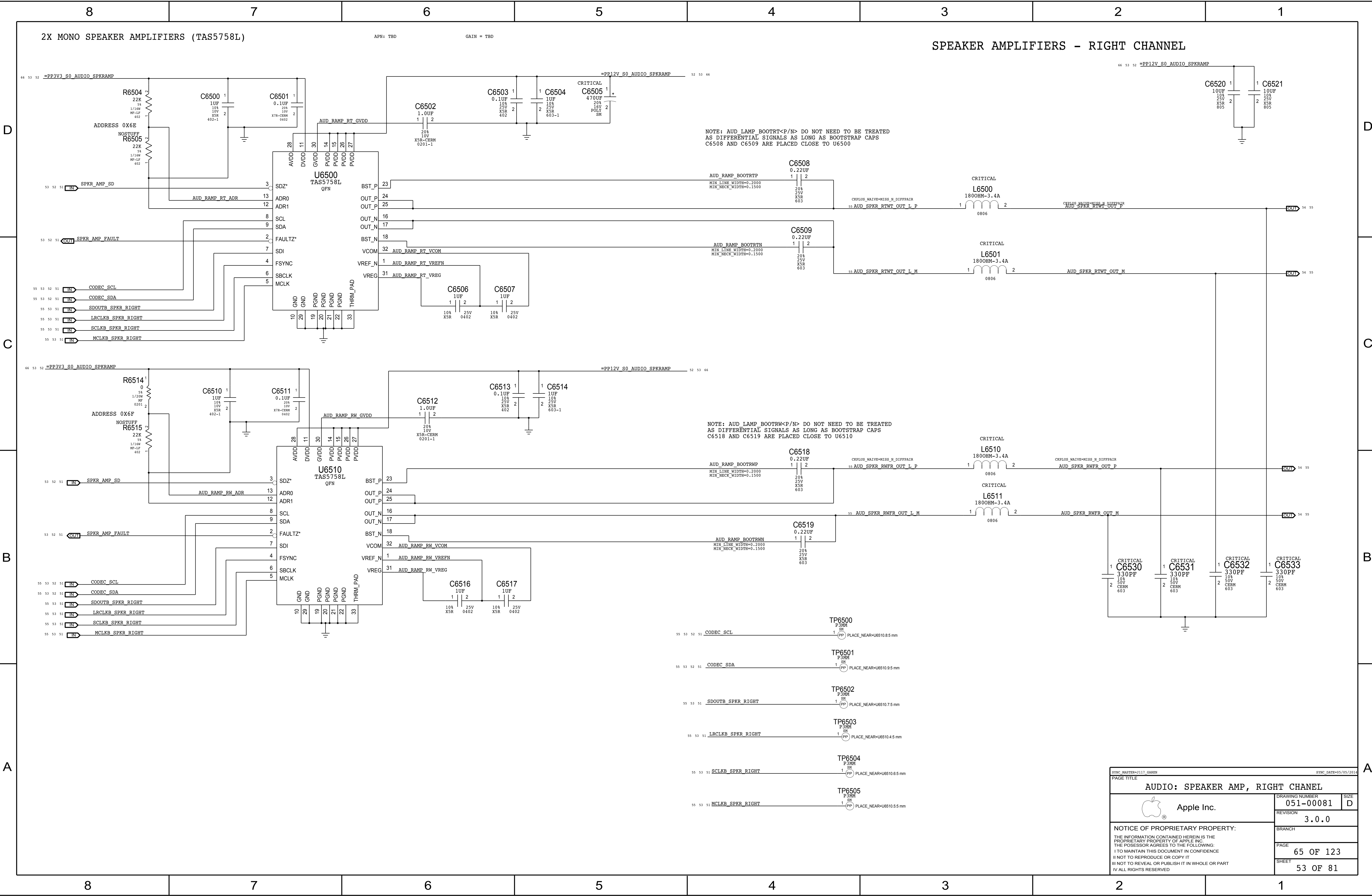


$$V_{ref} = 0.80V \cdot (R1+R2)/R2 = 4.25V$$

SYNC_MASTER=J117_GAREN		SYNC_DATE=05/05/2014	
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AUDIO: Codec (Analog)			
	Apple Inc.	DRAWING NUMBER	051-00081
		REVISION	3.0.0
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		PAGE	62 OF 123
		SHEET	50 OF 81





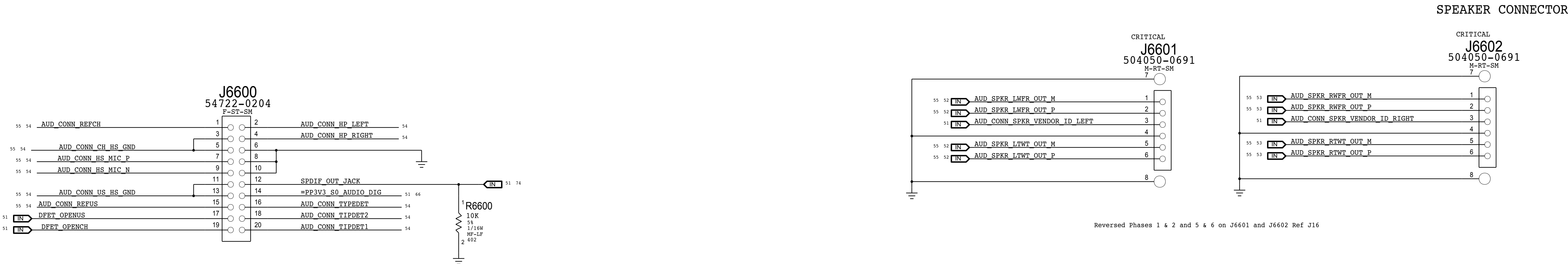


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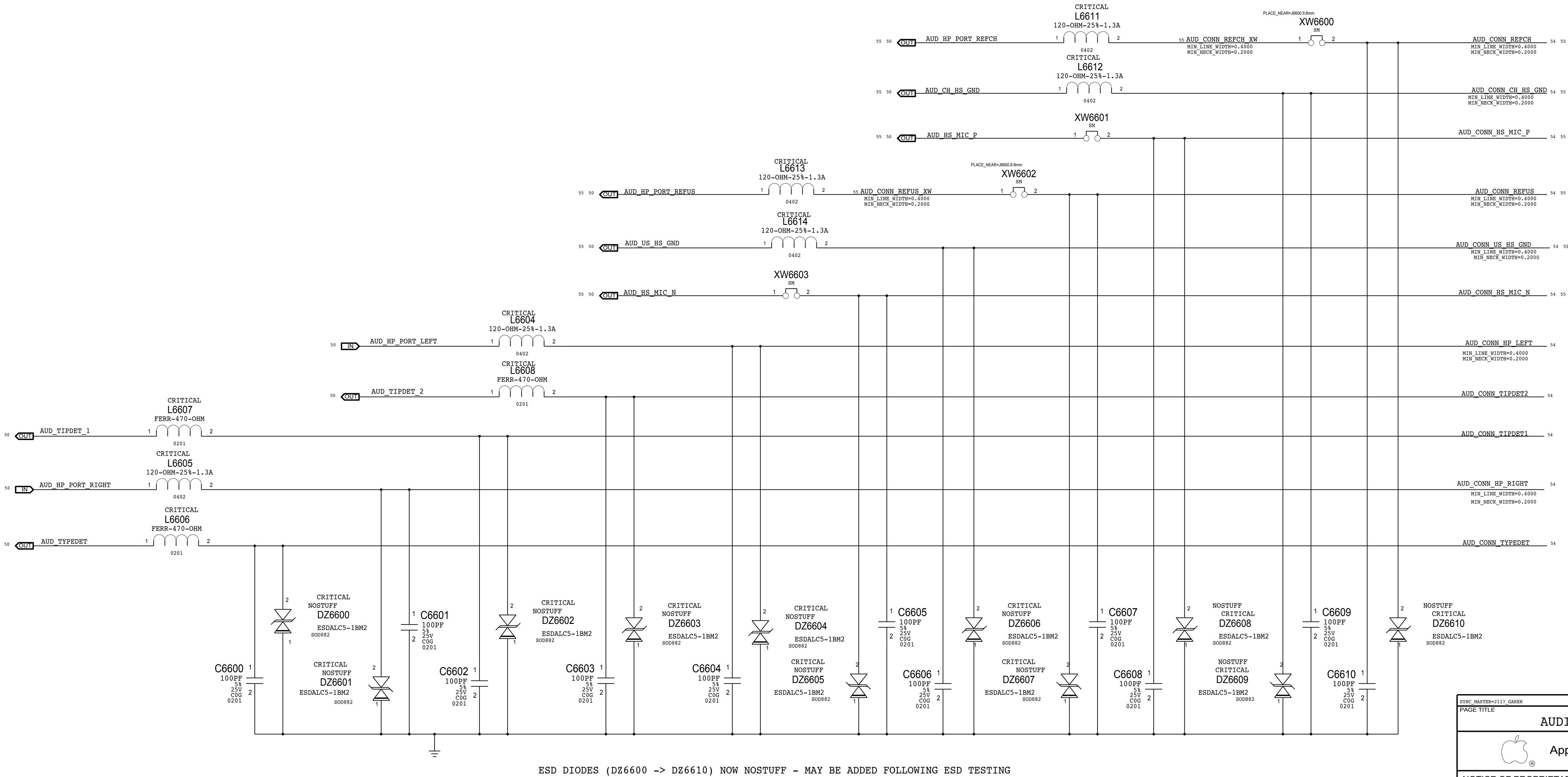


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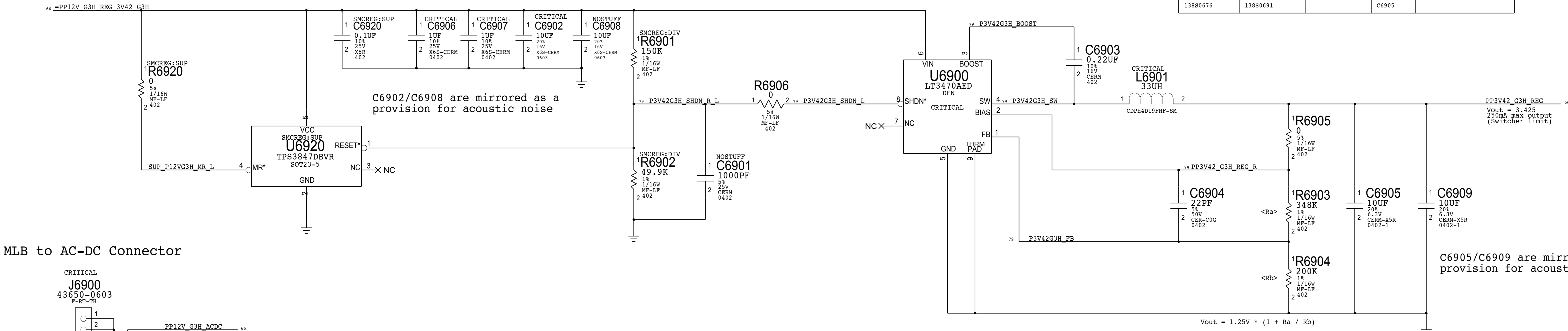
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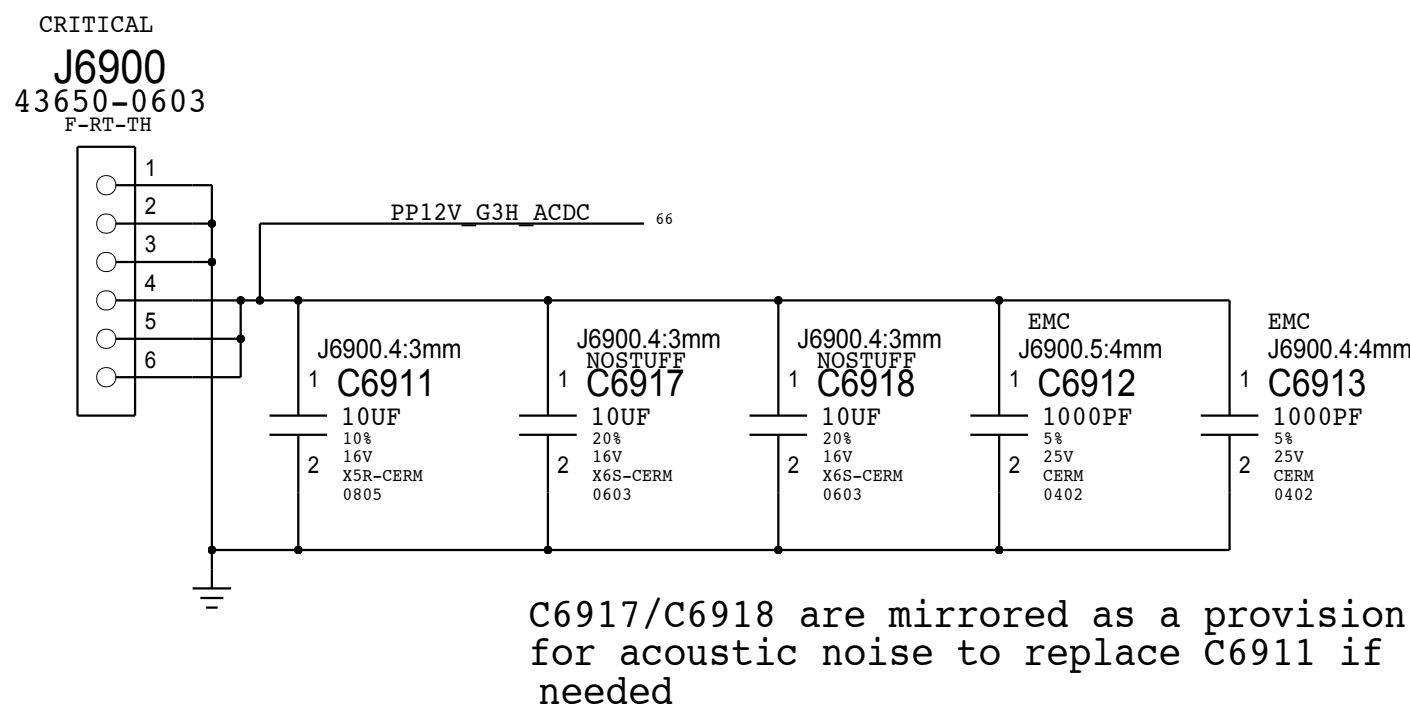
3.425V "G3Hot" Regulator

Switching freq: 409 kHz = $\frac{13.5}{L6901}$

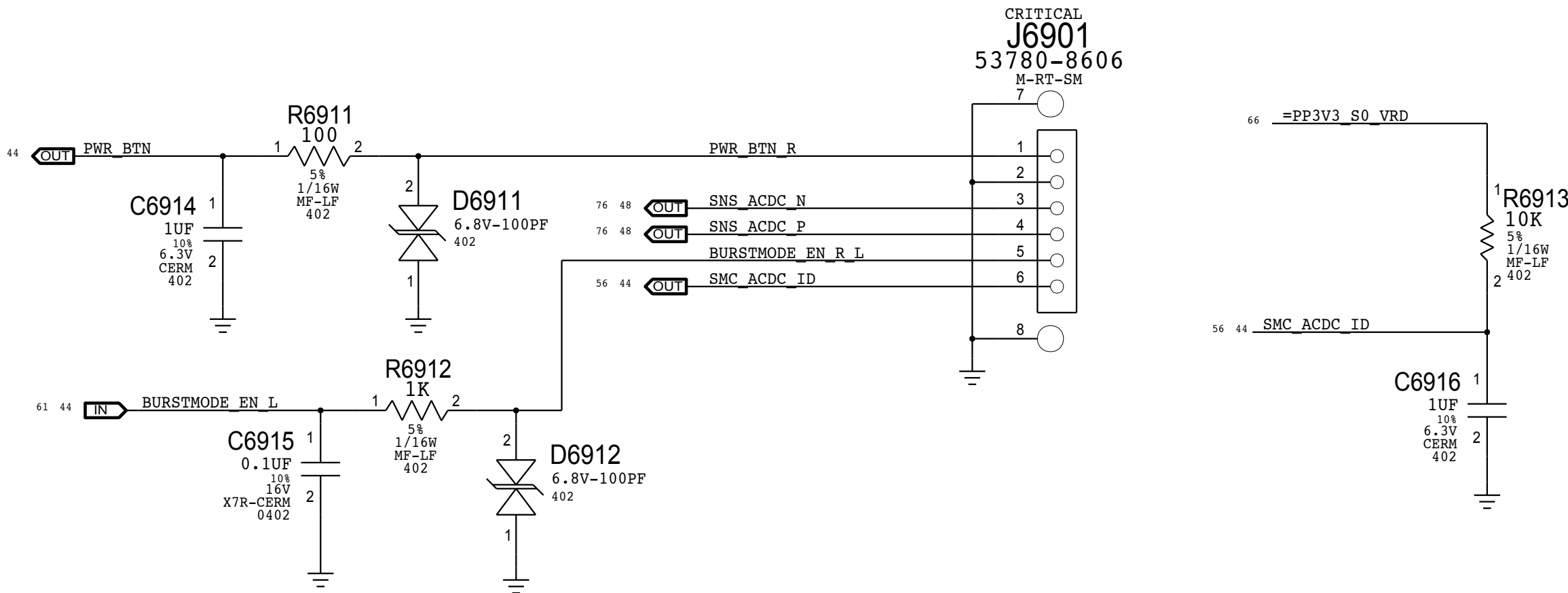
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0676	138S0691		C6905	



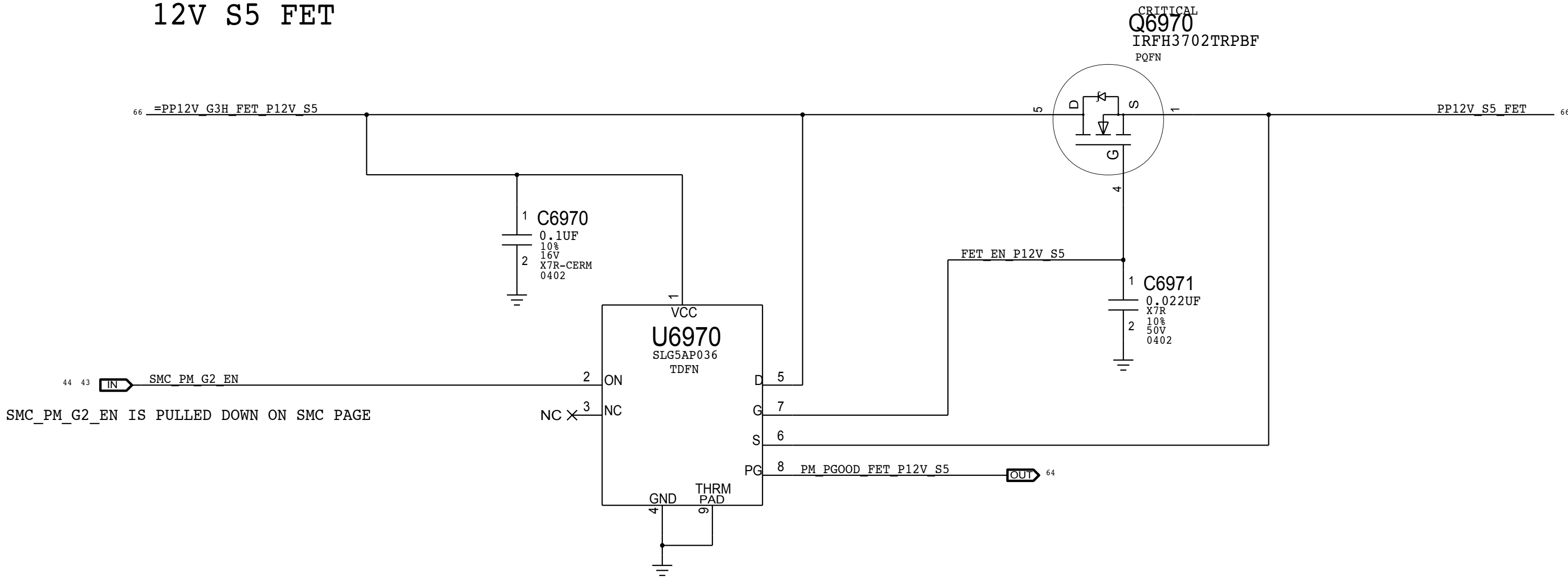
MLB to AC-DC Connector

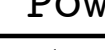


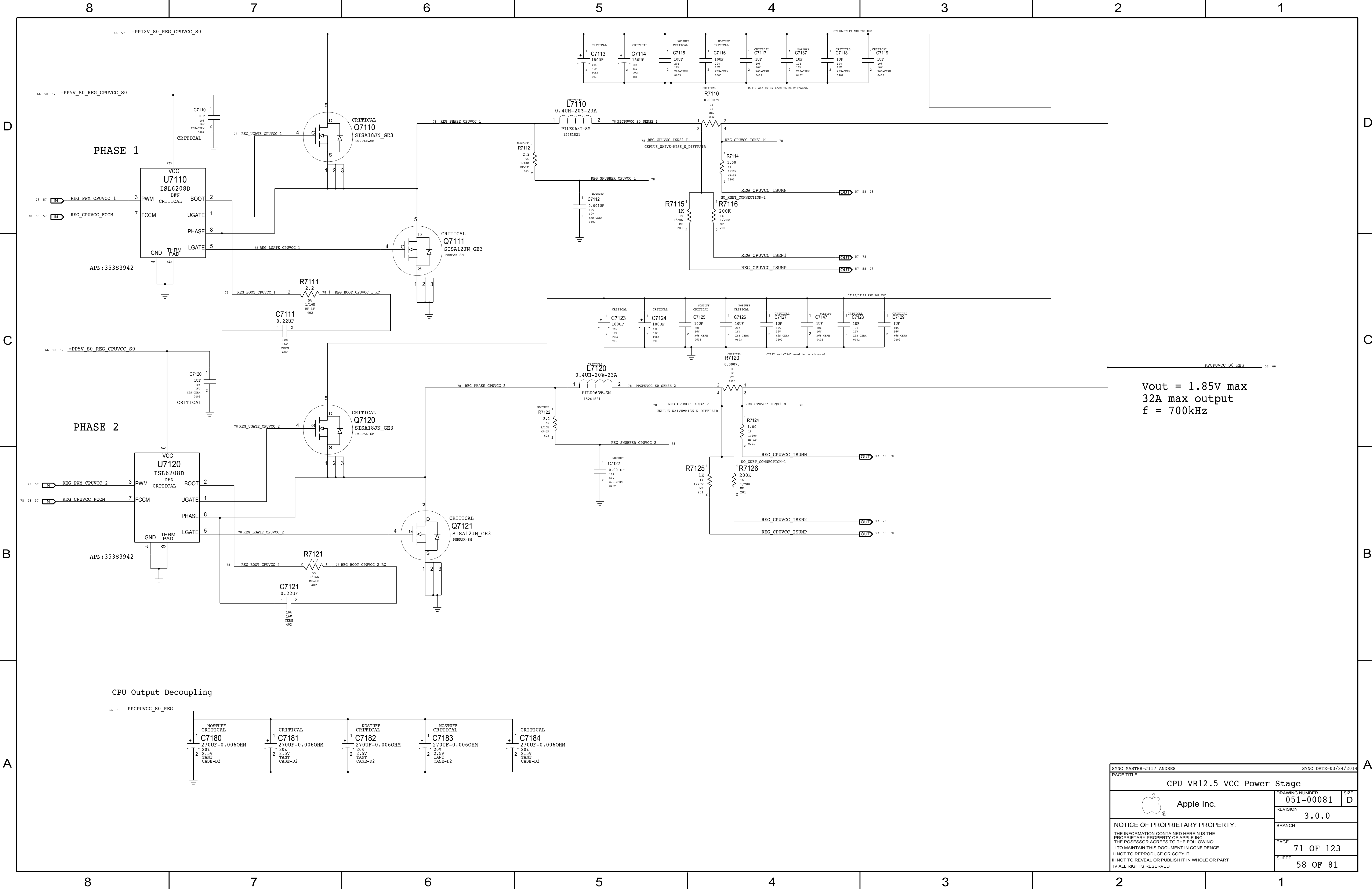
MLB to AC-DC Supplemental Signal Connector



12V S5 FET



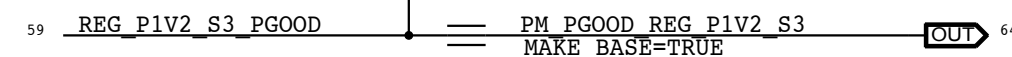
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PAGE TITLE			
Power Connectors / VReg G3Hot			
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Vout = 1.85V max
32A max output
f = 700kHz

SYNC_MASTER=J117_ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
CPU VR12.5 VCC Power Stage			
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Switching freq: 500 kHz


$$V_{out} = 0.8V * (1 + R_a / R_b)$$

SYNC MASTER=J70 ROSSANA		SYNC DATE=09/12/2013	
PAGE TITLE			
VReg VDDQ S3 / 1.8V S3			
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3.3V S5 Regulator

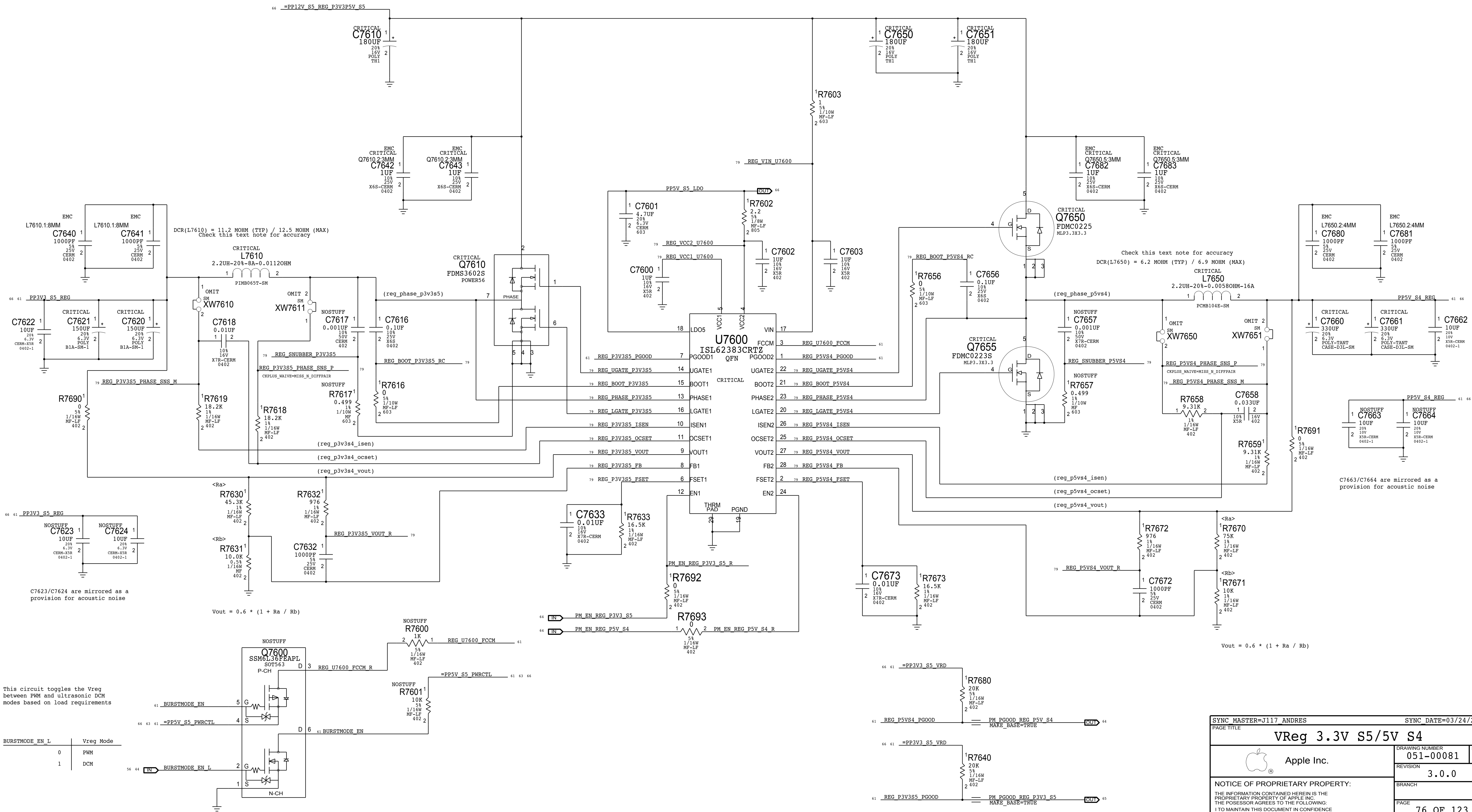
OC trip point: 12.5 A = $\frac{R7618 * 10 \text{ E-6}}{\text{DCR}(\text{L7610})}$

Switching freq: 356 kHz = $\frac{1}{170 \text{ E-12} * R7633}$

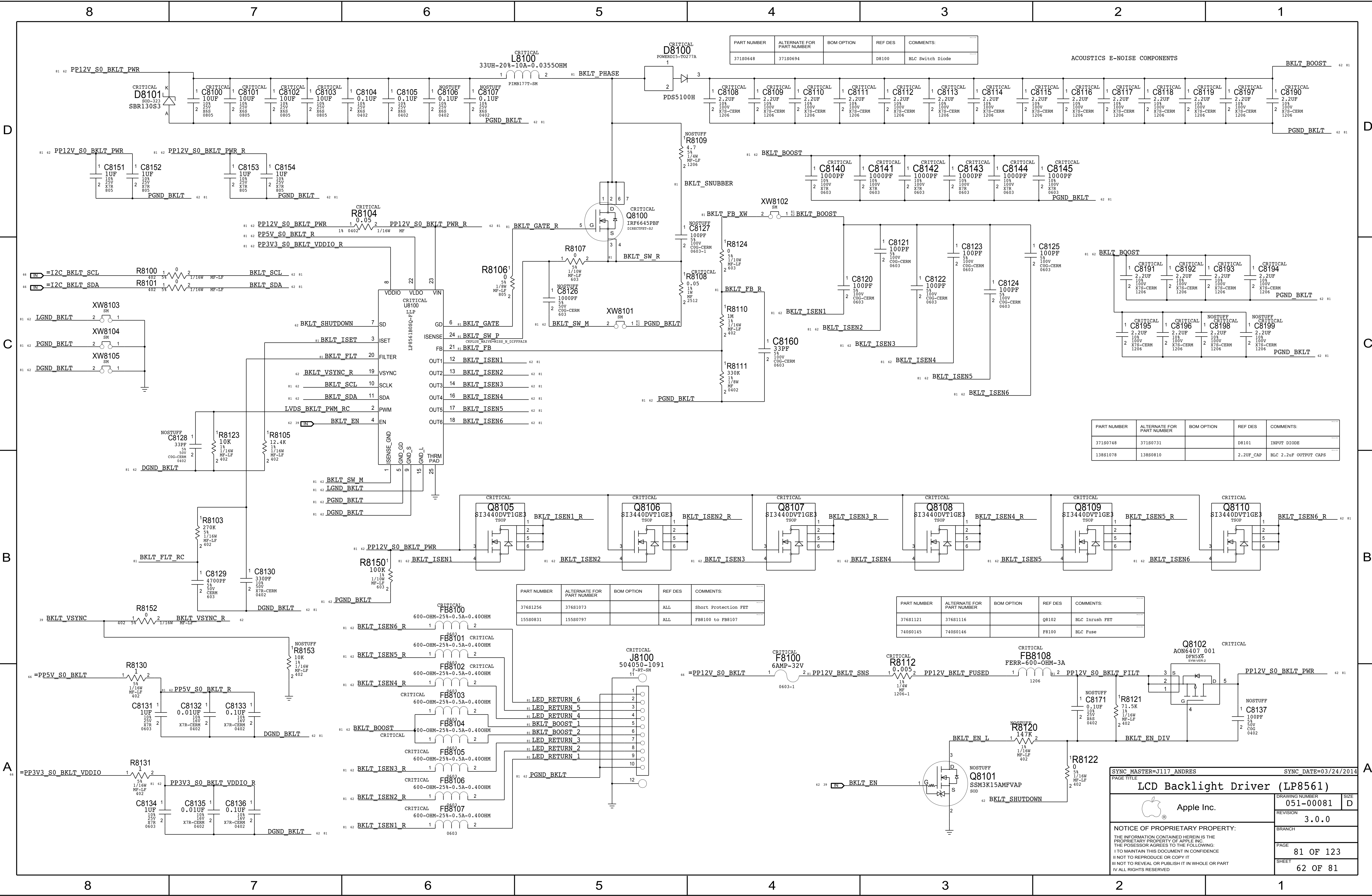
5V S4 Regulator

OC trip point: 14.1 A = $\frac{R7658 * 10 \text{ E-6}}{\text{DCR}(\text{L7650})}$

Switching freq: 356 kHz = $\frac{1}{170 \text{ E-12} * R7673}$



SYNC_MASTER=J117 ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
VReg 3.3V S5/5V S4			
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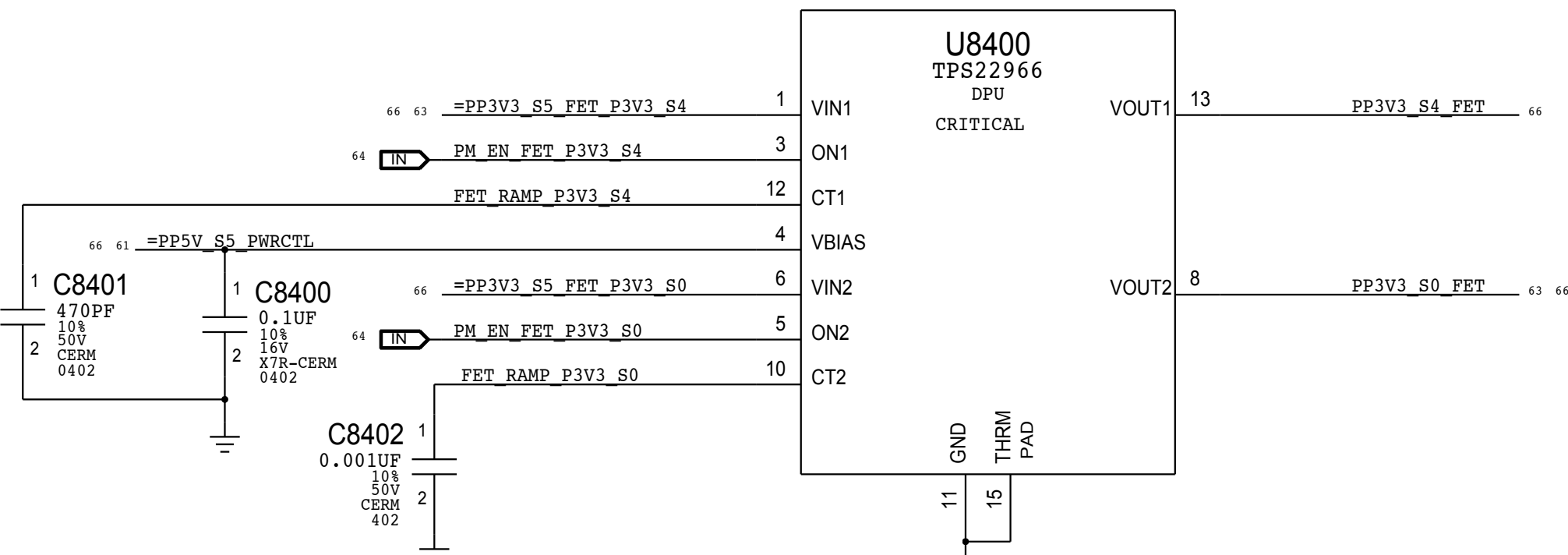
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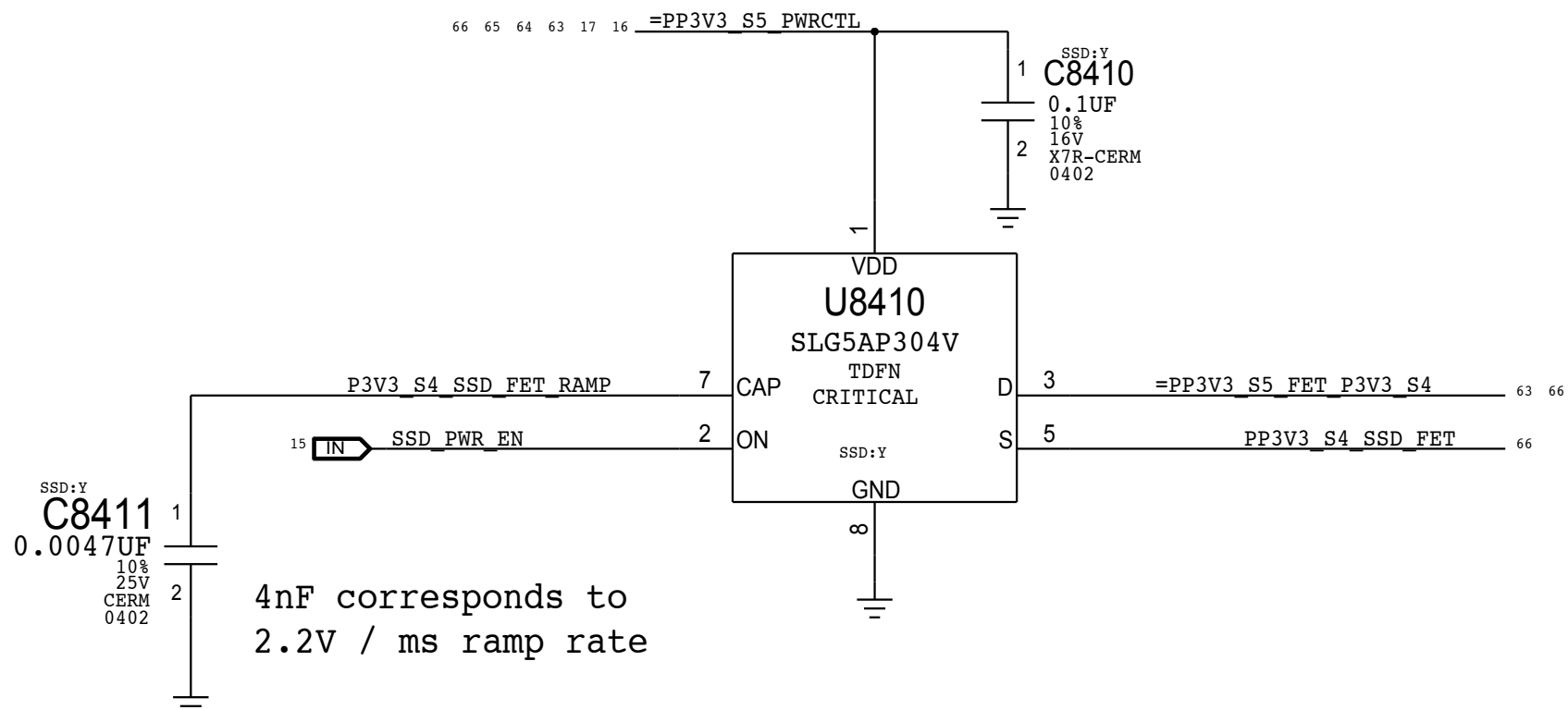
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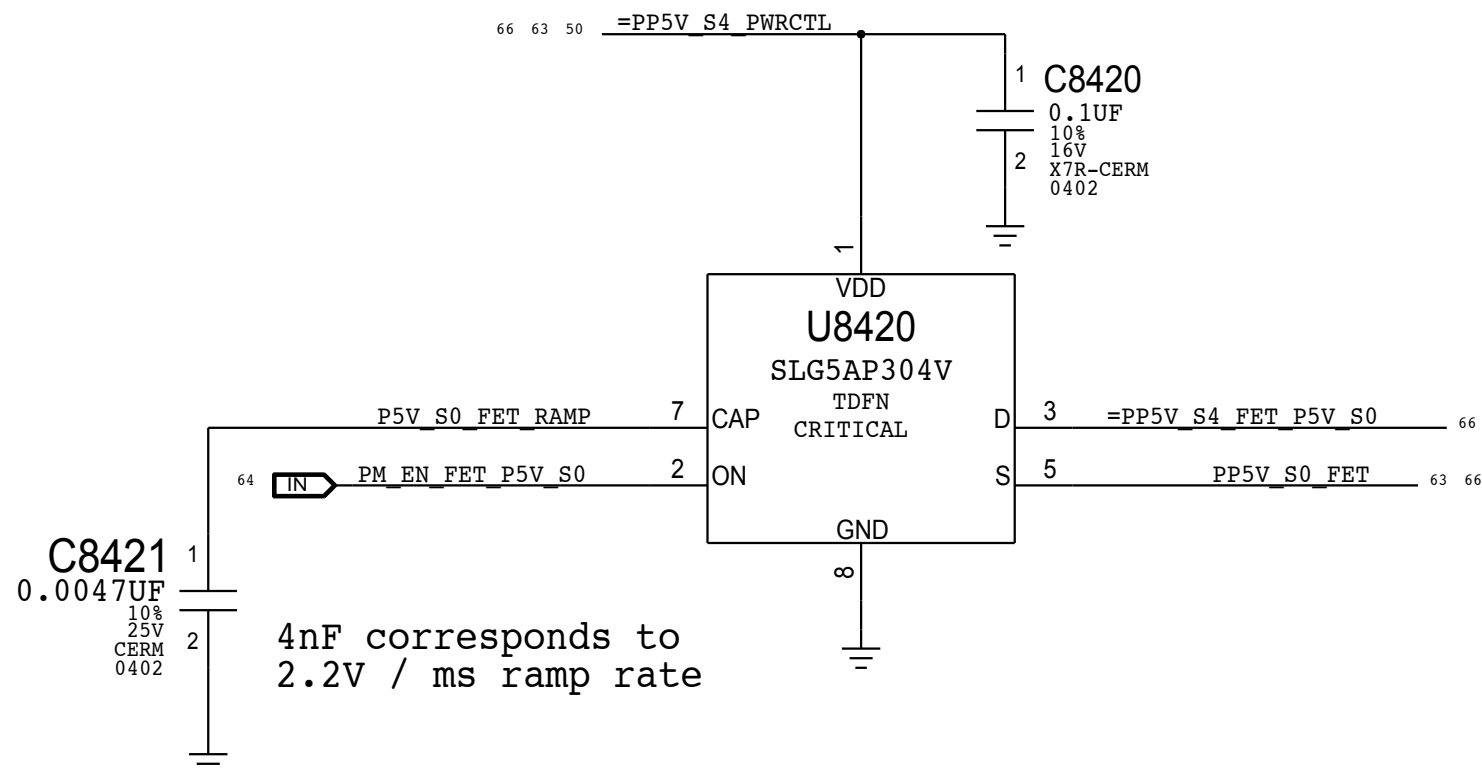
3.3V S4/S0 FET



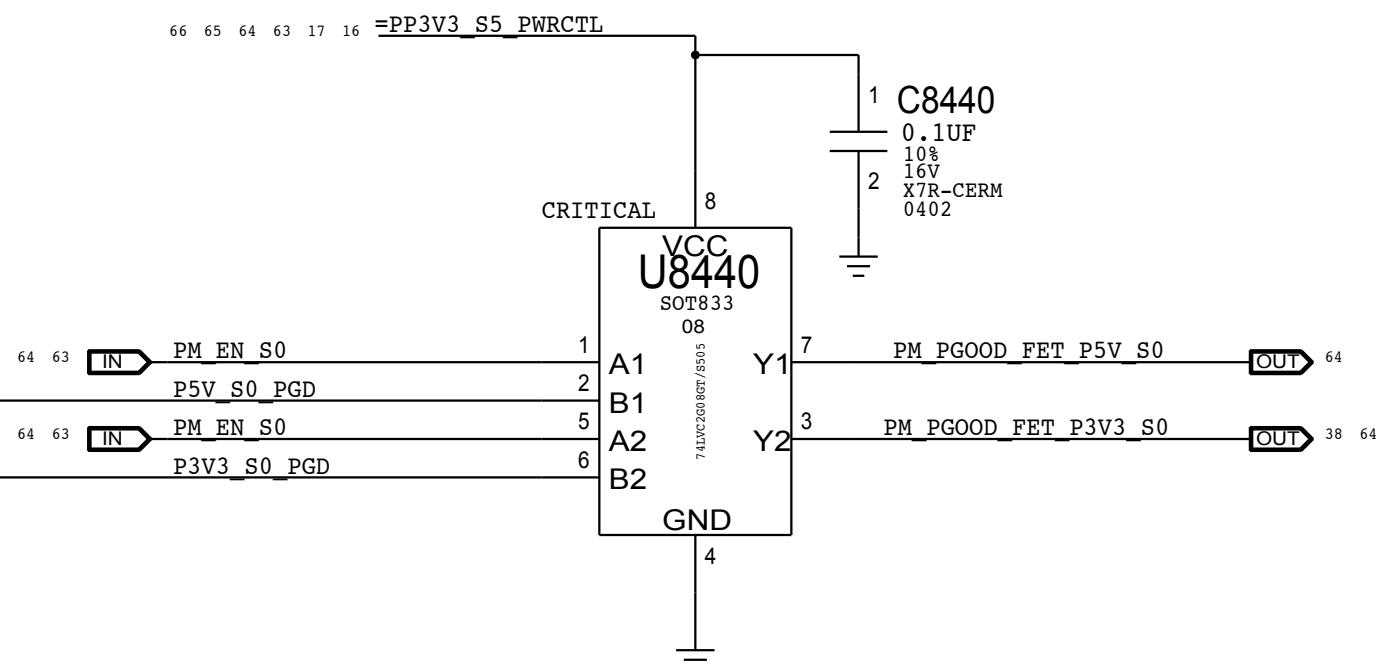
3V3 S4 SSD



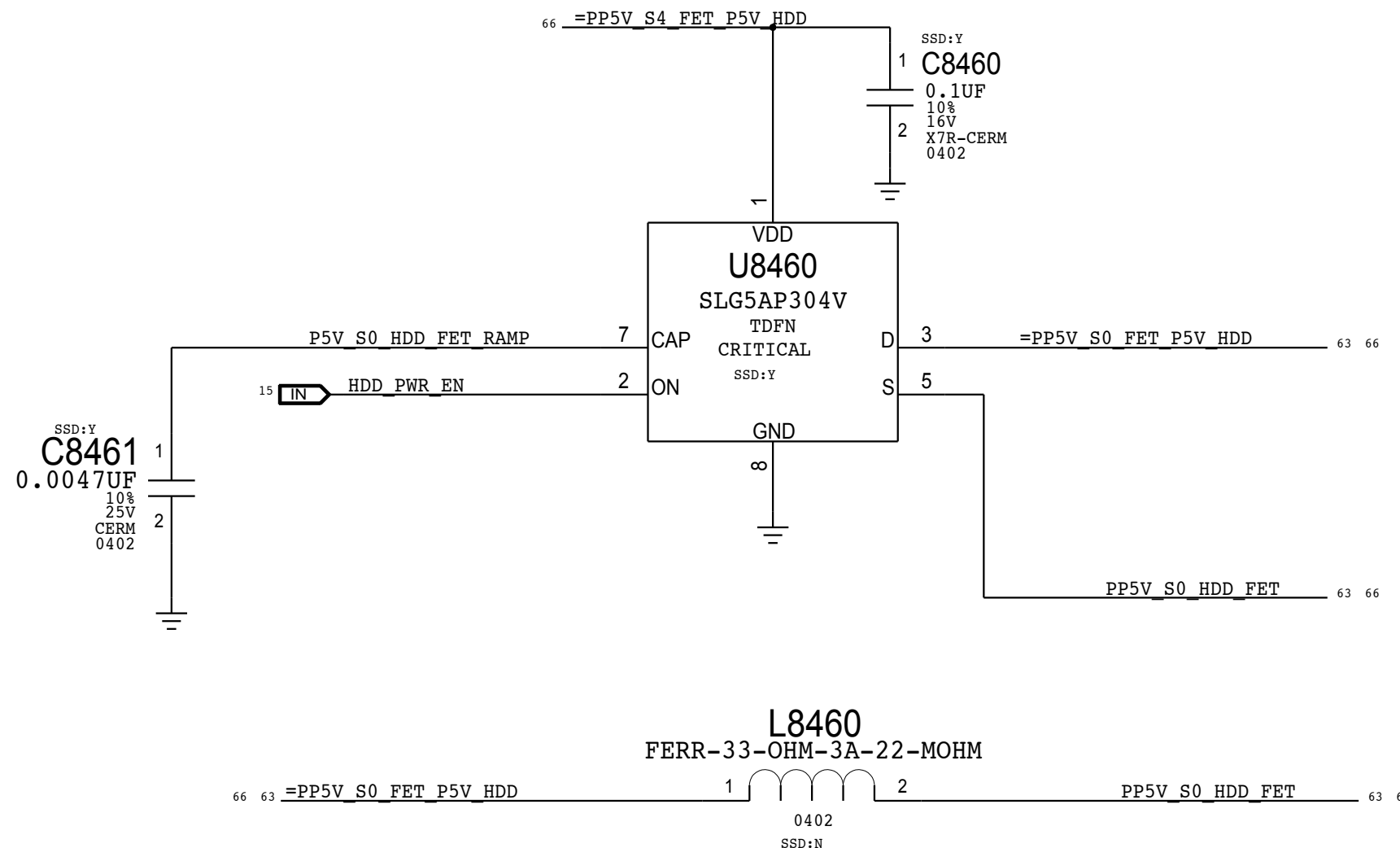
5V S0 FET



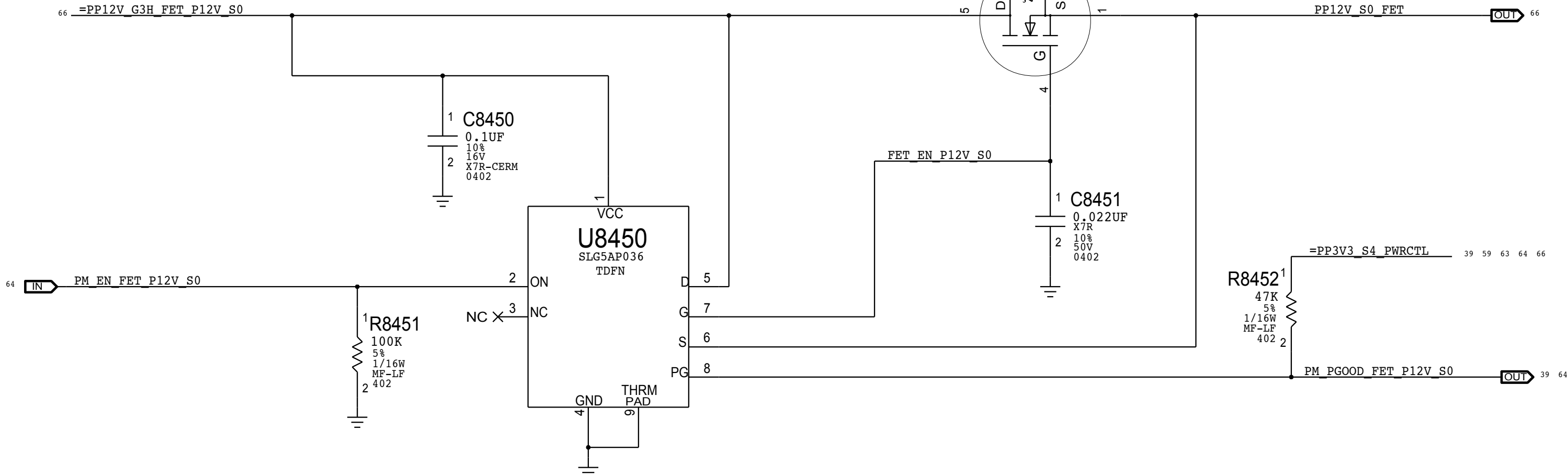
5V / 3V3 S0 PGOODs



5V HDD FET

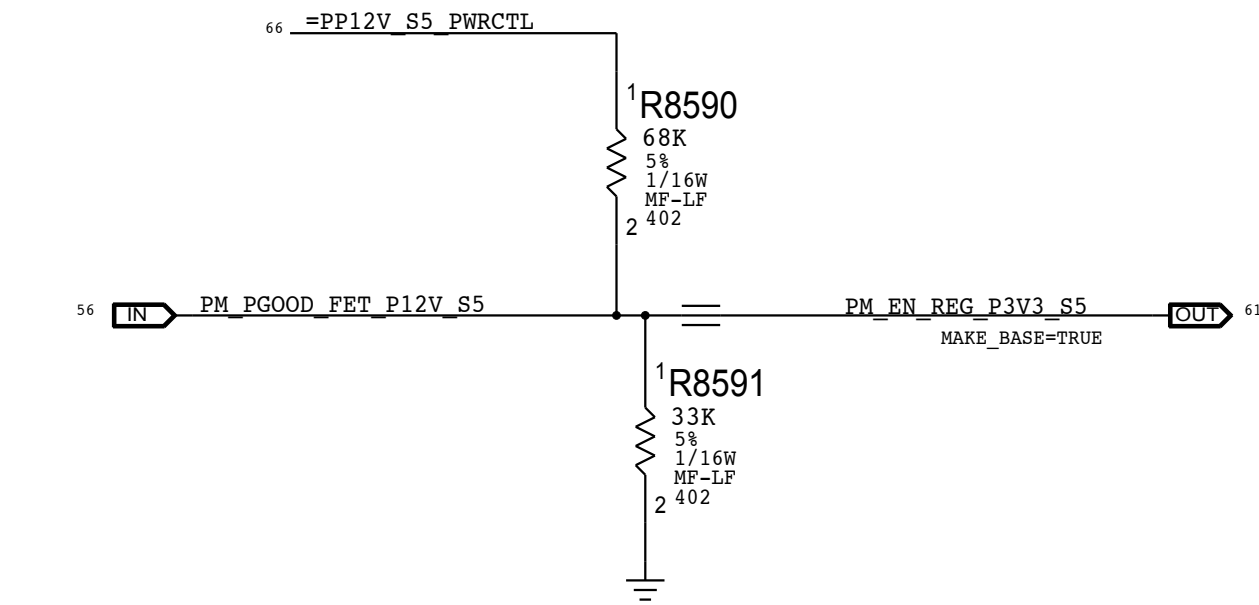


12V S0 FET

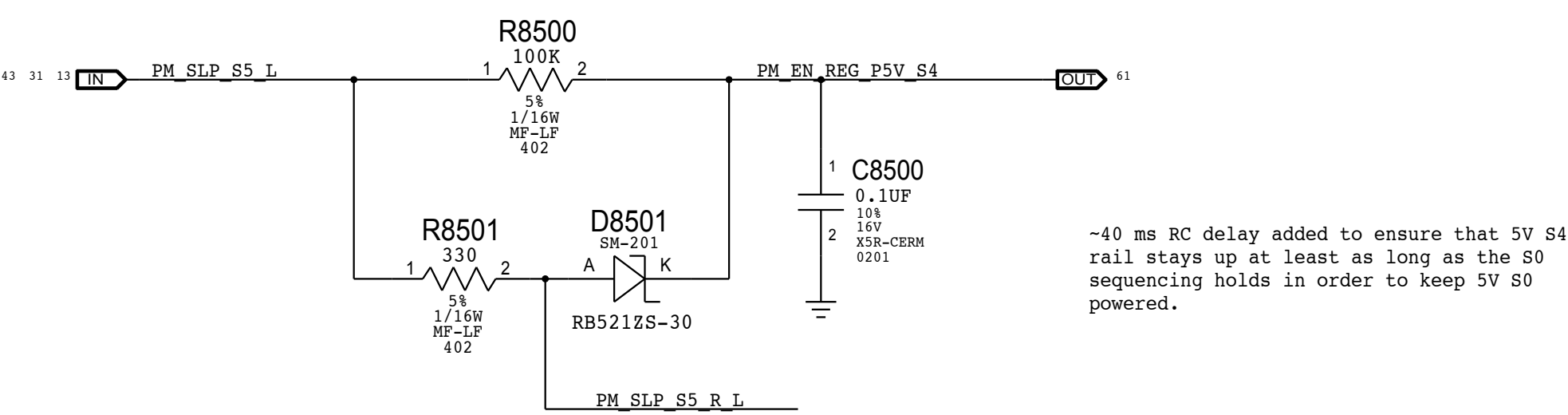


SYNC_MASTER=J16_MLB_IG		SYNC_DATE=08/27/2013	
PAGE TITLE			
PM FETs/LDOs			
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		REVISION	3.0.0
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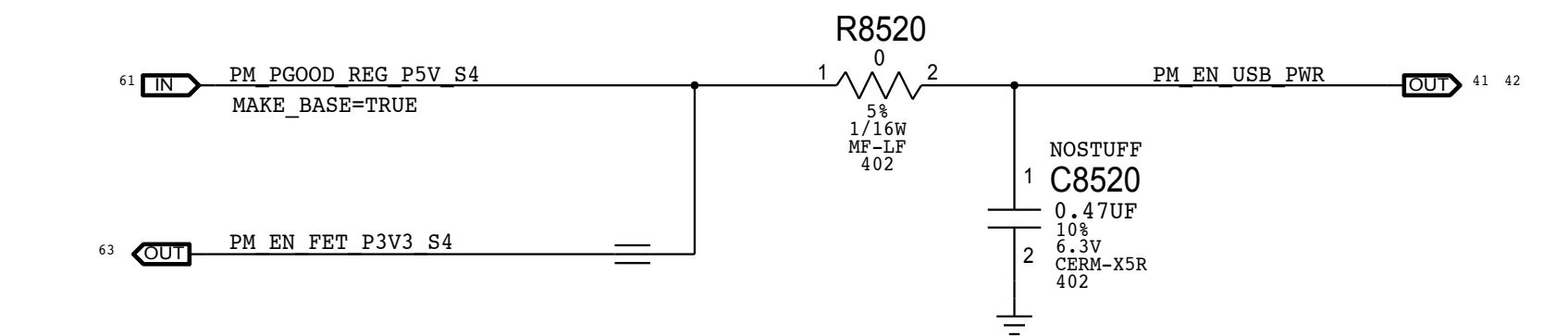
S5 Enable



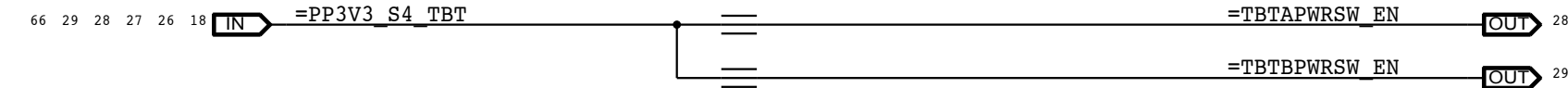
S4 Enables



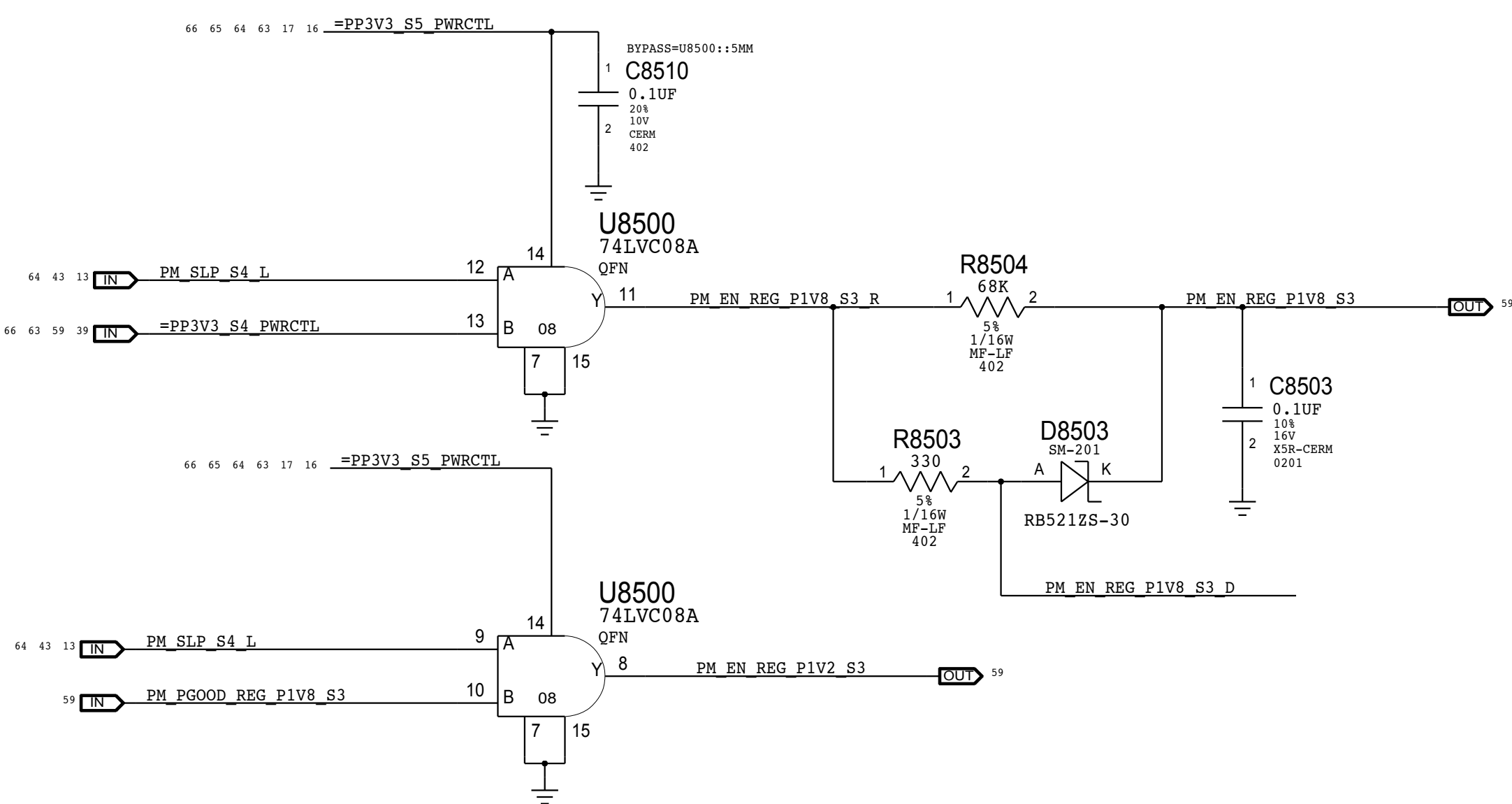
S4 USB Enable



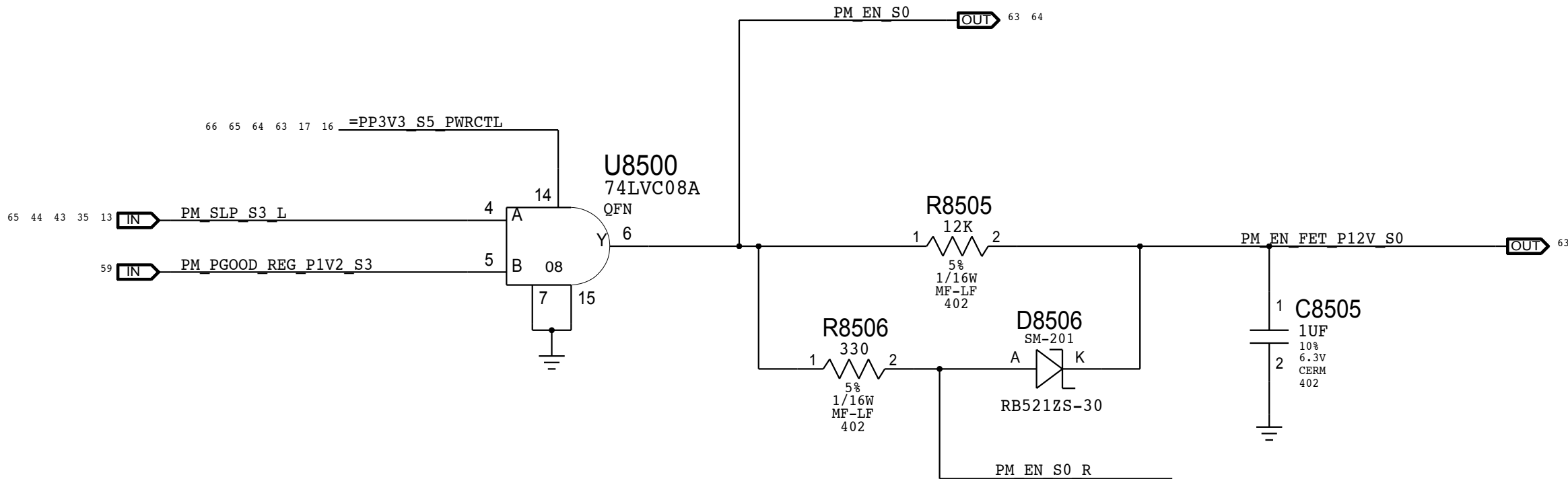
S4 TBT Port Enable



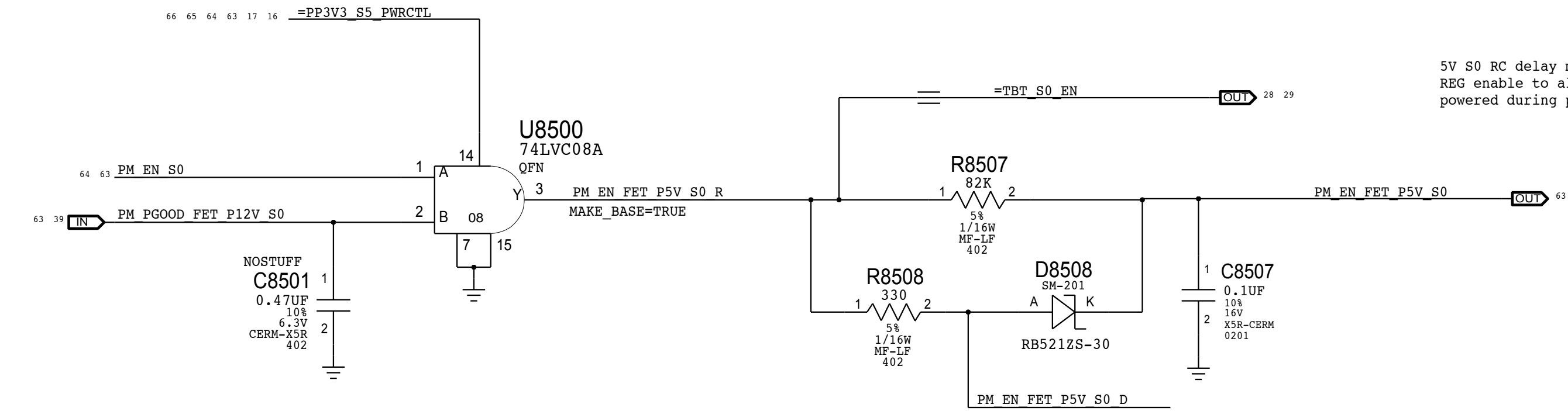
S3 Enables



S0 Enables

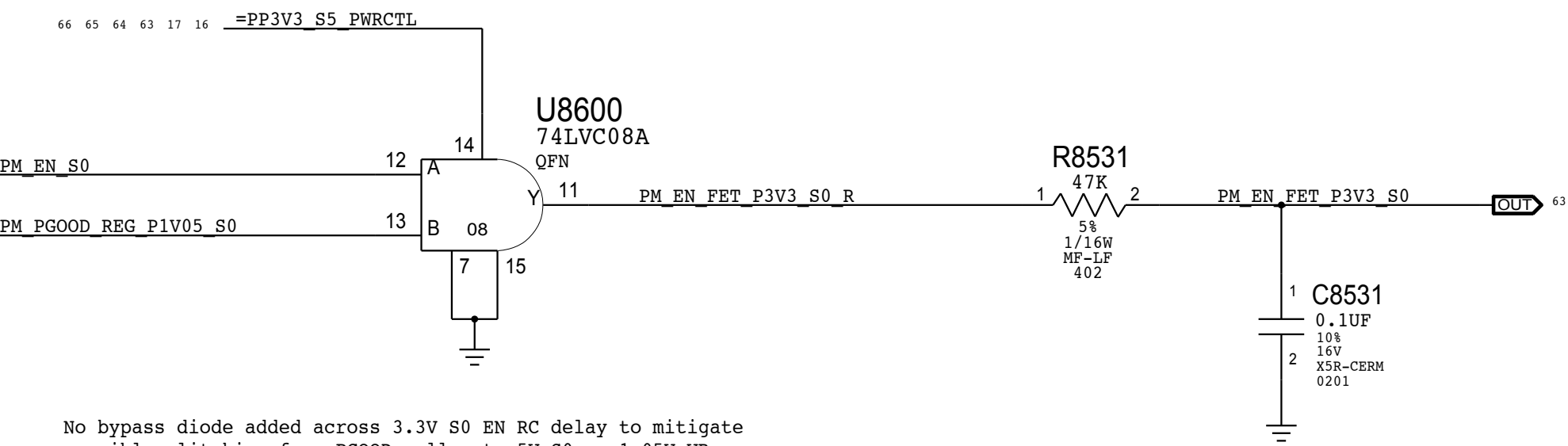
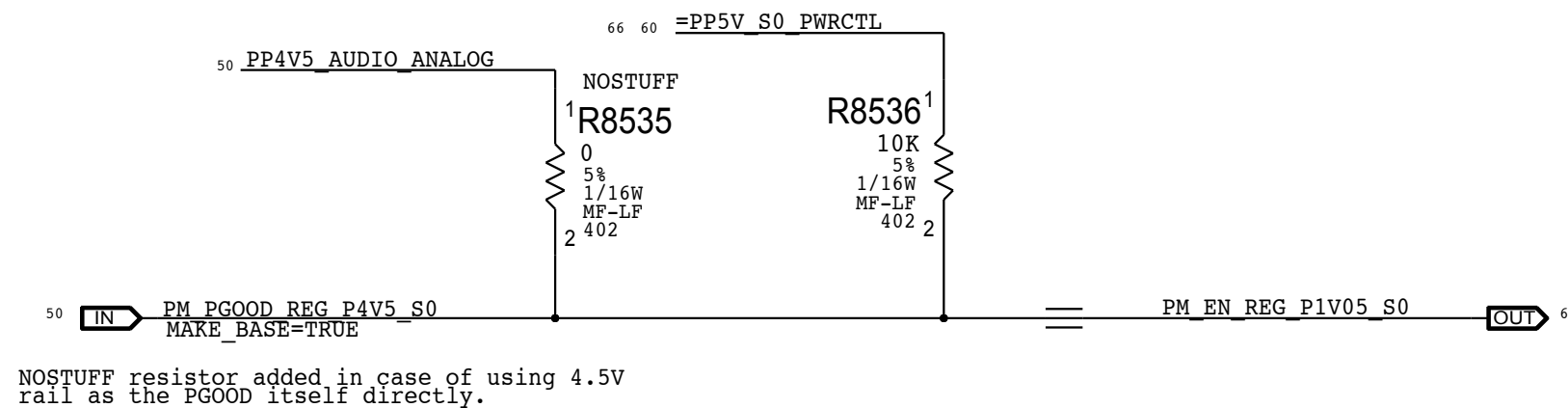
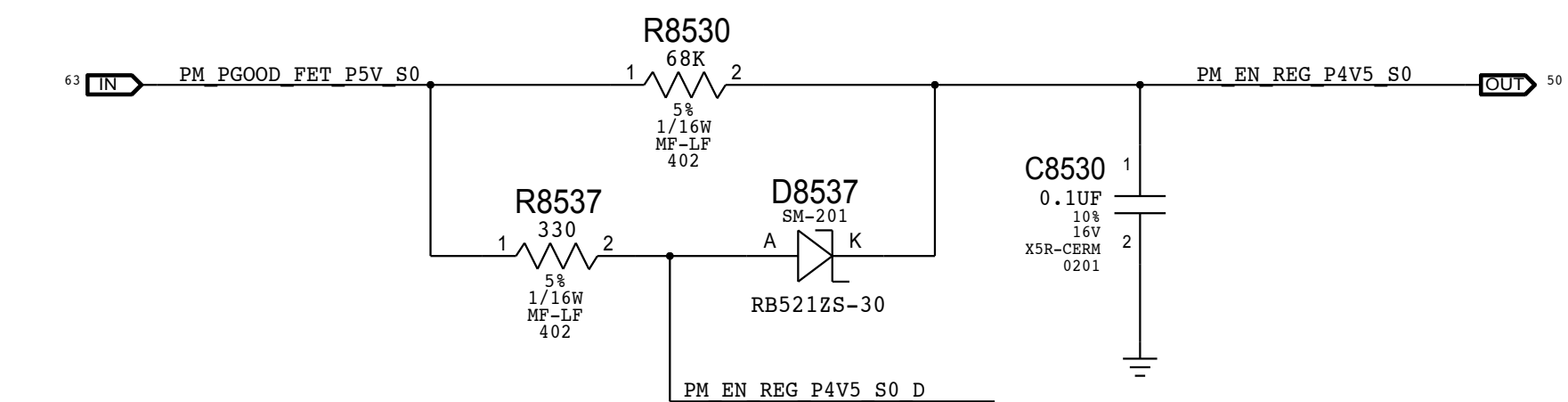


12V S0 EN RC delay must be >= downstream delay on 4.5V REG enable which in turn enables 1.05V S0. This allows for 12V S0 to hold as long as 1.05V S0 regulator is powered.

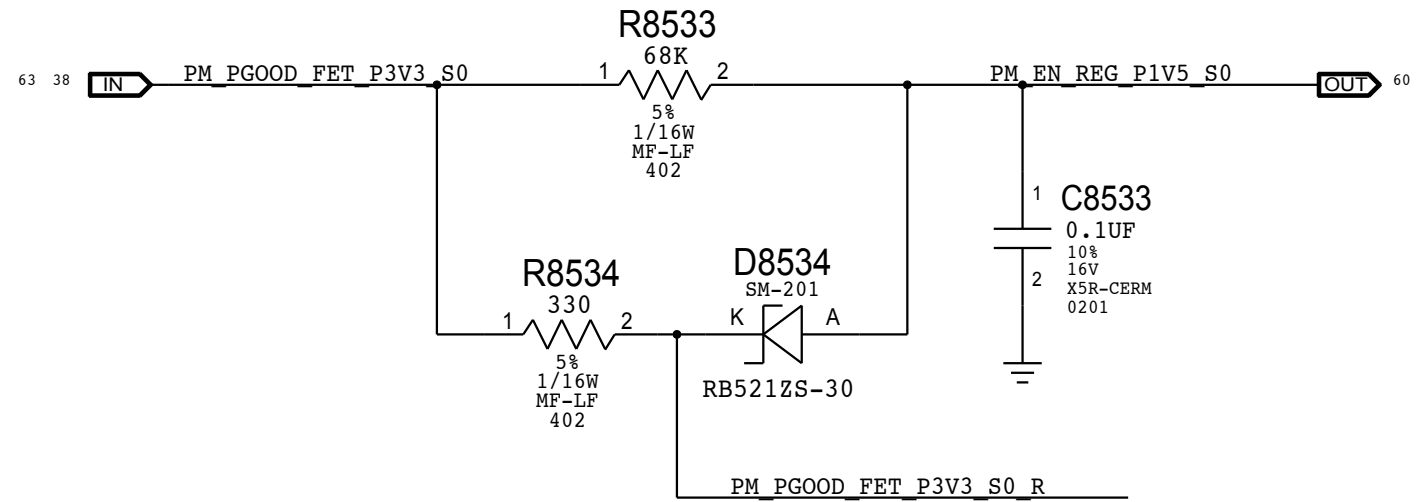


5V S0 RC delay must be >= downstream delay on 4.5V REG enable to allow for 4.5V regulator to remain powered during power down sequence.

Audio + PCH Sequencing Requirements:
4.5V -> 1.05V -> 3.3V -> 1.5V -> ALL SYS GOOD

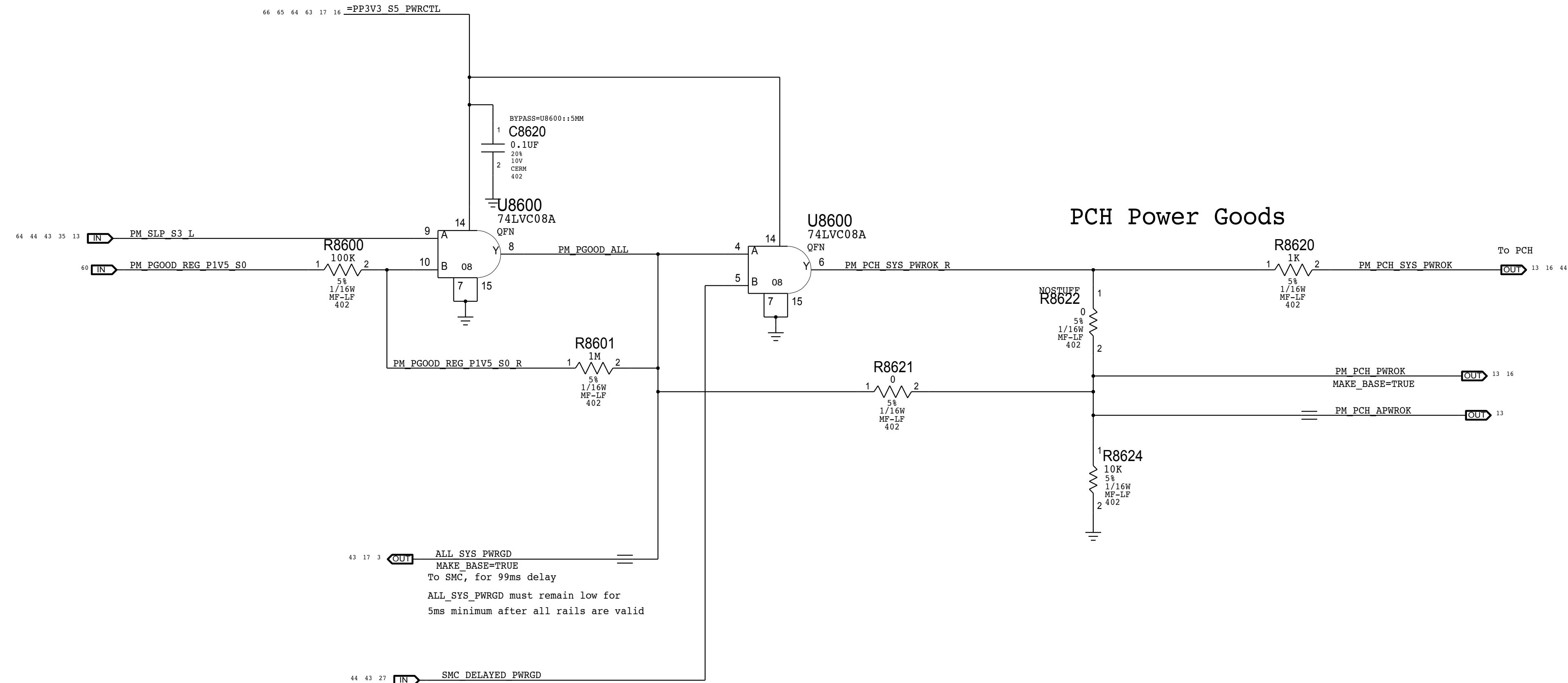


No bypass diode added across 3.3V S0 EN RC delay to mitigate possible glitching from PG00D pullup to 5V S0 on 1.05V VR page competing with logic turn on time.



SYNC_MASTER=J117_ANDRES		SYNC_DATE=03/24/2014	
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PM Regulator Enables			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00081	D
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ALL_SYS_PWRGD,PCH_PWROK & SYS_PWROK Generation



Rail definitions

Platform: All processor non-Core and non-Graphics (5V, 3.3V, 1.5V, 1.05V for PCH/TBT/GPU)
Uncore: 1.8V and 1.2V for DDR3

Notes on sequencing requirements

- Intel:
1. No hard specification on platform rails
 2. SMC guarantees timing on PCH DPWROK and PWROK
 3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
 4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
 5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
 6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

Resume Reset

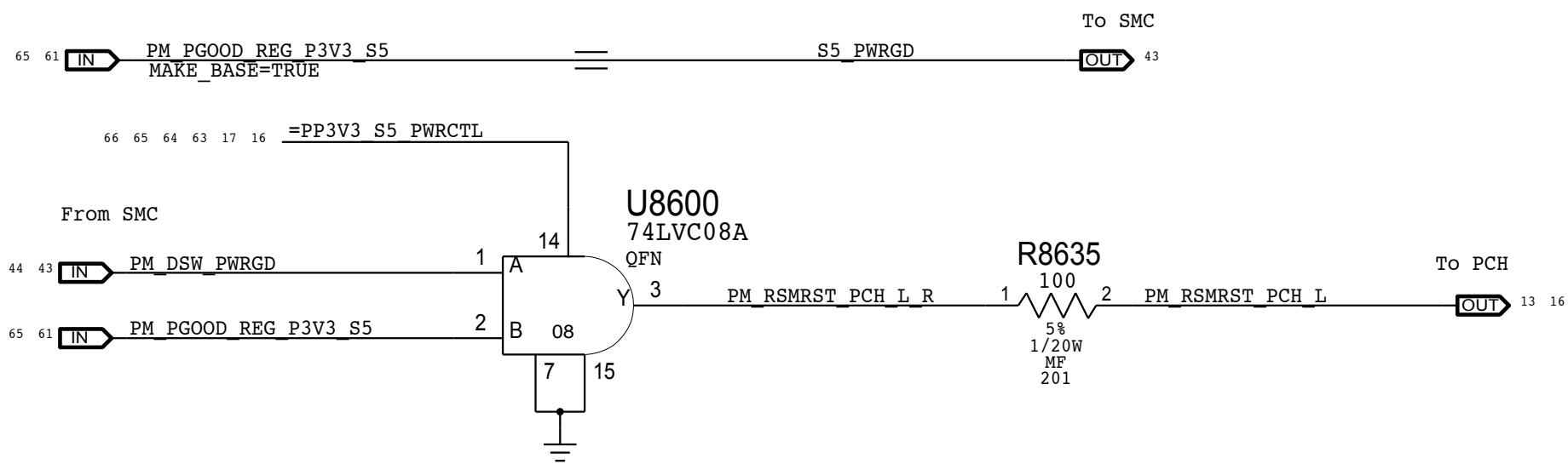
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

- Note:
- The iMac J70 design does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together
- Requirements:
- Power on:
- Asserted at least 10 ms after all suspend well power is valid
- Power off or loss of AC:
- Transition to 0.0V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

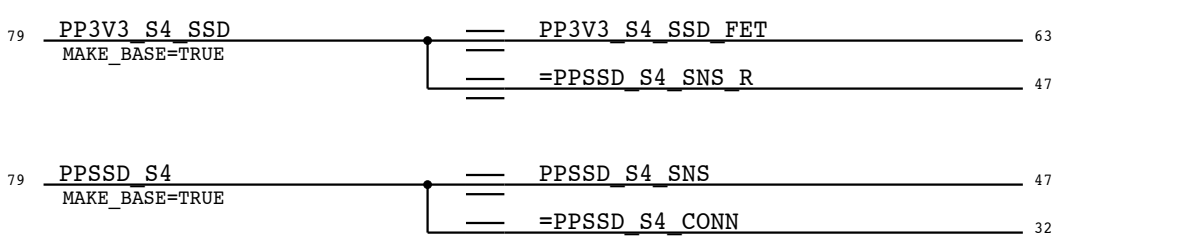
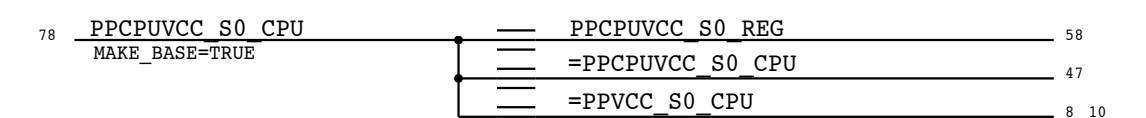
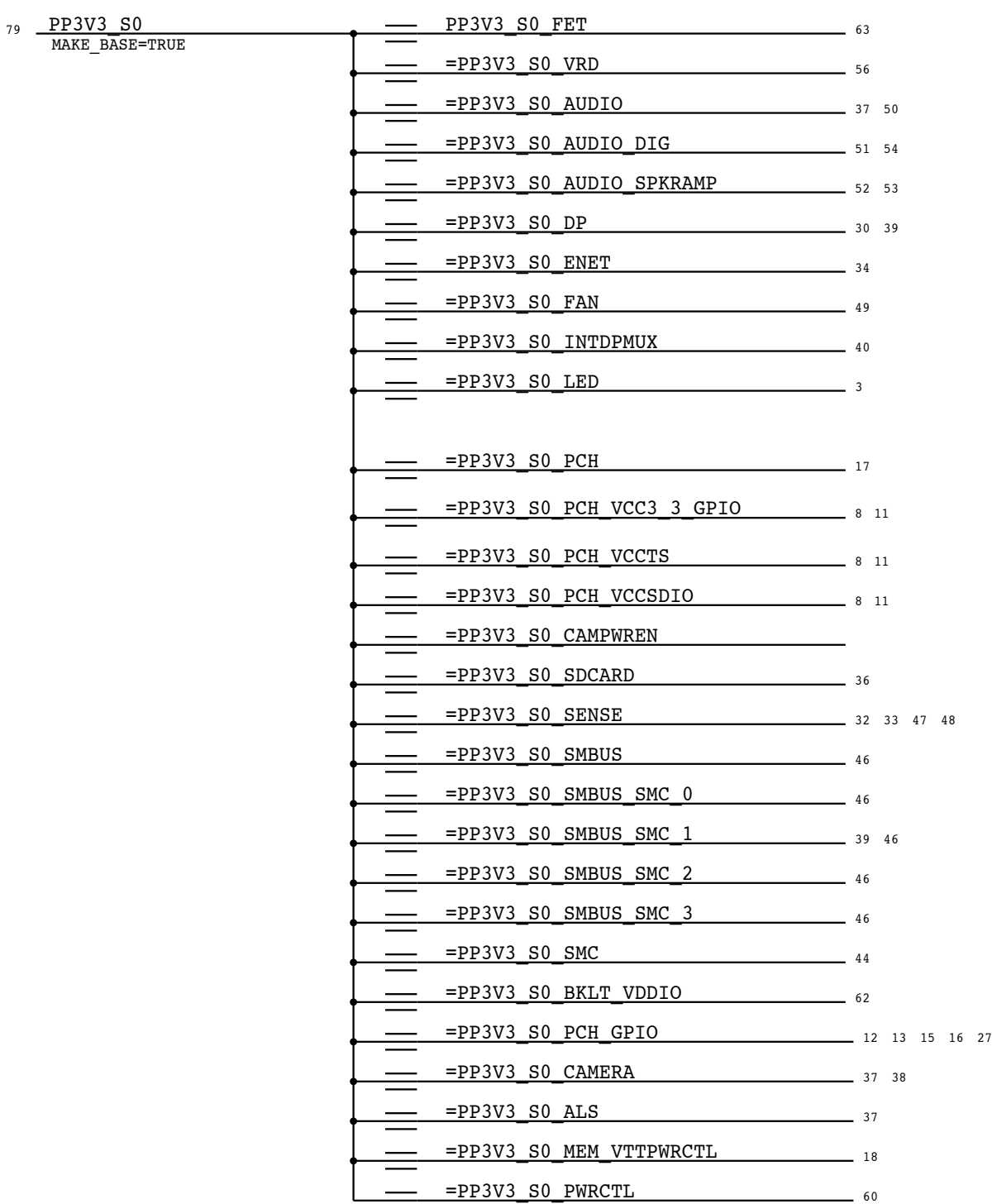
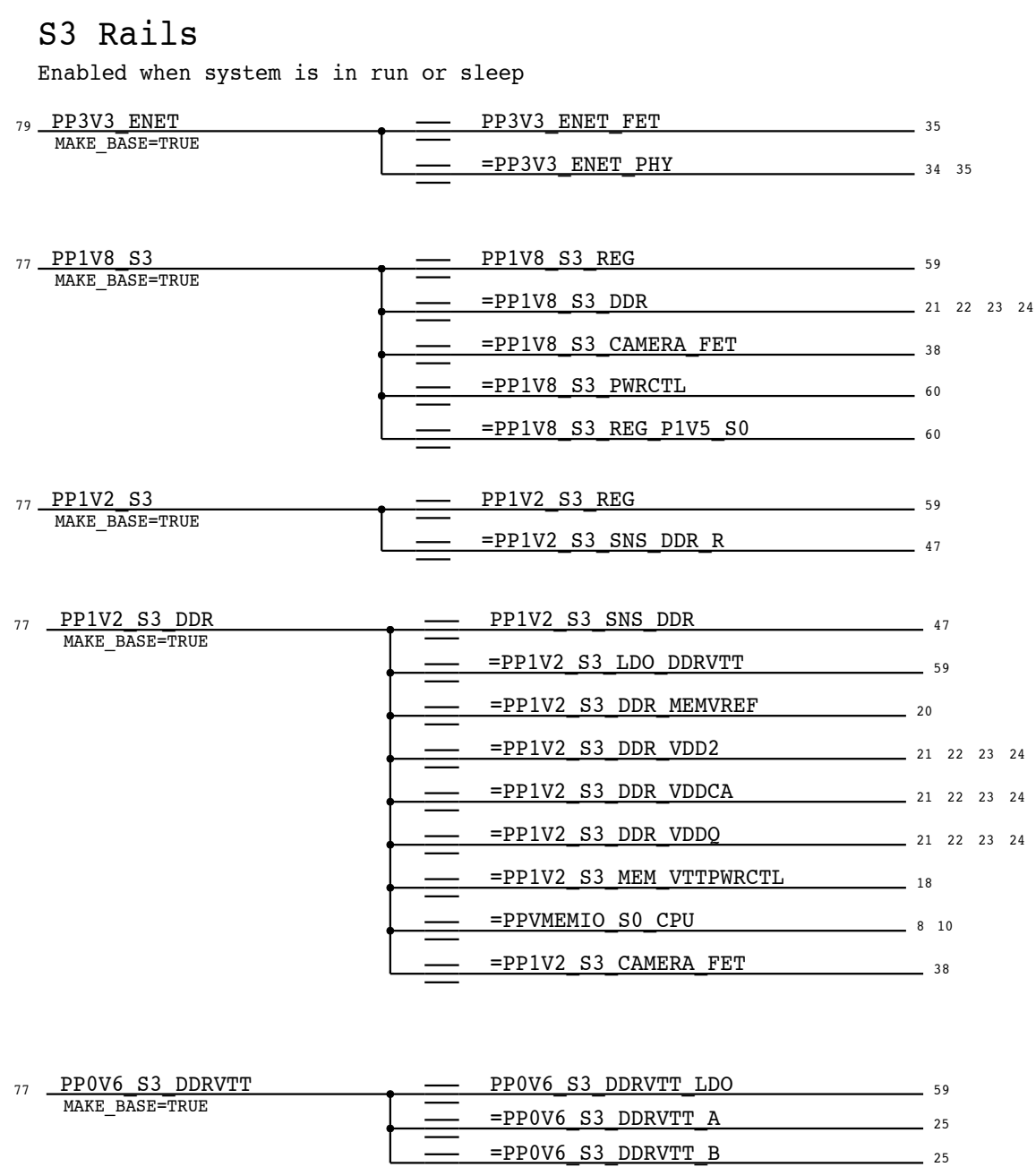
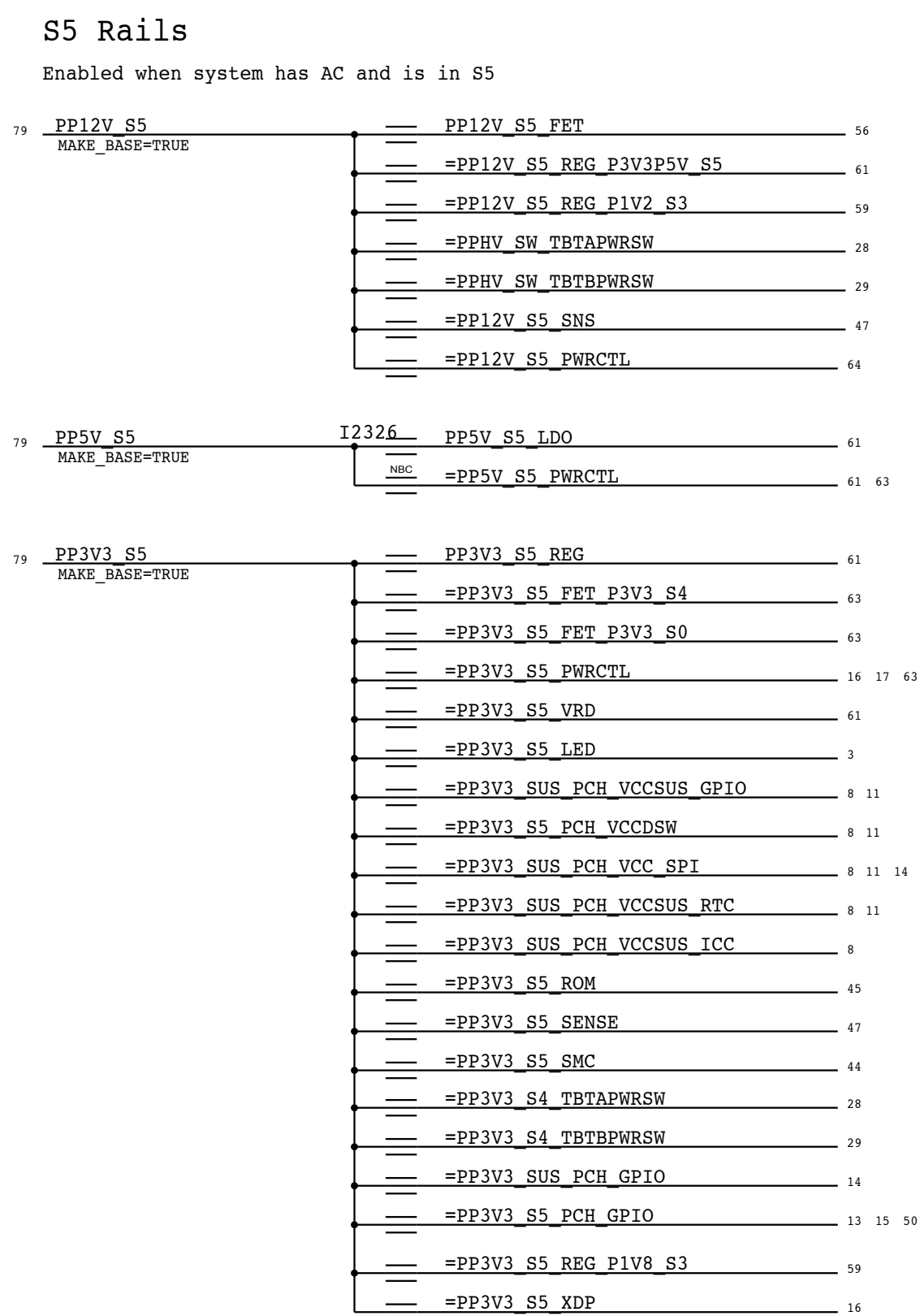
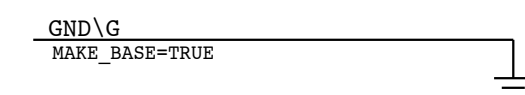
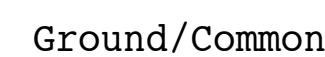
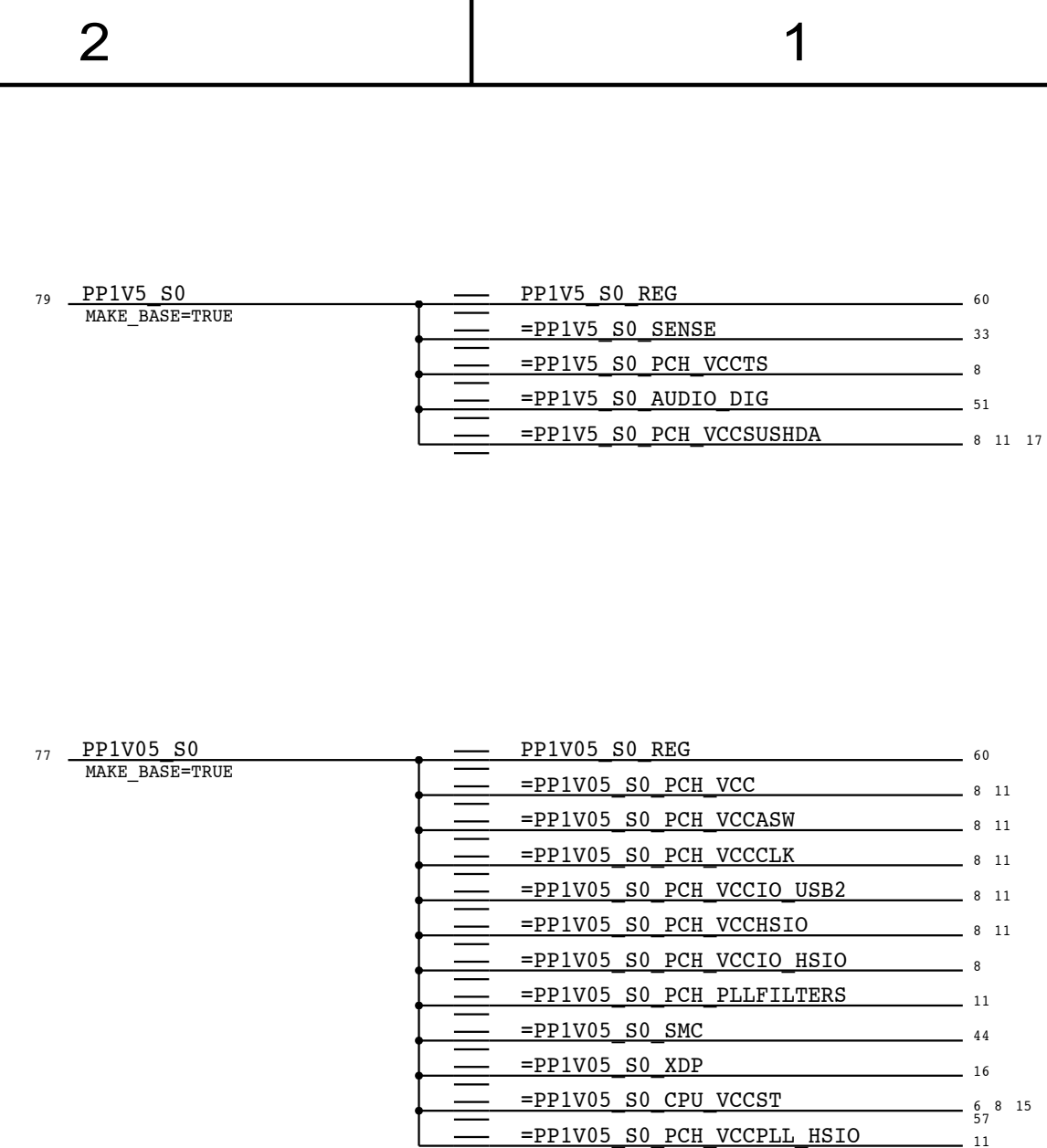
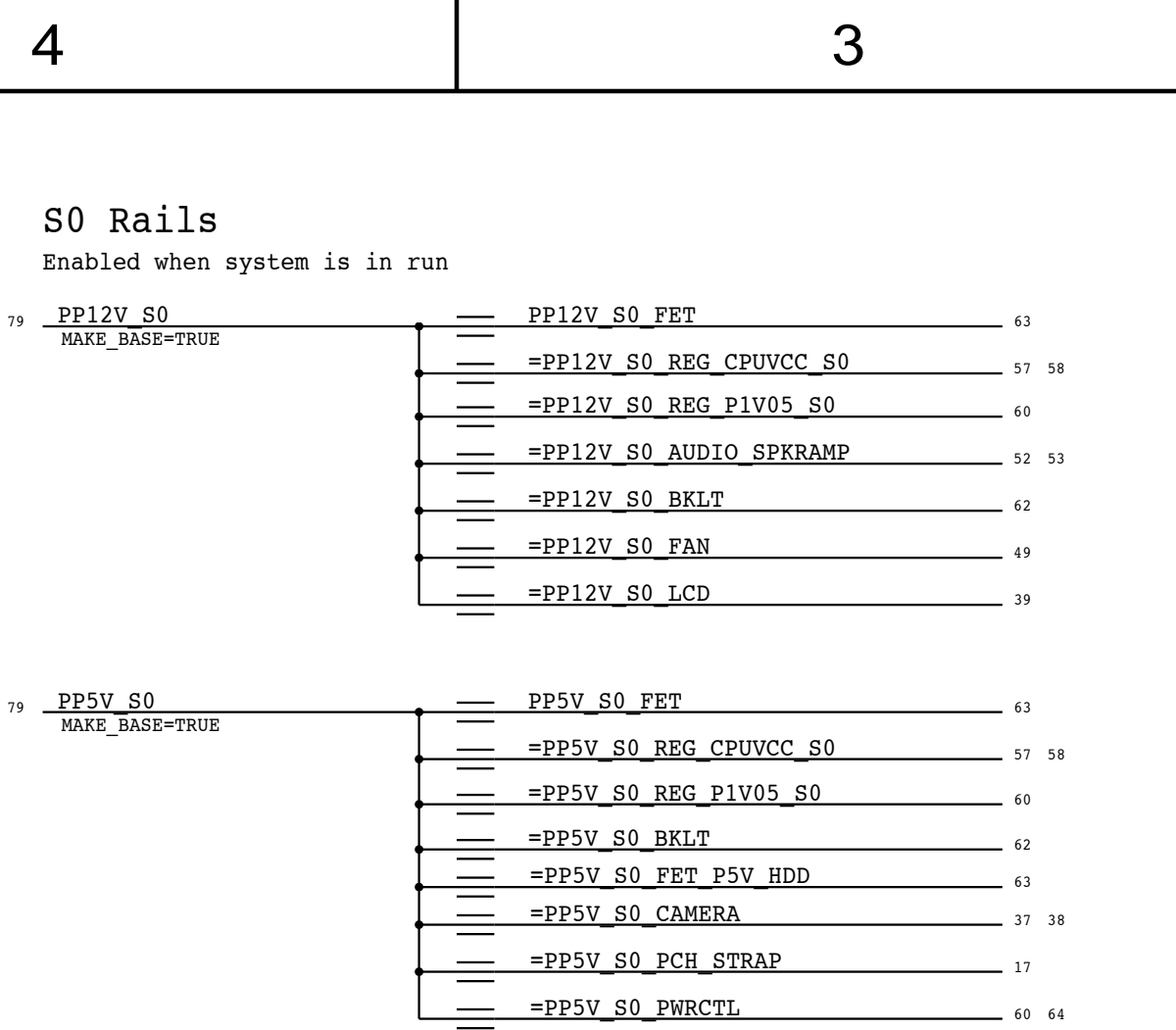
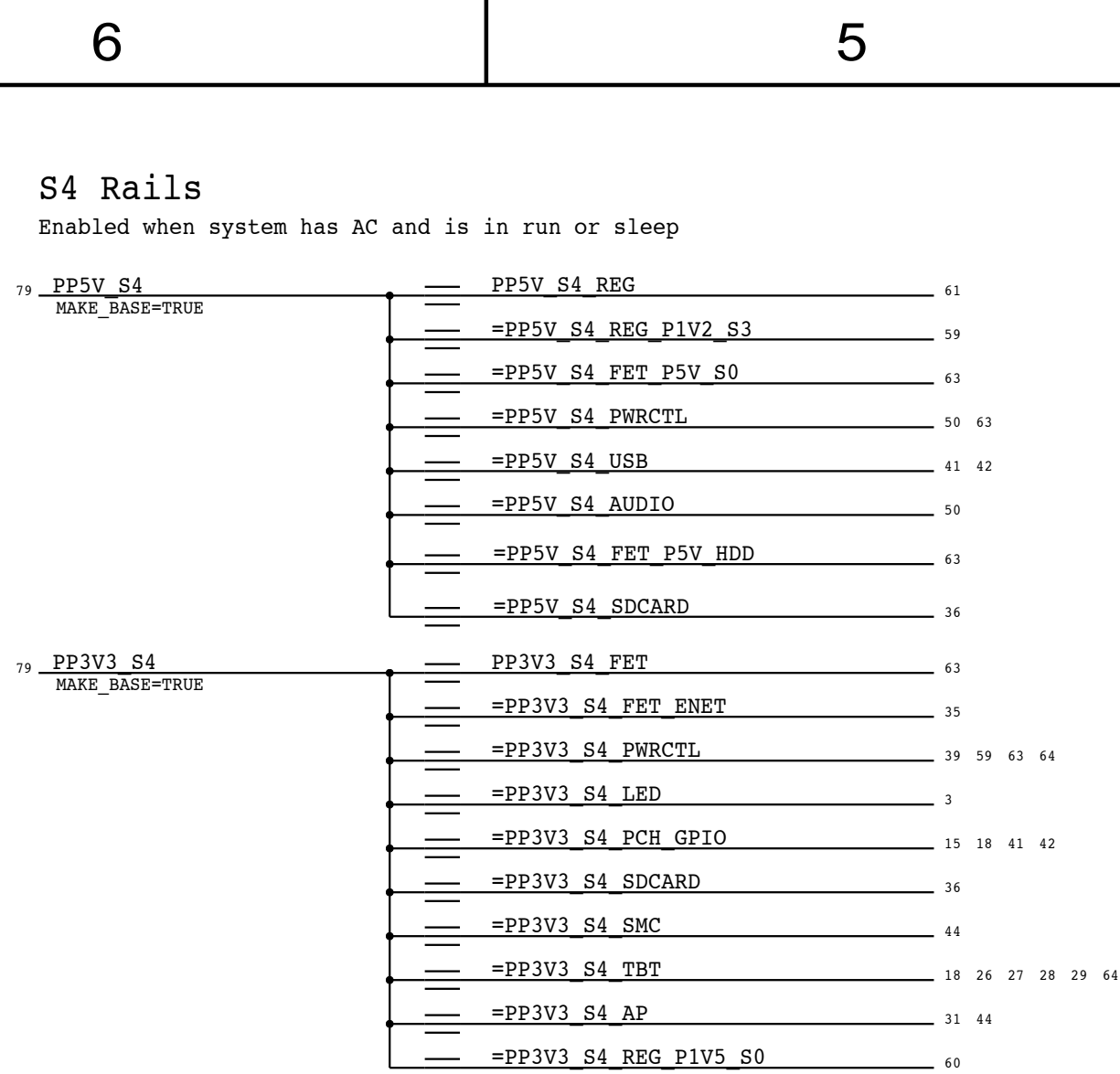
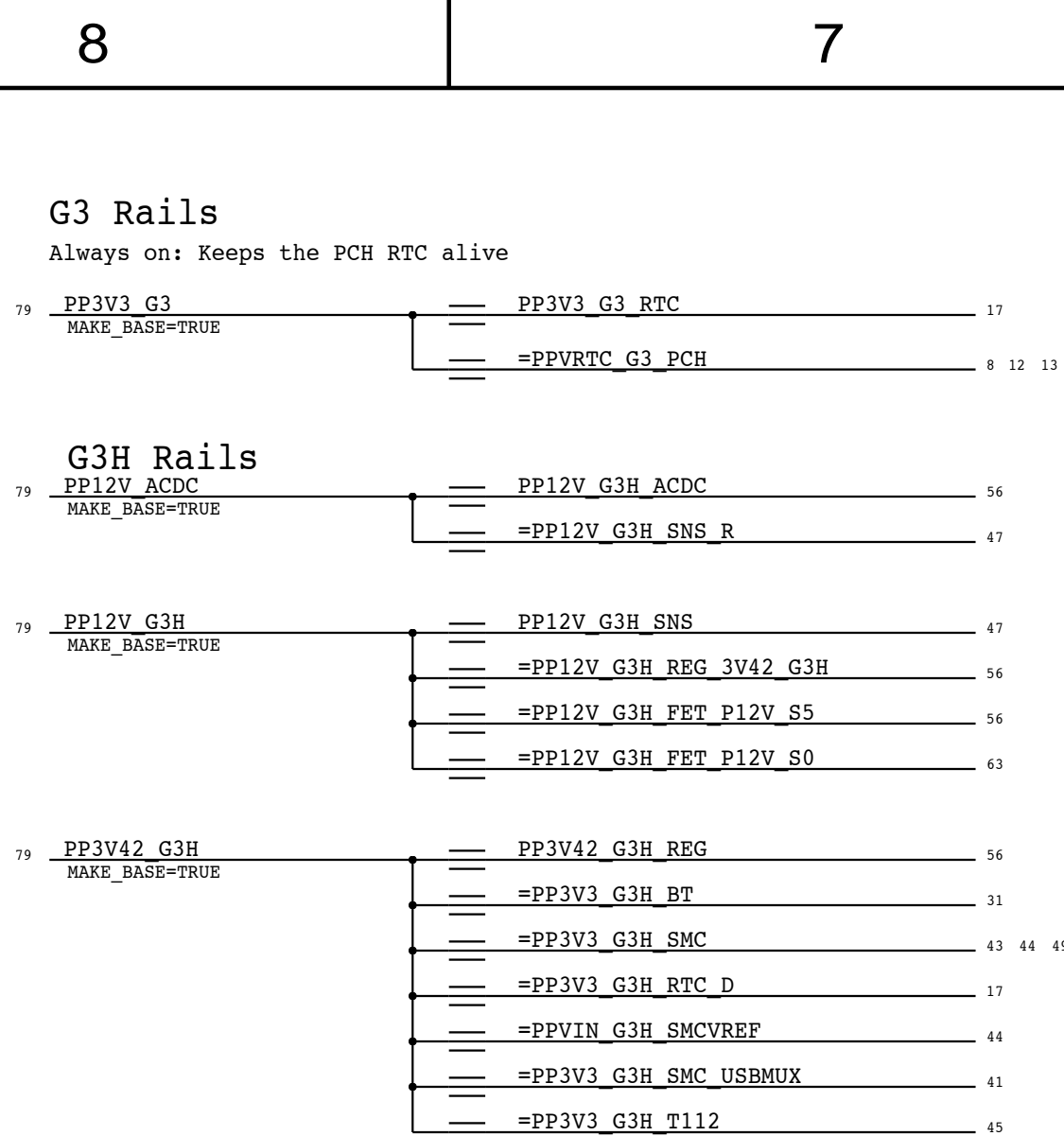
Method:

The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



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PM Power Good			
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PCH Miscellaneous

12	TP_HDA_SDI1	==	NC_HDA_SDI1	
			MAKE_BASE=TRUE	NO_TEST=1
12	TP_PCIE_CLK100M_CAMERAP	==	NC_PCIE_CLK100M_CAMERAP	
			MAKE_BASE=TRUE	NO_TEST=1
12	TP_PCIE_CLK100M_CAMERAN	==	NC_PCIE_CLK100M_CAMERAN	
			MAKE_BASE=TRUE	NO_TEST=1
12	TP_PCIE_CLK100M_FWP	==	NC_PCIE_CLK100M_FWP	
			MAKE_BASE=TRUE	NO_TEST=1
12	TP_PCIE_CLK100M_FWN	==	NC_PCIE_CLK100M_FWN	
			MAKE_BASE=TRUE	NO_TEST=1

Unused Thunderbolt Aliases

```

26  TP_TBT_PCIE RESET0 L      —      NC_TBT_PCIE RESET0 L
      —      MAKE BASE=TRUE      NO TEST=1

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SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
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Unused Signal Aliases			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00081		D
	REVISION		
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DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
DDR_73D	*	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF
DDR_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_70D
DDR_CTRL_PHY	*	DDR_40S
DDR_CMD_PHY	*	DDR_40S
DDR_DQ_PHY	*	DDR_40S
DDR_DQS_PHY	*	DDR_70D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=4:1_SPACING	?
DDR_STROBE_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?
DDR_COMP_ISO	*	0.381 MM	?

Main Segment Min Spacing Rules (mils) (HSW U/Y PDG, Intel Doc# 502636)

BDW does not have a separate power spec from HSW, so this document is still accurate

Table	Trace	Design	Iso	Design	Comments
6-14	4	(diff)	16	19.69	CLK trace spacing controlled by =70_OHM_DIFF.
6-14	7.5	7.87	12	13.78	
6-14	7.5	7.87	12	13.78	
6-14	7.5	7.87	16	11.81	DQ to other signals not in the same bytelane (but not ch)
			12	11.81	DQS to other signals of the same channel
			7.5	7.87	DQ to DQ in the same bytelane of the same channel
			12	11.81	DQ to DQS in the same bytelane of the same channel
			16	11.81	DQ or DQS in different bytelanes of the same channel
			?	25.59	DQ or DQS in different channels. ISO RULE NOT IN PDG
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_COMP	*	*	DDR_COMP_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_STROBE_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_STROBE_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):

In order for the constraints DDR_*_DQ_BYTE* to =SAME to win out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

DDR3

Electrical Constraint Set		Physical	Spacing		
Channel A					
E1170	DDR_A_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK P<0>	7 21 25
E1170	DDR_A_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK N<0>	7 21 25
E1250	DDR_A_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK P<1>	7 22 25
E1250	DDR_A_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK N<1>	7 22 25
E1260	DDR_A_CS0	NDR_CTRL_PHY	NDR_CTRL	MEM_A_CS I<0>	7 21 22 25
E1180	DDR_A_CS1	NDR_CTRL_PHY	NDR_CTRL	MEM_A_CS I<1>	7 21 22 25
E1180	DDR_A_ODT	NDR_CTRL_PHY	NDR_CTRL	MEM_A_ODT<0>	7 21 22 25
E1180	?	?	?	MEM_A_CKE<1..0>	7 21 25
E1240	?	?	?	MEM_A_CKE<3..2>	7 22 25
E1180	?	?	?	MEM_A_CAA<9..0>	7 21 25
E1180	?	?	?	MEM_A_CAB<9..0>	7 22 25
E1180	?	?	?	MEM_A_DQ<7..0>	7 19
E1180	?	?	?	MEM_A_DQ<15..8>	7 19
E1180	?	?	?	MEM_A_DQ<23..16>	7 19
E1180	?	?	?	MEM_A_DQ<31..24>	7 19
E1180	?	?	?	MEM_A_DQ<39..32>	7 19
E1180	?	?	?	MEM_A_DQ<47..40>	7 19
E1180	?	?	?	MEM_A_DQ<55..48>	7 19
E1180	?	?	?	MEM_A_DQ<63..56>	7 19
E1180	DDR_A_DQS0	NDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS P<0>	7 19
E1190	DDR_A_DQS0	NDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS N<0>	7 19
E1190	DDR_A_DQS1	NDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS P<1>	7 19
E1190	DDR_A_DQS1	NDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS N<1>	7 19
E2000	DDR_A_DQS2	NDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS P<2>	7 19
E1900	DDR_A_DQS2	NDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS N<2>	7 19
E2000	DDR_A_DQS3	NDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS P<3>	7 19
E1900	DDR_A_DQS3	NDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS N<3>	7 19
E2000	DDR_A_DQS4	NDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS P<4>	7 19
E2000	DDR_A_DQS4	NDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS N<4>	7 19
E2000	DDR_A_DQS5	NDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS P<5>	7 19
E2000	DDR_A_DQS5	NDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS N<5>	7 19
E2000	DDR_A_DQS6	NDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS P<6>	7 19
E2000	DDR_A_DQS6	NDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS N<6>	7 19
E2100	DDR_A_DQS7	NDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS P<7>	7 19
E2100	DDR_A_DQS7	NDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS N<7>	7 19
Channel B					
E2110	DDR_B_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK P<0>	7 23 25
E2110	DDR_B_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK N<0>	7 23 25
E2400	DDR_B_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK P<1>	7 24 25
E2400	DDR_B_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK N<1>	7 24 25
E2600	DDR_B_CS0	NDR_CTRL_PHY	NDR_CTRL	MEM_B_CS I<0>	7 23 24 25
E2100	DDR_B_CS1	NDR_CTRL_PHY	NDR_CTRL	MEM_B_CS I<1>	7 23 24 25
E2100	DDR_B_ODT	NDR_CTRL_PHY	NDR_CTRL	MEM_B_ODT<0>	7 23 24 25
E2110	?	?	?	MEM_B_CKE<1..0>	7 23 25
E2500	?	?	?	MEM_B_CKE<3..2>	7 24 25
E2100	?	?	?	MEM_B_CAA<9..0>	7 23 25
E2100	?	?	?	MEM_B_CAB<9..0>	7 24 25
E2220	?	?	?	MEM_B_DQ<7..0>	7 19
E2220	?	?	?	MEM_B_DQ<15..8>	7 19
E2220	?	?	?	MEM_B_DQ<23..16>	7 19
E2220	?	?	?	MEM_B_DQ<31..24>	7 19
E2220	?	?	?	MEM_B_DQ<39..32>	7 19
E2220	?	?	?	MEM_B_DQ<47..40>	7 19
E2220	?	?	?	MEM_B_DQ<55..48>	7 19
E2220	?	?	?	MEM_B_DQ<63..56>	7 19
E2300	DDR_B_DQS0	NDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS P<0>	7 19
E2310	DDR_B_DQS0	NDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS N<0>	7 19
E2320	DDR_B_DQS1	NDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS P<1>	7 19
E2330	DDR_B_DQS1	NDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS N<1>	7 19
E2340	DDR_B_DQS2	NDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS P<2>	7 19
E2350	DDR_B_DQS2	NDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS N<2>	7 19
E2360	DDR_B_DQS3	NDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS P<3>	7 19
E2370	DDR_B_DQS3	NDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS N<3>	7 19
E2380	DDR_B_DQS4	NDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS P<4>	7 19
E2390	DDR_B_DQS4	NDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS N<4>	7 19
E2400	DDR_B_DQS5	NDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS P<5>	7 19
E2410	DDR_B_DQS5	NDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS N<5>	7 19
E2420	DDR_B_DQS6	NDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS P<6>	7 19
E2430	DDR_B_DQS6	NDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS N<6>	7 19
E2440	DDR_B_DQS7	NDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS P<7>	7 19
E2450	DDR_B_DQS7	NDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS N<7>	7 19
SM COMP					
E2500	?	?		CPU_SM_RCOMP<0..2>	6

SYNC_MASTER=J70_NICK		SYNC_DATE=09/30/2013	
PAGE TITLE			
DDR3 Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
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SATA/FDI/XDP Constraints

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PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH_ISO	*	=4:1_SPACING	?	CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH_ISO	*	=2:1_SPACING	?	COMP_PCH	*	*	COMP_PCH_ISO

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC_ISO	*	=1.5:1_SPACING	?	LPC	*	*	LPC_ISO
CLK_LPC_ISO	*	=2:1_SPACING	?	CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA_ISO	*	=2X_DIELECTRIC	?	HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL_ISO	*	=4X_DIELECTRIC	?	XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI_ISO	*	=2:1_SPACING	?	SPI	*	*	SPI_ISO

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	2	2	LPC AD<3..0> 14 43
	2	2	LPC AD_R<3..0> 14
	LPC_55S	LPC	LPC FRAME_L 14 43
	LPC_55S	LPC	LPC FRAME_R_L 14
LPC Clocks			
	CLK_LPC_55S	CLK_LPC	LPC CLK24M_SMC 17 43
	CLK_LPC_55S	CLK_LPC	LPC CLK24M_SMC_R 12 17

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
	CLK_XTAL	XTAL	PCH CLK24M_XTAL_IN 12 17
	CLK_XTAL	XTAL	PCH CLK24M_XTAL_OUT 12 17
	CLK_XTAL	XTAL	PCH CLK24M_XTAL_OUT_R 17
PCH RTC 32K			
	CLK_XTAL	XTAL	PCH CLK32K_RTCX1 12 17
	CLK_XTAL	XTAL	PCH CLK32K_RTCX2 12 17
	CLK_XTAL	XTAL	PCH CLK32K_RTCX2_R 17
SMC 32K			
	CLK_PCH_55S	CLK_PCH	PM CLK32K_SUSCLK_R 13 44
	CLK_PCH_55S	CLK_PCH	SMC CLK32K 43 44

25 MHz XTALS

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
	CLK_XTAL	XTAL	TBT CLK25M_IN 17 26
	CLK_XTAL	XTAL	TBT CLK25M_OUT 17 26
	CLK_XTAL	XTAL	TBT CLK25M_OUT_R 17
	CLK_XTAL	XTAL	ENET XTAL_IN 17 34
	CLK_XTAL	XTAL	ENET XTAL_OUT 17 34
	CLK_XTAL	XTAL	ENET XTAL_OUT_R 17

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
	HDA	HDA	HDA BIT_CLK 12 51
	HDA	HDA	HDA BIT_CLK_R 12
	HDA	HDA	HDA_RST_L 12 51
	HDA	HDA	HDA_RST_R_L 12
	HDA	HDA	HDA_SDOUT 12 51
	HDA	HDA	HDA_SDOUT_R 12 17
	HDA	HDA	HDA_SYNC 12 51
	HDA	HDA	HDA_SYNC_R 12
	HDA	HDA	HDA_SDINO 12 51
SPDIF			
	HDA	HDA	DP_INT_SPDIF_AUDIO 39 51 80
	HDA_55S	2	SPDIF_OUT_JACK 51 54
	HDA_55S	2	CS4208_SPDIF_IN 51
	HDA_55S	2	CS4208_SPDIF_OUT 51

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
	SPI_50S	SPI	SPI_CLK_R 14 45
	SPI_50S	SPI	SPI_CLK 45
	SPI_50S	SPI	SPI_ALT_CLK 45
	SPI_50S	SPI	SPI_SMC_CLK 43 45
	SPI_50S	SPI	SPI_MLB_CLK 45
	SPI_50S	SPI	SPI_CS0_R_L 14 45
	SPI_50S	SPI	SPI_CS0_L 45
	SPI_50S	SPI	SPI_ALT_CS_L 45
	SPI_50S	SPI	SPI_SMC_CS_L 43 45
	SPI_50S	SPI	SPI_MLB_CS_L 45
	SPI_50S	SPI	SPI_MOSI_R 14 45
	SPI_50S	SPI	SPI_MOSI 45
	SPI_50S	SPI	SPI_ALT_MOSI 45
	SPI_50S	SPI	SPI_SMC_MOSI 43 45
	SPI_50S	SPI	SPI_MLB_MOSI 45
	SPI_50S	SPI	SPI_MISO 45
	SPI_50S	SPI	SPI_ALT_MISO 45
	SPI_50S	SPI	SPI_SMC_MISO 43 45
	SPI_50S	SPI	SPI_MLB_MISO 45
	SPI_50S	SPI	SPIROM_USE_MLB 15 45

SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=1.5:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SMC Generic Control Line Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMC_ISO	*	=1:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMC_CTRL	*	*	SMC_ISO

SMC Generic Control Line Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMC_GEN	*	SMC_50S

Current/Voltage Sense

Electrical Constraint Set	Physical	Spacing	
Common			
H1		SENSE	GND_SMC_AVSS 63 47
12V S5 (System Total)			
H10	SNS_CURRENT	SNS_DIFF_PHY	SNS_P12VG3H_P 67
H11	SNS_CURRENT	SNS_DIFF_PHY	SNS_P12VG3H_N 67
H12		SENSE	ISNS_P12VG3H_R 67
H13		SENSE	ISNS_P12VG3H 64 67
H14		SENSE	VSNS_P12VG3H 64 67
SSD			
H15	SNS_CURRENT	SNS_DIFF_PHY	SNS_SSD_P 67
H17	SNS_CURRENT	SNS_DIFF_PHY	SNS_SSD_N 67
H18		?	ISNS_SSdS4_R 67
H19		?	ISNS_SSdS4 64 67
H20		SENSE	VSNS_P3V3_SSD 64 67
VDDQ S3 (DDR)			
H23	SNS_CURRENT	SNS_DIFF_PHY	SNS_P1V2_S3_DDR_P 67
H27	SNS_CURRENT	SNS_DIFF_PHY	SNS_P1V2_S3_DDR_N 67
H28		SENSE	ISNS_P1V2_S3_DDR_R 67
H29		SENSE	ISNS_P1V2_S3_DDR 64 67
H30		SENSE	VSNS_P1V2_S3_DDR 64 67
CPU Core			
H32		SENSE	CPUVR_IMON 67 57
H45		SENSE	ISNS_CPUVCC 64 67
H46		SENSE	VSNS_CPUVCC 64 67

SMC

Electrical Constraint Set	Physical	Spacing	
SMC			
H1	CLF_XTAL1	XTAL	SMC_XTAL 63 44
H2	CLF_XTAL1	XTAL	SMC_EXTAL 63 44
H3	SMC_GEN	SMC_CTL1	SMC_LRESET_L 18 63
H4	SMC_GEN	SMC_CTL1	SMC_RUNTIME_SCI_L 13 43
H5	SMC_GEN	SMC_CTL1	SMC_WAKE_SCI_L 15 43
H6	SMC_GEN	SMC_CTL1	SMC_FAN_0_CTL 63 49
H7	SMC_GEN	SMC_CTL1	SMC_FAN_0_TACH 63 49

SMBus

Electrical Constraint Set	Physical	Spacing	
SMC			
	SMB_PHY	SMA	SMBUS_SMC_0_50_SCL 43 46
	SMB_PHY	SMA	SMBUS_SMC_0_50_SDA 43 46
	SMB_PHY	SMA	SMBUS_SMC_1_50_SCL 43 46
	SMA_PHY	SMA	SMBUS_SMC_1_50_SDA 43 46
	SMB_PHY	SMA	SMBUS_SMC_2_50_SCL 43 46
	SMB_PHY	SMA	SMBUS_SMC_2_50_SDA 43 46
	SMB_PHY	SMA	SMBUS_SMC_3_SCL 43 46
	SMA_PHY	SMA	SMBUS_SMC_3_SDA 43 46
	SMA_PHY	SMA	SMBUS_SMC_5_G3H_SCL 43 44
	SMA_PHY	SMA	SMBUS_SMC_5_G3H_SDA 43 44
PCH			
	TBT_12C_55S	TBT_12C	SMBUS_PCH_CLK 14 46
	TBT_12C_55S	TBT_12C	SMBUS_PCH_DATA 14 46
	SMB_PHY	SMA	SML_PCH_0_CLK 14 46
	SMA_PHY	SMA	SML_PCH_0_DATA 14 46

Temperature Sense

Electrical Constraint Set	Physical	Spacing	
EMC1414-1 (Production)			
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_1_P 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_1_N 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_3_P 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_3_N 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ADCD_P 48 56
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_ADCD_N 48 56
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_2_P 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T1_2_N 48
TMP423 (Development)			
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_1_P 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_1_N 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_2_P 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_2_N 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_3_P 48
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS_T2_3_N 48
HDD Out-of-Band			
 HDD		SENSE	HDD_OOB1_D2R_L 33
 HDD		SENSE	HDD_OOB1_D2R_F_L 33
 HDD		SENSE	HDD_OOB1_D2R_R_L 33
 HDD		SENSE	SMC_OOB1_D2R_L 33 43
 HDD		SENSE	SMC_OOB1_R2D_L 33 43
 HDD		SENSE	SMC_OOB1_R2D_R_L 33
SSD Out-of-Band			
 SSD		SENSE	SMC_OOB2_R2D_L 32 44
 SSD		SENSE	SMC_OOB2_D2R_L 32 44

8		7		6		5		4		3		2		1																																																																																																																																																																																																																																																																		
D	CPU VCC Phases														D																																																																																																																																																																																																																																																																	
	<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th></tr><tr><td colspan="7">Input Bus</td></tr><tr><td>E124D</td><td>POWER</td><td>POWER</td><td>12V</td><td></td><td></td><td>PP12V_S0_REG_CPUVCC_VIN</td><td>57</td></tr><tr><td>E180D</td><td>POWER</td><td>POWER</td><td>5V</td><td></td><td></td><td>PP5V_S0_REG_CPUVCC_VDD</td><td>57</td></tr><tr><td colspan="7">Phase 1</td></tr><tr><td>E184D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUVCC_1</td><td>57 58</td></tr><tr><td>E187D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUVCC_1</td><td>58</td></tr><tr><td>E188D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUVCC_1</td><td>58</td></tr><tr><td>E189D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUVCC_1_RC</td><td>58</td></tr><tr><td>E190D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUVCC_1</td><td>58</td></tr><tr><td>E192D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUVCC_1</td><td>58</td></tr><tr><td>E192D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_SNUBBER_CPUVCC_1</td><td>58</td></tr><tr><td>E193D</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td><td>PPCPUVCC_S0_SENSE_1</td><td>58</td></tr><tr><td>E194D</td><td>ISNS_CPU_CORR</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_CPUVCC_ISNS1_M</td><td>58</td></tr><tr><td>E195D</td><td>ISNS_CPU_CORR</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_CPUVCC_ISNS1_P</td><td>58</td></tr><tr><td>E196D</td><td></td><td></td><td></td><td></td><td></td><td>REG_CPUVCC_ISEN1</td><td>57 58</td></tr><tr><td>E198D</td><td></td><td></td><td></td><td></td><td></td><td>REG_CPUVCC_ISUMN</td><td>57 58</td></tr><tr><td>E197D</td><td></td><td></td><td></td><td></td><td></td><td>REG_CPUVCC_ISUMP</td><td>57 58</td></tr><tr><td colspan="7">Phase 2</td></tr><tr><td>E110D</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUVCC_2</td><td>57 58</td></tr><tr><td>E110D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUVCC_2</td><td>58</td></tr><tr><td>E110D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUVCC_2</td><td>58</td></tr><tr><td>E110D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUVCC_2_RC</td><td>58</td></tr><tr><td>E194D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUVCC_2</td><td>58</td></tr><tr><td>E197D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUVCC_2</td><td>58</td></tr><tr><td>E110D</td><td>VR_DIDT_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_SNUBBER_CPUVCC_2</td><td>58</td></tr><tr><td>E114D</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td><td>PPCPUVCC_S0_SENSE_2</td><td>58</td></tr><tr><td>E114D</td><td>ISNS_CPU_CORR</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_CPUVCC_ISNS2_M</td><td>58</td></tr><tr><td>E115D</td><td>ISNS_CPU_CORR</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_CPUVCC_ISNS2_P</td><td>58</td></tr><tr><td>E115D</td><td></td><td></td><td></td><td></td><td></td><td>REG_CPUVCC_ISEN2</td><td>57 58</td></tr></table>															Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		Input Bus							E124D	POWER	POWER	12V			PP12V_S0_REG_CPUVCC_VIN	57	E180D	POWER	POWER	5V			PP5V_S0_REG_CPUVCC_VDD	57	Phase 1							E184D	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1	57 58	E187D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_1	58	E188D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1	58	E189D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1_RC	58	E190D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_1	58	E192D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_1	58	E192D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_1	58	E193D	POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_1	58	E194D	ISNS_CPU_CORR	SNS_DIFF_PHY				REG_CPUVCC_ISNS1_M	58	E195D	ISNS_CPU_CORR	SNS_DIFF_PHY				REG_CPUVCC_ISNS1_P	58	E196D						REG_CPUVCC_ISEN1	57 58	E198D						REG_CPUVCC_ISUMN	57 58	E197D						REG_CPUVCC_ISUMP	57 58	Phase 2							E110D	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2	57 58	E110D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUVCC_2	58	E110D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2	58	E110D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2_RC	58	E194D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUVCC_2	58	E197D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_2	58	E110D	VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_2	58	E114D	POWER	POWER	1.8V			PPCPUVCC_S0_SENSE_2	58	E114D	ISNS_CPU_CORR	SNS_DIFF_PHY				REG_CPUVCC_ISNS2_M	58	E115D	ISNS_CPU_CORR	SNS_DIFF_PHY				REG_CPUVCC_ISNS2_P	58	E115D						REG_CPUVCC_ISEN2	57 58	C																				
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Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTDP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBT_I2C_ISO	*	=2X_DIELECTRIC	?	TBTDP	*	*	TBTDP_ISO
TBT_SPI_ISO	*	=2X_DIELECTRIC	?	TBT_SPI	*	*	TBT_SPI_ISO
TBTDP_ISO	*	=5X_DIELECTRIC	?	TBT_I2C	*	*	TBT_I2C_ISO
TBTDP_ISO	TOP,BOTTOM	=7X_DIELECTRIC	?				

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DP_ISO	*	=3:1_SPACING	?	DISPLAYPORT	*	*	DP_ISO

Pairs should be within 100 mils of clock length.
Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

Electrical Constraint Set	Physical	Spacing			
E631	?	?	DP_TBTSNK0_ML_C P<3..0>	5 26	
E633	?	?	DP_TBTSNK0_ML_C N<3..0>	5 26	
E632	?	?	DP_TBTSNK0_ML_P<3..0>	26	
E634	?	?	DP_TBTSNK0_ML_N<3..0>	26	
E636	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P	13 26	
E635	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N	13 26	
E637	DP_TBTSNK0_AUX	DP_85D	DP_TBTSNK0_AUXCH_P	26	
E638	DP_TBTSNK0_AUX	DP_85D	DP_TBTSNK0_AUXCH_N	26	
E639	?	?	DP_TBTSNK1_ML_C P<3..0>	5 26	
E621	?	?	DP_TBTSNK1_ML_C N<3..0>	5 26	
E620	?	?	DP_TBTSNK1_ML_P<3..0>	26	
E623	?	?	DP_TBTSNK1_ML_N<3..0>	26	
E622	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P	13 26	
E624	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N	13 26	
E626	DP_TBTSNK1_AUX	DP_85D	DP_TBTSNK1_AUXCH_P	26	
E625	DP_TBTSNK1_AUX	DP_85D	DP_TBTSNK1_AUXCH_N	26	
E603	?	?	DP_TBTSRC_ML_P<1..0>	26 40	
E604	?	?	DP_TBTSRC_ML_N<1..0>	26 40	
E686	?	?	DP_TBTSRC_ML_C P<1..0>	26	
E693	?	?	DP_TBTSRC_ML_C N<1..0>	26	
E629	DP_INTPNL_TBT_AUX_MUX	DP_85D	DP_TBTSRC_AUX_P	26 40	
E630	DP_INTPNL_TBT_AUX_MUX	DP_85D	DP_TBTSRC_AUX_N	26 40	
E681	?	?	DP_TBTSRC_AUX_C_P	26	
E692	?	?	DP_TBTSRC_AUX_C_N	26	
E633	TBT_SPI_CLK	TBT_SPI_55S	TBT_SPI	TBT_SPI_CLK	26
E634	TBT_SPI_MOSI	TBT_SPI_55S	TBT_SPI	TBT_SPI_MOSI	26
E636	TBT_SPI_MISO	TBT_SPI_55S	TBT_SPI	TBT_SPI_MISO	26
E635	TBT_SPI_CS_L	TBT_SPI_55S	TBT_SPI	TBT_SPI_CS_L	26

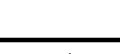
*: Only used on hosts supporting T29 video-in

TBT/DP Net Properties

Electrical Constraint Set		Physical	Spacing		
Port A					
E641	TBT_R2D_RVSD	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<1>	26 28
E642	TBT_R2D_RVSD	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<1>	26 28
E643		TBTDP_90D	TBTDP	TBT_A_R2D_P<1>	28
E644		TBTDP_90D	TBTDP	TBT_A_R2D_N<1>	28
E600	TBT_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_P<0>	26 28
E601	TBT_R2D	TBTDP_90D	TBTDP	TBT_A_R2D_C_N<0>	26 28
E602		TBTDP_90D	TBTDP	TBT_A_R2D_P<0>	28
E603		TBTDP_90D	TBTDP	TBT_A_R2D_N<0>	28
E695	DP_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	26 28
E696	DP_ML1	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	26 28
E645		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	28
E647		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	28
E680	DP_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	26 28
E681	DP_ML3	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	26 28
E678		DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	28
E679		DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	28
E646	DP_L5X	DP_85D	DISPLAYPORT	DP_A_L5X_ML_P<1>	28
E648	DP_L5X	DP_85D	DISPLAYPORT	DP_A_L5X_ML_N<1>	28
E644	TBT_D2R1_RVSD	TBTDP_90D	TBTDP	TBT_A_D2R_P<1>	26 28
E683	TBT_D2R1_RVSD	TBTDP_90D	TBTDP	TBT_A_D2R_N<1>	26 28
E611		TBTDP_90D	TBTDP	TBT_A_D2R_C_P<1>	28
E610		TBTDP_90D	TBTDP	TBT_A_D2R_C_N<1>	28
E675	TBT_D2R0_RVSD	TBTDP_90D	TBTDP	TBT_A_D2R_P<0>	26 28
E674	TBT_D2R0_RVSD	TBTDP_90D	TBTDP	TBT_A_D2R_N<0>	26 28
E690		TBTDP_90D	TBTDP	TBT_A_D2R_C_P<0>	28
E690		TBTDP_90D	TBTDP	TBT_A_D2R_C_N<0>	28
E650	TBT_AUXDDC	TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_P	28
E651	TBT_AUXDDC	TBTDP_90D	TBTDP	TBT_A_D2R1_AUXDDC_N	28
E659	TBT_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_P	26 28
E600	TBT_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_N	26 28
E655		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_P	28
E656		DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_N	28
Port B					
E692	TBT_R2D_RVSD	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<1>	26 29
E693	TBT_R2D_RVSD	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<1>	26 29
E695		TBTDP_90D	TBTDP	TBT_B_R2D_P<1>	29
E694		TBTDP_90D	TBTDP	TBT_B_R2D_N<1>	29
E604	TBT_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_P<0>	26 29
E605	TBT_R2D	TBTDP_90D	TBTDP	TBT_B_R2D_C_N<0>	26 29
E607		TBTDP_90D	TBTDP	TBT_B_R2D_P<0>	29
E606		TBTDP_90D	TBTDP	TBT_B_R2D_N<0>	29
E697	DP_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	26 29
E698	DP_ML1	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	26 29
E689		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	29
E682		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	29
E685	DP_ML3_RVSD	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	26 29
E684	DP_ML3_RVSD	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	26 29
E696		DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	29
E690		DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	29
E697	DP_L5X	DP_85D	DISPLAYPORT	DP_B_L5X_ML_P<1>	29
E698	DP_L5X	DP_85D	DISPLAYPORT	DP_B_L5X_ML_N<1>	29
E695	TBT_D2R1_RVSD	TBTDP_90D	TBTDP	TBT_B_D2R_P<1>	26 29
E664	TBT_D2R1_RVSD	TBTDP_90D	TBTDP	TBT_B_D2R_N<1>	26 29
E608		TBTDP_90D	TBTDP	TBT_B_D2R_C_P<1>	29
E609		TBTDP_90D	TBTDP	TBT_B_D2R_C_N<1>	29
E676	TBT_D2R0_RVSD	TBTDP_90D	TBTDP	TBT_B_D2R_P<0>	26 29
E697	TBT_D2R0_RVSD	TBTDP_90D	TBTDP	TBT_B_D2R_N<0>	26 29
E681		TBTDP_90D	TBTDP	TBT_B_D2R_C_P<0>	29
E683		TBTDP_90D	TBTDP	TBT_B_D2R_C_N<0>	29
E643	TBT_AUXDDC	TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_P	29
E648	TBT_AUXDDC	TBTDP_90D	TBTDP	TBT_B_D2R1_AUXDDC_N	29
E666	TBT_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_P	26 29
E697	TBT_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_N	26 29
E689		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_P	29
E690		DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_N	29

DisplayPort

Electrical Constraint Set	Physical	Spacing		
Graphics Source				
E642	?	?	DP_INT_ML_P<1..0>	5 40
E693	?	?	DP_INT_ML_N<1..0>	5 40
E644	DP_INTPNL_IG_AUX_MUX	DP_85D	DP_INT_AUX_P	5 40
E645	DP_INTPNL_IG_AUX_MUX	DP_85D	DP_INT_AUX_N	5 40
E646	?	?	DP_INT_AUX_C_P	
E647	?	?	DP_INT_AUX_C_N	
Internal Panel				
E605	?	?	DP_INTPNL_ML_C_P<1..0>	40
E606	?	?	DP_INTPNL_ML_C_N<1..0>	40
E607	?	?	DP_INTPNL_ML_P<1..0>	39 40
E608	?	?	DP_INTPNL_ML_N<1..0>	39 40
E609	DP_INTPNL_AUX_CONN	DP_85D	DP_INTPNL_AUX_P	39 40
E610	DP_INTPNL_AUX_CONN	DP_85D	DP_INTPNL_AUX_N	39 40
Internal DP SPDIF				
E611		HDA	DP_INT_SPDIF_AUDIO	39 51 74
DDC				
E692	TBT_I2C_55S	TBT_I2C	DP_TBTSNK0_DDC_CLK	13 30
E693	TBT_I2C_55S	TBT_I2C	DP_TBTSNK0_DDC_DATA	13 30
E694	TBT_I2C_55S	TBT_I2C	DP_TBTSNK1_DDC_CLK	13 30
E695	TBT_I2C_55S	TBT_I2C	DP_TBTSNK1_DDC_DATA	13 30
E698	TBT_I2C_55S	TBT_I2C	DP_TBTPA_DDC_CLK	28 30
E698	TBT_I2C_55S	TBT_I2C	DP_TBTPA_DDC_DATA	28 30
E697	TBT_I2C_55S	TBT_I2C	DP_TBTPB_DDC_CLK	29 30
E698	TBT_I2C_55S	TBT_I2C	DP_TBTPB_DDC_DATA	29 30

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TBT/DP Constraints			
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Backlight Controller

BLC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BLC_P6MM	*	Y	0.600 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_P3MM	*	Y	0.300 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_BLC	*	BLC_P6MM
POWER_BLC_RET	*	BLC_P3MM
BLC_CTL_PHY	*	BLC_P3MM

BLC-specific Spacing Definitions

BLC High Voltage Output

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_HV_ISO	*	0.45mm	1000

BLC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PHASE_ISO	*	=8:1_SPACING	2000
PHASE_SW2SW	*	=1:1_SPACING	?
PHASE_SW2PWR	*	=2:1_SPACING	?
PHASE_SW2GND	*	=2:1_SPACING	?

BLC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_CTL_ISO	*	=3:1_SPACING	?

Constraints

BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_CTL	*	BLC_CTL_ISO
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Is it chel'oh or sel'oh?

Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus				
U692 POWER	POWER	12V		PP12V_BKLT_SNS
U695 POWER	POWER	12V		PP12V_BKLT_FUSED
U690 POWER	POWER	12V		PP12V_S0_BKLT_FILT
U672 POWER	POWER	12V		PP12V_S0_BKLT_PWR
U681 POWER	POWER	12V		PP12V_S0_BKLT_PWR_R
U695 POWER	POWER	5V		PP5V_S0_BKLT_R
U696 POWER	POWER	3.3V		PP3V3_S0_BKLT_VDDIO_R
Local Ground				
U681 BLC_CTL_PHY	BLC_PHASE	0V		PGND_BKLT
U683 BLC_CTL_PHY	BLC_PHASE	0V		DGND_BKLT
U690 BLC_CTL_PHY	BLC_PHASE	0V		LGND_BKLT
Backlight				
U695 POWER_BLC	BLC_PHASE	80V	TRUE	BKLT_PHASE
U693 BLC_CTL_PHY	BLC_PHASE	80V	TRUE	BKLT_GATE
U684 BLC_CTL_PHY	BLC_PHASE	80V	TRUE	BKLT_GATE_R
U685 BLC_CTL_PHY	BLC_PHASE	80V	TRUE	BKLT_SHUBBER
U697 BLC_CTL_PHY	BLC_PHASE	12V	TRUE	BKLT_SW_R
U695 BLC_CTL_PHY	BLC_CTL			BKLT_ISET
U693 BLC_CTL_PHY	BLC_CTL			BKLT_FLT
U685 BLC_CTL_PHY	BLC_CTL			BKLT_FLT_RC
U697 SNS_DIFP_PHY	SENSE			BKLT_SW_P
U698 SNS_DIFP_PHY	SENSE			BKLT_SW_M
U682	SENSE			BKLT_FB
U681	BLC_HV	67V		BKLT_FB_XW
U684	BLC_HV	67V		BKLT_FB_R
U676 POWER_BLC_RET	BLC_CTL			BKLT_ISEN1
U675 POWER_BLC_RET	BLC_CTL			BKLT_ISEN2
U678 POWER_BLC_RET	BLC_CTL			BKLT_ISEN3
U677 POWER_BLC_RET	BLC_CTL			BKLT_ISEN4
U680 POWER_BLC_RET	BLC_CTL			BKLT_ISEN5
U679 POWER_BLC_RET	BLC_CTL			BKLT_ISEN6
U689 POWER_BLC_RET	BLC_HV			BKLT_ISEN1_R
U688 POWER_BLC_RET	BLC_HV			BKLT_ISEN2_R
U690 POWER_BLC_RET	BLC_HV			BKLT_ISEN3_R
U691 POWER_BLC_RET	BLC_HV			BKLT_ISEN4_R
U692 POWER_BLC_RET	BLC_HV			BKLT_ISEN5_R
U693 POWER_BLC_RET	BLC_HV			BKLT_ISEN6_R
U681 POWER_BLC_RET	BLC_HV			LED_RETURN_1
U682 POWER_BLC_RET	BLC_HV			LED_RETURN_2
U684 POWER_BLC_RET	BLC_HV			LED_RETURN_3
U685 POWER_BLC_RET	BLC_HV			LED_RETURN_4
U686 POWER_BLC_RET	BLC_HV			LED_RETURN_5
U687 POWER_BLC_RET	BLC_HV			LED_RETURN_6
Output Bus				
U670 POWER_BLC	BLC_HV	67V		BKLT_BOOST
U693 POWER_BLC	BLC_HV	67V		BKLT_BOOST_1
U674 POWER_BLC	BLC_HV	67V		BKLT_BOOST_2

Cello Miscellaneous

Electrical Constraint Set	Physical	Spacing
SPI		
U684	SMB_PHY	SMB
U683	SMB_PHY	SMB

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